Finite Rings

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The Sage Development Team

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CHAPTER

ONE

FINITE RINGS

1.1 Ring $\mathbf{Z}/n\mathbf{Z}$ of integers modulo n

EXAMPLES:

```
sage: R = Integers(97)
sage: a = R(5)
sage: a**1000000000000000000000000000000000000000000000000000000000000000
61
```

```
>>> from sage.all import *
>>> R = Integers(Integer(97))
>>> a = R(Integer(5))
>>> a**Integer(1000000000000000000000000000000000000000000000000000000000000000)
61
```

This example illustrates the relation between $\mathbf{Z}/p\mathbf{Z}$ and $\mathbf{F}_p$. In particular, there is a canonical map to $\mathbf{F}_p$, but not in the other direction.

```
sage: r = Integers(7)
sage: s = GF(7)
sage: r.has_coerce_map_from(s)
False
sage: s.has_coerce_map_from(r)
True
sage: s(1) + r(1)
2
sage: parent(s(1) + r(1))
Finite Field of size 7
sage: parent(r(1) + s(1))
Finite Field of size 7
```

```
>>> from sage.all import *
>>> r = Integers(Integer(7))
>>> s = GF(Integer(7))
>>> r.has_coerce_map_from(s)
False
>>> s.has_coerce_map_from(r)
True
>>> s(Integer(1)) + r(Integer(1))
2
```

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```
>>> parent(s(Integer(1)) + r(Integer(1)))
Finite Field of size 7
>>> parent(r(Integer(1)) + s(Integer(1)))
Finite Field of size 7
```

We list the elements of $\mathbf{Z}/3\mathbf{Z}$:

```
sage: R = Integers(3)
sage: list(R)
[0, 1, 2]
```

```
>>> from sage.all import *
>>> R = Integers(Integer(3))
>>> list(R)
[0, 1, 2]
```

AUTHORS:

- William Stein (initial code)
- David Joyner (2005-12-22): most examples
- Robert Bradshaw (2006-08-24): convert to SageX (Cython)
- William Stein (2007-04-29): square_roots_of_one
- Simon King (2011-04-21): allow to prescribe a category
- Simon King (2013-09): Only allow to prescribe the category of fields
- Kyle Hofmann (2024-02): New implementation of root-finding

class `sage.rings.finite_rings.integer_mod_ring.`**`IntegerModFactory`**

Bases: `UniqueFactory`

Return the quotient ring $\mathbf{Z}/n\mathbf{Z}$.

INPUT:

- `order` – integer (default: 0); positive or negative
- `is_field` – boolean (default: `False`); assert that the order is prime and hence the quotient ring belongs to the category of fields
- `category` – (optional) the category that the quotient ring belongs to

> **Note**
>
> The optional argument `is_field` is not part of the cache key. Hence, this factory will create precisely one instance of $\mathbf{Z}/n\mathbf{Z}$. However, if `is_field` is true, then a previously created instance of the quotient ring will be updated to be in the category of fields.
>
> **Use with care!** Erroneously putting $\mathbf{Z}/n\mathbf{Z}$ into the category of fields may have consequences that can compromise a whole Sage session, so that a restart will be needed.

EXAMPLES:

```
sage: IntegerModRing(15)
Ring of integers modulo 15
sage: IntegerModRing(7)
Ring of integers modulo 7
sage: IntegerModRing(-100)
Ring of integers modulo 100
```

```
>>> from sage.all import *
>>> IntegerModRing(Integer(15))
Ring of integers modulo 15
>>> IntegerModRing(Integer(7))
Ring of integers modulo 7
>>> IntegerModRing(-Integer(100))
Ring of integers modulo 100
```

Note that you can also use `Integers`, which is a synonym for `IntegerModRing`.

```
sage: Integers(18)
Ring of integers modulo 18
sage: Integers() is Integers(0) is ZZ
True
```

```
>>> from sage.all import *
>>> Integers(Integer(18))
Ring of integers modulo 18
>>> Integers() is Integers(Integer(0)) is ZZ
True
```

Note

Testing whether a quotient ring $\mathbf{Z}/n\mathbf{Z}$ is a field can of course be very costly. By default, it is not tested whether n is prime or not, in contrast to `GF()`. If the user is sure that the modulus is prime and wants to avoid a primality test, (s)he can provide `category=Fields()` when constructing the quotient ring, and then the result will behave like a field. If the category is not provided during initialisation, and it is found out later that the ring is in fact a field, then the category will be changed at runtime, having the same effect as providing `Fields()` during initialisation.

EXAMPLES:

```
sage: R = IntegerModRing(5)
sage: R.category()
Join of Category of finite commutative rings
    and Category of subquotients of monoids
    and Category of quotients of semigroups
    and Category of finite enumerated sets
sage: R in Fields()
True
sage: R.category()
Join of Category of finite enumerated fields
    and Category of subquotients of monoids
    and Category of quotients of semigroups
```

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```
sage: S = IntegerModRing(5, is_field=True)
sage: S is R
True
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(5))
>>> R.category()
Join of Category of finite commutative rings
    and Category of subquotients of monoids
    and Category of quotients of semigroups
    and Category of finite enumerated sets
>>> R in Fields()
True
>>> R.category()
Join of Category of finite enumerated fields
    and Category of subquotients of monoids
    and Category of quotients of semigroups
>>> S = IntegerModRing(Integer(5), is_field=True)
>>> S is R
True
```

Warning

If the optional argument `is_field` was used by mistake, there is currently no way to revert its impact, even though *`IntegerModRing_generic.is_field()`* with the optional argument `proof=True` would return the correct answer. So, prescribe `is_field=True` only if you know what your are doing!

EXAMPLES:

```
sage: R = IntegerModRing(33, is_field=True)
sage: R in Fields()
True
sage: R.is_field()
True
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(33), is_field=True)
>>> R in Fields()
True
>>> R.is_field()
True
```

If the optional argument $proof = True$ is provided, primality is tested and the mistaken category assignment is reported:

```
sage: R.is_field(proof=True)
Traceback (most recent call last):
...
ValueError: THIS SAGE SESSION MIGHT BE SERIOUSLY COMPROMISED!
The order 33 is not prime, but this ring has been put
```

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```
into the category of fields. This may already have consequences
in other parts of Sage. Either it was a mistake of the user,
or a probabilistic primality test has failed.
In the latter case, please inform the developers.
```

```
>>> from sage.all import *
>>> R.is_field(proof=True)
Traceback (most recent call last):
...
ValueError: THIS SAGE SESSION MIGHT BE SERIOUSLY COMPROMISED!
The order 33 is not prime, but this ring has been put
into the category of fields. This may already have consequences
in other parts of Sage. Either it was a mistake of the user,
or a probabilistic primality test has failed.
In the latter case, please inform the developers.
```

However, the mistaken assignment is not automatically corrected:

```
sage: R in Fields()
True
```

```
>>> from sage.all import *
>>> R in Fields()
True
```

To avoid side-effects of this test on other tests, we clear the cache of the ring factory:

```
sage: IntegerModRing._cache.clear()
```

```
>>> from sage.all import *
>>> IntegerModRing._cache.clear()
```

create_key_and_extra_args(*order=0*, *is_field=False*, *category=None*)

An integer mod ring is specified uniquely by its order.

EXAMPLES:

```
sage: Zmod.create_key_and_extra_args(7)
(7, {})
sage: Zmod.create_key_and_extra_args(7, True)
(7, {'category': Category of fields})
```

```
>>> from sage.all import *
>>> Zmod.create_key_and_extra_args(Integer(7))
(7, {})
>>> Zmod.create_key_and_extra_args(Integer(7), True)
(7, {'category': Category of fields})
```

create_object(*version*, *order*, ***kwds*)

EXAMPLES:

```
sage: R = Integers(10)
sage: TestSuite(R).run() # indirect doctest
```

```
>>> from sage.all import *
>>> R = Integers(Integer(10))
>>> TestSuite(R).run() # indirect doctest
```

get_object(*version*, *key*, *extra_args*)

class sage.rings.finite_rings.integer_mod_ring.**IntegerModRing_generic**(*order*, *cache=None*, *category=None*)

Bases: QuotientRing_generic, IntegerModRing

The ring of integers modulo N.

INPUT:

- order – integer
- category – a subcategory of CommutativeRings() (the default)

OUTPUT: the ring of integers modulo N

EXAMPLES:

First we compute with integers modulo 29.

```
sage: FF = IntegerModRing(29)
sage: FF
Ring of integers modulo 29
sage: FF.category()
Join of Category of finite commutative rings
    and Category of subquotients of monoids
    and Category of quotients of semigroups
    and Category of finite enumerated sets
sage: FF.is_field()
True
sage: FF.characteristic()
29
sage: FF.order()
29

sage: # needs sage.groups
sage: gens = FF.unit_gens()
sage: a = gens[0]
sage: a
2
sage: a.is_square()
False
sage: def pow(i): return a**i
sage: [pow(i) for i in range(16)]
[1, 2, 4, 8, 16, 3, 6, 12, 24, 19, 9, 18, 7, 14, 28, 27]
sage: TestSuite(FF).run()
```

```
>>> from sage.all import *
>>> FF = IntegerModRing(Integer(29))
```

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```
>>> FF
Ring of integers modulo 29
>>> FF.category()
Join of Category of finite commutative rings
    and Category of subquotients of monoids
    and Category of quotients of semigroups
    and Category of finite enumerated sets
>>> FF.is_field()
True
>>> FF.characteristic()
29
>>> FF.order()
29

>>> # needs sage.groups
>>> gens = FF.unit_gens()
>>> a = gens[Integer(0)]
>>> a
2
>>> a.is_square()
False
>>> def pow(i): return a**i
>>> [pow(i) for i in range(Integer(16))]
[1, 2, 4, 8, 16, 3, 6, 12, 24, 19, 9, 18, 7, 14, 28, 27]
>>> TestSuite(FF).run()
```

We have seen above that an integer mod ring is, by default, not initialised as an object in the category of fields. However, one can force it to be. Moreover, testing containment in the category of fields my re-initialise the category of the integer mod ring:

```
sage: F19 = IntegerModRing(19, is_field=True)
sage: F19.category().is_subcategory(Fields())
True
sage: F23 = IntegerModRing(23)
sage: F23.category().is_subcategory(Fields())
False
sage: F23 in Fields()
True
sage: F23.category().is_subcategory(Fields())
True
sage: TestSuite(F19).run()
sage: TestSuite(F23).run()
```

```
>>> from sage.all import *
>>> F19 = IntegerModRing(Integer(19), is_field=True)
>>> F19.category().is_subcategory(Fields())
True
>>> F23 = IntegerModRing(Integer(23))
>>> F23.category().is_subcategory(Fields())
False
>>> F23 in Fields()
True
```

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```
>>> F23.category().is_subcategory(Fields())
True
>>> TestSuite(F19).run()
>>> TestSuite(F23).run()
```

By Issue #15229, there is a unique instance of the integral quotient ring of a given order. Using the `IntegerModRing()` factory twice, and using `is_field=True` the second time, will update the category of the unique instance:

```
sage: F31a = IntegerModRing(31)
sage: F31a.category().is_subcategory(Fields())
False
sage: F31b = IntegerModRing(31, is_field=True)
sage: F31a is F31b
True
sage: F31a.category().is_subcategory(Fields())
True
```

```
>>> from sage.all import *
>>> F31a = IntegerModRing(Integer(31))
>>> F31a.category().is_subcategory(Fields())
False
>>> F31b = IntegerModRing(Integer(31), is_field=True)
>>> F31a is F31b
True
>>> F31a.category().is_subcategory(Fields())
True
```

Next we compute with the integers modulo 16.

```
sage: Z16 = IntegerModRing(16)
sage: Z16.category()
Join of Category of finite commutative rings
    and Category of subquotients of monoids
    and Category of quotients of semigroups
    and Category of finite enumerated sets
sage: Z16.is_field()
False
sage: Z16.order()
16
sage: Z16.characteristic()
16

sage: # needs sage.groups
sage: gens = Z16.unit_gens()
sage: gens
(15, 5)
sage: a = gens[0]
sage: b = gens[1]
sage: def powa(i): return a**i
sage: def powb(i): return b**i
sage: gp_exp = FF.unit_group_exponent()
sage: gp_exp
```

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```
28
sage: [powa(i) for i in range(15)]
[1, 15, 1, 15, 1, 15, 1, 15, 1, 15, 1, 15, 1, 15, 1]
sage: [powb(i) for i in range(15)]
[1, 5, 9, 13, 1, 5, 9, 13, 1, 5, 9, 13, 1, 5, 9]
sage: a.multiplicative_order()
2
sage: b.multiplicative_order()
4

sage: TestSuite(Z16).run()
```

```
>>> from sage.all import *
>>> Z16 = IntegerModRing(Integer(16))
>>> Z16.category()
Join of Category of finite commutative rings
    and Category of subquotients of monoids
    and Category of quotients of semigroups
    and Category of finite enumerated sets
>>> Z16.is_field()
False
>>> Z16.order()
16
>>> Z16.characteristic()
16

>>> # needs sage.groups
>>> gens = Z16.unit_gens()
>>> gens
(15, 5)
>>> a = gens[Integer(0)]
>>> b = gens[Integer(1)]
>>> def powa(i): return a**i
>>> def powb(i): return b**i
>>> gp_exp = FF.unit_group_exponent()
>>> gp_exp
28
>>> [powa(i) for i in range(Integer(15))]
[1, 15, 1, 15, 1, 15, 1, 15, 1, 15, 1, 15, 1, 15, 1]
>>> [powb(i) for i in range(Integer(15))]
[1, 5, 9, 13, 1, 5, 9, 13, 1, 5, 9, 13, 1, 5, 9]
>>> a.multiplicative_order()
2
>>> b.multiplicative_order()
4

>>> TestSuite(Z16).run()
```

Saving and loading:

```
sage: R = Integers(100000)
sage: TestSuite(R).run()  # long time (17s on sage.math, 2011)
```

```
>>> from sage.all import *
>>> R = Integers(Integer(100000))
>>> TestSuite(R).run()  # long time (17s on sage.math, 2011)
```

Testing ideals and quotients:

```
sage: Z10 = Integers(10)
sage: I = Z10.principal_ideal(0)
sage: Z10.quotient(I) == Z10
True
sage: I = Z10.principal_ideal(2)
sage: Z10.quotient(I) == Z10
False
sage: I.is_prime()
True
```

```
>>> from sage.all import *
>>> Z10 = Integers(Integer(10))
>>> I = Z10.principal_ideal(Integer(0))
>>> Z10.quotient(I) == Z10
True
>>> I = Z10.principal_ideal(Integer(2))
>>> Z10.quotient(I) == Z10
False
>>> I.is_prime()
True
```

```
sage: R = IntegerModRing(97)
sage: a = R(5)
sage: a**(10^62)
61
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(97))
>>> a = R(Integer(5))
>>> a**(Integer(10)**Integer(62))
61
```

cardinality()

Return the cardinality of this ring.

EXAMPLES:

```
sage: Zmod(87).cardinality()
87
```

```
>>> from sage.all import *
>>> Zmod(Integer(87)).cardinality()
87
```

characteristic()

EXAMPLES:

```
sage: R = IntegerModRing(18)
sage: FF = IntegerModRing(17)
sage: FF.characteristic()
17
sage: R.characteristic()
18
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(18))
>>> FF = IntegerModRing(Integer(17))
>>> FF.characteristic()
17
>>> R.characteristic()
18
```

degree()

Return 1.

EXAMPLES:

```
sage: R = Integers(12345678900)
sage: R.degree()
1
```

```
>>> from sage.all import *
>>> R = Integers(Integer(12345678900))
>>> R.degree()
1
```

extension(*poly*, *name=None*, *names=None*, ***kwds*)

Return an algebraic extension of `self`. See `sage.rings.ring.CommutativeRing.extension()` for more information.

EXAMPLES:

```
sage: R.<t> = QQ[]
sage: Integers(8).extension(t^2 - 3)
Univariate Quotient Polynomial Ring in t
 over Ring of integers modulo 8 with modulus t^2 + 5
```

```
>>> from sage.all import *
>>> R = QQ['t']; (t,) = R._first_ngens(1)
>>> Integers(Integer(8)).extension(t**Integer(2) - Integer(3))
Univariate Quotient Polynomial Ring in t
 over Ring of integers modulo 8 with modulus t^2 + 5
```

factored_order()

EXAMPLES:

```
sage: R = IntegerModRing(18)
sage: FF = IntegerModRing(17)
sage: R.factored_order()
2 * 3^2
```

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```
sage: FF.factored_order()
17
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(18))
>>> FF = IntegerModRing(Integer(17))
>>> R.factored_order()
2 * 3^2
>>> FF.factored_order()
17
```

factored_unit_order()

Return a list of `Factorization` objects, each the factorization of the order of the units in a $\mathbf{Z}/p^n\mathbf{Z}$ component of this group (using the Chinese Remainder Theorem).

EXAMPLES:

```
sage: R = Integers(8*9*25*17*29)
sage: R.factored_unit_order()
[2^2, 2 * 3, 2^2 * 5, 2^4, 2^2 * 7]
```

```
>>> from sage.all import *
>>> R = Integers(Integer(8)*Integer(9)*Integer(25)*Integer(17)*Integer(29))
>>> R.factored_unit_order()
[2^2, 2 * 3, 2^2 * 5, 2^4, 2^2 * 7]
```

field()

If this ring is a field, return the corresponding field as a finite field, which may have extra functionality and structure. Otherwise, raise a `ValueError`.

EXAMPLES:

```
sage: R = Integers(7); R
Ring of integers modulo 7
sage: R.field()
Finite Field of size 7
sage: R = Integers(9)
sage: R.field()
Traceback (most recent call last):
...
ValueError: self must be a field
```

```
>>> from sage.all import *
>>> R = Integers(Integer(7)); R
Ring of integers modulo 7
>>> R.field()
Finite Field of size 7
>>> R = Integers(Integer(9))
>>> R.field()
Traceback (most recent call last):
...
ValueError: self must be a field
```

is_field(*proof=None*)

Return `True` precisely if the order is prime.

INPUT:

- `proof` – boolean or `None` (default). If `False`, then test whether the category of the quotient is a subcategory of `Fields()`, or do a probabilistic primality test. If `None`, then test the category and then do a primality test according to the global arithmetic proof settings. If `True`, do a deterministic primality test.

If it is found (perhaps probabilistically) that the ring is a field, then the category of the ring is refined to include the category of fields. This may change the Python class of the ring!

EXAMPLES:

```
sage: R = IntegerModRing(18)
sage: R.is_field()
False
sage: FF = IntegerModRing(17)
sage: FF.is_field()
True
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(18))
>>> R.is_field()
False
>>> FF = IntegerModRing(Integer(17))
>>> FF.is_field()
True
```

By Issue #15229, the category of the ring is refined, if it is found that the ring is in fact a field:

```
sage: R = IntegerModRing(127)
sage: R.category()
Join of Category of finite commutative rings
    and Category of subquotients of monoids
    and Category of quotients of semigroups
    and Category of finite enumerated sets
sage: R.is_field()
True
sage: R.category()
Join of Category of finite enumerated fields
    and Category of subquotients of monoids
    and Category of quotients of semigroups
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(127))
>>> R.category()
Join of Category of finite commutative rings
    and Category of subquotients of monoids
    and Category of quotients of semigroups
    and Category of finite enumerated sets
>>> R.is_field()
True
>>> R.category()
```

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```
Join of Category of finite enumerated fields
    and Category of subquotients of monoids
    and Category of quotients of semigroups
```

It is possible to mistakenly put $\mathbf{Z}/n\mathbf{Z}$ into the category of fields. In this case, `is_field()` will return `True` without performing a primality check. However, if the optional argument `proof=True` is provided, primality is tested and the mistake is uncovered in a warning message:

```
sage: R = IntegerModRing(21, is_field=True)
sage: R.is_field()
True
sage: R.is_field(proof=True)
Traceback (most recent call last):
...
ValueError: THIS SAGE SESSION MIGHT BE SERIOUSLY COMPROMISED!
The order 21 is not prime, but this ring has been put
into the category of fields. This may already have consequences
in other parts of Sage. Either it was a mistake of the user,
or a probabilistic primality test has failed.
In the latter case, please inform the developers.
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(21), is_field=True)
>>> R.is_field()
True
>>> R.is_field(proof=True)
Traceback (most recent call last):
...
ValueError: THIS SAGE SESSION MIGHT BE SERIOUSLY COMPROMISED!
The order 21 is not prime, but this ring has been put
into the category of fields. This may already have consequences
in other parts of Sage. Either it was a mistake of the user,
or a probabilistic primality test has failed.
In the latter case, please inform the developers.
```

To avoid side-effects of this test on other tests, we clear the cache of the ring factory:

```
sage: IntegerModRing._cache.clear()
```

```
>>> from sage.all import *
>>> IntegerModRing._cache.clear()
```

is_integral_domain(*proof=None*)

Return `True` if and only if the order of `self` is prime.

EXAMPLES:

```
sage: Integers(389).is_integral_domain()
True
sage: Integers(389^2).is_integral_domain()                                  #␣
↪needs sage.libs.pari
False
```

```
>>> from sage.all import *
>>> Integers(Integer(389)).is_integral_domain()
True
>>> Integers(Integer(389)**Integer(2)).is_integral_domain()
↪                # needs sage.libs.pari
False
```

is_noetherian()

Check if `self` is a Noetherian ring.

EXAMPLES:

```
sage: Integers(8).is_noetherian()
True
```

```
>>> from sage.all import *
>>> Integers(Integer(8)).is_noetherian()
True
```

is_prime_field()

Return `True` if the order is prime.

EXAMPLES:

```
sage: Zmod(7).is_prime_field()
True
sage: Zmod(8).is_prime_field()
False
```

```
>>> from sage.all import *
>>> Zmod(Integer(7)).is_prime_field()
True
>>> Zmod(Integer(8)).is_prime_field()
False
```

is_unique_factorization_domain(*proof=None*)

Return `True` if and only if the order of `self` is prime.

EXAMPLES:

```
sage: Integers(389).is_unique_factorization_domain()
True
sage: Integers(389^2).is_unique_factorization_domain()                    #
↪needs sage.libs.pari
False
```

```
>>> from sage.all import *
>>> Integers(Integer(389)).is_unique_factorization_domain()
True
>>> Integers(Integer(389)**Integer(2)).is_unique_factorization_domain()
↪                # needs sage.libs.pari
False
```

krull_dimension()

Return the Krull dimension of `self`.

EXAMPLES:

```
sage: Integers(18).krull_dimension()
0
```

```
>>> from sage.all import *
>>> Integers(Integer(18)).krull_dimension()
0
```

list_of_elements_of_multiplicative_group()

Return a list of all invertible elements, as python ints.

EXAMPLES:

```
sage: R = Zmod(12)
sage: L = R.list_of_elements_of_multiplicative_group(); L
[1, 5, 7, 11]
sage: type(L[0])
<... 'int'>
sage: Zmod(1).list_of_elements_of_multiplicative_group()
[0]
```

```
>>> from sage.all import *
>>> R = Zmod(Integer(12))
>>> L = R.list_of_elements_of_multiplicative_group(); L
[1, 5, 7, 11]
>>> type(L[Integer(0)])
<... 'int'>
>>> Zmod(Integer(1)).list_of_elements_of_multiplicative_group()
[0]
```

modulus()

Return the polynomial $x - 1$ over this ring.

> **Note**
>
> This function exists for consistency with the finite-field modulus function.

EXAMPLES:

```
sage: R = IntegerModRing(18)
sage: R.modulus()
x + 17
sage: R = IntegerModRing(17)
sage: R.modulus()
x + 16
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(18))
```

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```
>>> R.modulus()
x + 17
>>> R = IntegerModRing(Integer(17))
>>> R.modulus()
x + 16
```

multiplicative_generator()

Return a generator for the multiplicative group of this ring, assuming the multiplicative group is cyclic.

Use the unit_gens function to obtain generators even in the non-cyclic case.

EXAMPLES:

```
sage: # needs sage.groups sage.libs.pari
sage: R = Integers(7); R
Ring of integers modulo 7
sage: R.multiplicative_generator()
3
sage: R = Integers(9)
sage: R.multiplicative_generator()
2
sage: Integers(8).multiplicative_generator()
Traceback (most recent call last):
...
ValueError: multiplicative group of this ring is not cyclic
sage: Integers(4).multiplicative_generator()
3
sage: Integers(25*3).multiplicative_generator()
Traceback (most recent call last):
...
ValueError: multiplicative group of this ring is not cyclic
sage: Integers(25*3).unit_gens()
(26, 52)
sage: Integers(162).unit_gens()
(83,)
```

```
>>> from sage.all import *
>>> # needs sage.groups sage.libs.pari
>>> R = Integers(Integer(7)); R
Ring of integers modulo 7
>>> R.multiplicative_generator()
3
>>> R = Integers(Integer(9))
>>> R.multiplicative_generator()
2
>>> Integers(Integer(8)).multiplicative_generator()
Traceback (most recent call last):
...
ValueError: multiplicative group of this ring is not cyclic
>>> Integers(Integer(4)).multiplicative_generator()
3
>>> Integers(Integer(25)*Integer(3)).multiplicative_generator()
Traceback (most recent call last):
```

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```
...
ValueError: multiplicative group of this ring is not cyclic
>>> Integers(Integer(25)*Integer(3)).unit_gens()
(26, 52)
>>> Integers(Integer(162)).unit_gens()
(83,)
```

multiplicative_group_is_cyclic()

Return `True` if the multiplicative group of this field is cyclic. This is the case exactly when the order is less than 8, a power of an odd prime, or twice a power of an odd prime.

EXAMPLES:

```
sage: R = Integers(7); R
Ring of integers modulo 7
sage: R.multiplicative_group_is_cyclic()
True
sage: R = Integers(9)
sage: R.multiplicative_group_is_cyclic()                                    #␣
↪needs sage.libs.pari
True
sage: Integers(8).multiplicative_group_is_cyclic()
False
sage: Integers(4).multiplicative_group_is_cyclic()
True
sage: Integers(25*3).multiplicative_group_is_cyclic()                       #␣
↪needs sage.libs.pari
False
```

```
>>> from sage.all import *
>>> R = Integers(Integer(7)); R
Ring of integers modulo 7
>>> R.multiplicative_group_is_cyclic()
True
>>> R = Integers(Integer(9))
>>> R.multiplicative_group_is_cyclic()                                     #␣
↪needs sage.libs.pari
True
>>> Integers(Integer(8)).multiplicative_group_is_cyclic()
False
>>> Integers(Integer(4)).multiplicative_group_is_cyclic()
True
>>> Integers(Integer(25)*Integer(3)).multiplicative_group_is_cyclic()        ␣
↪              # needs sage.libs.pari
False
```

We test that Issue #5250 is fixed:

```
sage: Integers(162).multiplicative_group_is_cyclic()                        #␣
↪needs sage.libs.pari
True
```

```
>>> from sage.all import *
>>> Integers(Integer(162)).multiplicative_group_is_cyclic()
↪     # needs sage.libs.pari
True
```

multiplicative_subgroups()

Return generators for each subgroup of $(\mathbf{Z}/N\mathbf{Z})^*$.

EXAMPLES:

```
sage: # optional - gap_package_polycyclic, needs sage.groups
sage: Integers(5).multiplicative_subgroups()
((2,), (4,), ())
sage: Integers(15).multiplicative_subgroups()
((11, 7), (11, 4), (2,), (11,), (14,), (7,), (4,), ())
sage: Integers(2).multiplicative_subgroups()
((),)
sage: len(Integers(341).multiplicative_subgroups())
80
```

```
>>> from sage.all import *
>>> # optional - gap_package_polycyclic, needs sage.groups
>>> Integers(Integer(5)).multiplicative_subgroups()
((2,), (4,), ())
>>> Integers(Integer(15)).multiplicative_subgroups()
((11, 7), (11, 4), (2,), (11,), (14,), (7,), (4,), ())
>>> Integers(Integer(2)).multiplicative_subgroups()
((),)
>>> len(Integers(Integer(341)).multiplicative_subgroups())
80
```

order()

Return the order of this ring.

EXAMPLES:

```
sage: Zmod(87).order()
87
```

```
>>> from sage.all import *
>>> Zmod(Integer(87)).order()
87
```

quadratic_nonresidue()

Return a quadratic non-residue in `self`.

EXAMPLES:

```
sage: R = Integers(17)
sage: R.quadratic_nonresidue()                                          #
↪needs sage.libs.pari
3
sage: R(3).is_square()
False
```

```
>>> from sage.all import *
>>> R = Integers(Integer(17))
>>> R.quadratic_nonresidue()                                                #␣
↪needs sage.libs.pari
3
>>> R(Integer(3)).is_square()
False
```

random_element(*bound=None*)

Return a random element of this ring.

INPUT:

- `bound` – positive integer or `None` (the default); if given, return the coercion of an integer in the interval `[-bound, bound]` into this ring

EXAMPLES:

```
sage: R = IntegerModRing(18)
sage: R.random_element().parent() is R
True
sage: found = [False]*18
sage: while not all(found):
....:     found[R.random_element()] = True
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(18))
>>> R.random_element().parent() is R
True
>>> found = [False]*Integer(18)
>>> while not all(found):
...     found[R.random_element()] = True
```

We test the `bound` option:

```
sage: R.random_element(2) in [R(-2), R(-1), R(0), R(1), R(2)]
True
```

```
>>> from sage.all import *
>>> R.random_element(Integer(2)) in [R(-Integer(2)), R(-Integer(1)),␣
↪R(Integer(0)), R(Integer(1)), R(Integer(2))]
True
```

square_roots_of_one()

Return all square roots of 1 in self, i.e., all solutions to $x^2 - 1 = 0$.

OUTPUT: the square roots of 1 in `self` as a tuple

EXAMPLES:

```
sage: R = Integers(2^10)
sage: [x for x in R if x^2 == 1]
[1, 511, 513, 1023]
sage: R.square_roots_of_one()
(1, 511, 513, 1023)
```

```
>>> from sage.all import *
>>> R = Integers(Integer(2)**Integer(10))
>>> [x for x in R if x**Integer(2) == Integer(1)]
[1, 511, 513, 1023]
>>> R.square_roots_of_one()
(1, 511, 513, 1023)
```

```
sage: # needs sage.libs.pari
sage: v = Integers(9*5).square_roots_of_one(); v
(1, 19, 26, 44)
sage: [x^2 for x in v]
[1, 1, 1, 1]
sage: v = Integers(9*5*8).square_roots_of_one(); v
(1, 19, 71, 89, 91, 109, 161, 179, 181, 199, 251, 269, 271, 289, 341, 359)
sage: [x^2 for x in v]
[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
```

```
>>> from sage.all import *
>>> # needs sage.libs.pari
>>> v = Integers(Integer(9)*Integer(5)).square_roots_of_one(); v
(1, 19, 26, 44)
>>> [x**Integer(2) for x in v]
[1, 1, 1, 1]
>>> v = Integers(Integer(9)*Integer(5)*Integer(8)).square_roots_of_one(); v
(1, 19, 71, 89, 91, 109, 161, 179, 181, 199, 251, 269, 271, 289, 341, 359)
>>> [x**Integer(2) for x in v]
[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
```

unit_gens(***kwds*)

Return generators for the unit group $(\mathbf{Z}/N\mathbf{Z})^*$.

We compute the list of generators using a deterministic algorithm, so the generators list will always be the same. For each odd prime divisor of N there will be exactly one corresponding generator; if N is even there will be 0, 1 or 2 generators according to whether 2 divides N to order 1, 2 or ≥ 3.

OUTPUT: a tuple containing the units of `self`

EXAMPLES:

```
sage: R = IntegerModRing(18)
sage: R.unit_gens()                                                    #␣
↪needs sage.groups
(11,)
sage: R = IntegerModRing(17)
sage: R.unit_gens()                                                    #␣
↪needs sage.groups
(3,)
sage: IntegerModRing(next_prime(10^30)).unit_gens()                    #␣
↪needs sage.groups
(5,)
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(18))
```

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```
>>> R.unit_gens()                                                          #␣
↪needs sage.groups
(11,)
>>> R = IntegerModRing(Integer(17))
>>> R.unit_gens()                                                          #␣
↪needs sage.groups
(3,)
>>> IntegerModRing(next_prime(Integer(10)**Integer(30))).unit_gens()        ␣
↪                 # needs sage.groups
(5,)
```

The choice of generators is affected by the optional keyword `algorithm`; this can be `'sage'` (default) or `'pari'`. See *unit_group()* for details.

```
sage: A = Zmod(55)
sage: A.unit_gens(algorithm='sage')                                        #␣
↪needs sage.groups
(12, 46)
sage: A.unit_gens(algorithm='pari')                                        #␣
↪needs sage.groups sage.libs.pari
(2, 21)
```

```
>>> from sage.all import *
>>> A = Zmod(Integer(55))
>>> A.unit_gens(algorithm='sage')                                          #␣
↪needs sage.groups
(12, 46)
>>> A.unit_gens(algorithm='pari')                                          #␣
↪needs sage.groups sage.libs.pari
(2, 21)
```

unit_group(*algorithm='sage'*)

Return the unit group of `self`.

INPUT:

- `self` – the ring $\mathbf{Z}/n\mathbf{Z}$ for a positive integer n
- `algorithm` – either `'sage'` (default) or `'pari'`

OUTPUT:

The unit group of `self`. This is a finite Abelian group equipped with a distinguished set of generators, which is computed using a deterministic algorithm depending on the `algorithm` parameter.

- If `algorithm == 'sage'`, the generators correspond to the prime factors $p \mid n$ (one generator for each odd p; the number of generators for $p = 2$ is 0, 1 or 2 depending on the order to which 2 divides n).
- If `algorithm == 'pari'`, the generators are chosen such that their orders form a decreasing sequence with respect to divisibility.

EXAMPLES:

The output of the algorithms `'sage'` and `'pari'` can differ in various ways. In the following example, the same cyclic factors are computed, but in a different order:

```
sage: # needs sage.groups
sage: A = Zmod(15)
sage: G = A.unit_group(); G
Multiplicative Abelian group isomorphic to C2 x C4
sage: G.gens_values()
(11, 7)
sage: H = A.unit_group(algorithm='pari'); H                                #␣
↪needs sage.libs.pari
Multiplicative Abelian group isomorphic to C4 x C2
sage: H.gens_values()                                                      #␣
↪needs sage.libs.pari
(7, 11)
```

```
>>> from sage.all import *
>>> # needs sage.groups
>>> A = Zmod(Integer(15))
>>> G = A.unit_group(); G
Multiplicative Abelian group isomorphic to C2 x C4
>>> G.gens_values()
(11, 7)
>>> H = A.unit_group(algorithm='pari'); H                                #␣
↪needs sage.libs.pari
Multiplicative Abelian group isomorphic to C4 x C2
>>> H.gens_values()                                                      #␣
↪needs sage.libs.pari
(7, 11)
```

Here are two examples where the cyclic factors are isomorphic, but are ordered differently and have different generators:

```
sage: # needs sage.groups
sage: A = Zmod(40)
sage: G = A.unit_group(); G
Multiplicative Abelian group isomorphic to C2 x C2 x C4
sage: G.gens_values()
(31, 21, 17)
sage: H = A.unit_group(algorithm='pari'); H                                #␣
↪needs sage.libs.pari
Multiplicative Abelian group isomorphic to C4 x C2 x C2
sage: H.gens_values()                                                      #␣
↪needs sage.libs.pari
(17, 31, 21)

sage: # needs sage.groups
sage: A = Zmod(192)
sage: G = A.unit_group(); G
Multiplicative Abelian group isomorphic to C2 x C16 x C2
sage: G.gens_values()
(127, 133, 65)
sage: H = A.unit_group(algorithm='pari'); H                                #␣
↪needs sage.libs.pari
Multiplicative Abelian group isomorphic to C16 x C2 x C2
sage: H.gens_values()                                                      #␣
```

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```
↪needs sage.libs.pari
(133, 127, 65)
```

```
>>> from sage.all import *
>>> # needs sage.groups
>>> A = Zmod(Integer(40))
>>> G = A.unit_group(); G
Multiplicative Abelian group isomorphic to C2 x C2 x C4
>>> G.gens_values()
(31, 21, 17)
>>> H = A.unit_group(algorithm='pari'); H                                  #␣
↪needs sage.libs.pari
Multiplicative Abelian group isomorphic to C4 x C2 x C2
>>> H.gens_values()                                                        #␣
↪needs sage.libs.pari
(17, 31, 21)

>>> # needs sage.groups
>>> A = Zmod(Integer(192))
>>> G = A.unit_group(); G
Multiplicative Abelian group isomorphic to C2 x C16 x C2
>>> G.gens_values()
(127, 133, 65)
>>> H = A.unit_group(algorithm='pari'); H                                  #␣
↪needs sage.libs.pari
Multiplicative Abelian group isomorphic to C16 x C2 x C2
>>> H.gens_values()                                                        #␣
↪needs sage.libs.pari
(133, 127, 65)
```

In the following examples, the cyclic factors are not even isomorphic:

```
sage: A = Zmod(319)
sage: A.unit_group()                                                       #␣
↪needs sage.groups
Multiplicative Abelian group isomorphic to C10 x C28
sage: A.unit_group(algorithm='pari')                                       #␣
↪needs sage.groups sage.libs.pari
Multiplicative Abelian group isomorphic to C140 x C2

sage: A = Zmod(30.factorial())
sage: A.unit_group()                                                       #␣
↪needs sage.groups
Multiplicative Abelian group isomorphic to
 C2 x C16777216 x C3188646 x C62500 x C2058 x C110 x C156 x C16 x C18 x C22 x␣
↪C28
sage: A.unit_group(algorithm='pari')                                       #␣
↪needs sage.groups sage.libs.pari
Multiplicative Abelian group isomorphic to
 C20499647385305088000000 x C55440 x C12 x C12 x C4 x C2 x C2 x C2 x C2 x C2␣
↪x C2
```

```
>>> from sage.all import *
>>> A = Zmod(Integer(319))
>>> A.unit_group()                                                          #␣
↪needs sage.groups
Multiplicative Abelian group isomorphic to C10 x C28
>>> A.unit_group(algorithm='pari')                                          #␣
↪needs sage.groups sage.libs.pari
Multiplicative Abelian group isomorphic to C140 x C2

>>> A = Zmod(Integer(30).factorial())
>>> A.unit_group()                                                          #␣
↪needs sage.groups
Multiplicative Abelian group isomorphic to
 C2 x C16777216 x C3188646 x C62500 x C2058 x C110 x C156 x C16 x C18 x C22 x␣
↪C28
>>> A.unit_group(algorithm='pari')                                          #␣
↪needs sage.groups sage.libs.pari
Multiplicative Abelian group isomorphic to
 C2049964738530508800000 x C55440 x C12 x C12 x C4 x C2 x C2 x C2 x C2 x C2␣
↪x C2
```

unit_group_exponent()

EXAMPLES:

```
sage: R = IntegerModRing(17)
sage: R.unit_group_exponent()                                               #␣
↪needs sage.groups
16
sage: R = IntegerModRing(18)
sage: R.unit_group_exponent()                                               #␣
↪needs sage.groups
6
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(17))
>>> R.unit_group_exponent()                                                 #␣
↪needs sage.groups
16
>>> R = IntegerModRing(Integer(18))
>>> R.unit_group_exponent()                                                 #␣
↪needs sage.groups
6
```

unit_group_order()

Return the order of the unit group of this residue class ring.

EXAMPLES:

```
sage: R = Integers(500)
sage: R.unit_group_order()                                                  #␣
↪needs sage.groups
200
```

```
>>> from sage.all import *
>>> R = Integers(Integer(500))
>>> R.unit_group_order()                                                    #␣
↪needs sage.groups
200
```

sage.rings.finite_rings.integer_mod_ring.**crt**(*v*)

INPUT:

- v – (list) a lift of elements of rings.IntegerMod(n), for various coprime moduli n

EXAMPLES:

```
sage: from sage.rings.finite_rings.integer_mod_ring import crt
sage: crt([mod(3, 8), mod(1,19), mod(7, 15)])
1027
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.integer_mod_ring import crt
>>> crt([mod(Integer(3), Integer(8)), mod(Integer(1),Integer(19)), mod(Integer(7),
↪ Integer(15))])
1027
```

1.2 Elements of $\mathbf{Z}/n\mathbf{Z}$

An element of the integers modulo n.

There are three types of integer_mod classes, depending on the size of the modulus.

- IntegerMod_int stores its value in a int_fast32_t (typically an int); this is used if the modulus is less than $\sqrt{2^{31}-1}$.
- IntegerMod_int64 stores its value in a int_fast64_t (typically a long long); this is used if the modulus is less than $2^{31}-1$. In many places, we assume that the values and the modulus actually fit inside an unsigned long.
- IntegerMod_gmp stores its value in a mpz_t; this can be used for an arbitrarily large modulus.

All extend IntegerMod_abstract.

For efficiency reasons, it stores the modulus (in all three forms, if possible) in a common (cdef) class NativeIntStruct rather than in the parent.

AUTHORS:

- Robert Bradshaw: most of the work
- Didier Deshommes: bit shifting
- William Stein: editing and polishing; new arith architecture
- Robert Bradshaw: implement native is_square and square_root
- William Stein: sqrt
- Maarten Derickx: moved the valuation code from the global valuation function to here

class sage.rings.finite_rings.integer_mod.**Int_to_IntegerMod**

Bases: *IntegerMod_hom*

EXAMPLES:

We make sure it works for every type.

```
sage: from sage.rings.finite_rings.integer_mod import Int_to_IntegerMod
sage: Rs = [Integers(2**k) for k in range(1,50,10)]
sage: [type(R(0)) for R in Rs]
[<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>]
sage: fs = [Int_to_IntegerMod(R) for R in Rs]
sage: [f(-1) for f in fs]
[1, 2047, 2097151, 2147483647, 2199023255551]
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.integer_mod import Int_to_IntegerMod
>>> Rs = [Integers(Integer(2)**k) for k in range(Integer(1),Integer(50),
↪Integer(10))]
>>> [type(R(Integer(0))) for R in Rs]
[<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>]
>>> fs = [Int_to_IntegerMod(R) for R in Rs]
>>> [f(-Integer(1)) for f in fs]
[1, 2047, 2097151, 2147483647, 2199023255551]
```

sage.rings.finite_rings.integer_mod.**IntegerMod**(*parent*, *value*)

Create an integer modulo n with the given parent.

This is mainly for internal use.

EXAMPLES:

```
sage: from sage.rings.finite_rings.integer_mod import IntegerMod
sage: R = IntegerModRing(100)
sage: type(R._pyx_order.table)
<class 'list'>
sage: IntegerMod(R, 42)
42
sage: IntegerMod(R, 142)
42
sage: IntegerMod(R, 10^100 + 42)
42
sage: IntegerMod(R, -9158)
42
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.integer_mod import IntegerMod
```

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```
>>> R = IntegerModRing(Integer(100))
>>> type(R._pyx_order.table)
<class 'list'>
>>> IntegerMod(R, Integer(42))
42
>>> IntegerMod(R, Integer(142))
42
>>> IntegerMod(R, Integer(10)**Integer(100) + Integer(42))
42
>>> IntegerMod(R, -Integer(9158))
42
```

class sage.rings.finite_rings.integer_mod.**IntegerMod_abstract**

Bases: *FiniteRingElement*

EXAMPLES:

```
sage: a = Mod(10, 30^10); a
10
sage: type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>
sage: loads(a.dumps()) == a
True
```

```
>>> from sage.all import *
>>> a = Mod(Integer(10), Integer(30)**Integer(10)); a
10
>>> type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>
>>> loads(a.dumps()) == a
True
```

additive_order()

Return the additive order of `self`.

This is the same as `self.order()`.

EXAMPLES:

```
sage: Integers(20)(2).additive_order()
10
sage: Integers(20)(7).additive_order()
20
sage: Integers(90308402384902)(2).additive_order()
45154201192451
```

```
>>> from sage.all import *
>>> Integers(Integer(20))(Integer(2)).additive_order()
10
>>> Integers(Integer(20))(Integer(7)).additive_order()
20
>>> Integers(Integer(90308402384902))(Integer(2)).additive_order()
45154201192451
```

charpoly(*var='x'*)

Return the characteristic polynomial of this element.

EXAMPLES:

```
sage: k = GF(3)
sage: a = k.gen()
sage: a.charpoly('x')
x + 2
sage: a + 2
0
```

```
>>> from sage.all import *
>>> k = GF(Integer(3))
>>> a = k.gen()
>>> a.charpoly('x')
x + 2
>>> a + Integer(2)
0
```

AUTHORS:

- Craig Citro

crt(*other*)

Use the Chinese Remainder Theorem to find an element of the integers modulo the product of the moduli that reduces to `self` and to `other`. The modulus of `other` must be coprime to the modulus of `self`.

EXAMPLES:

```
sage: a = mod(3, 5)
sage: b = mod(2, 7)
sage: a.crt(b)
23
```

```
>>> from sage.all import *
>>> a = mod(Integer(3), Integer(5))
>>> b = mod(Integer(2), Integer(7))
>>> a.crt(b)
23
```

```
sage: a = mod(37, 10^8)
sage: b = mod(9, 3^8)
sage: a.crt(b)
125900000037
```

```
>>> from sage.all import *
>>> a = mod(Integer(37), Integer(10)**Integer(8))
>>> b = mod(Integer(9), Integer(3)**Integer(8))
>>> a.crt(b)
125900000037
```

```
sage: b = mod(0, 1)
sage: a.crt(b) == a
```

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```
True
sage: a.crt(b).modulus()
100000000
```

```
>>> from sage.all import *
>>> b = mod(Integer(0), Integer(1))
>>> a.crt(b) == a
True
>>> a.crt(b).modulus()
100000000
```

AUTHORS:

- Robert Bradshaw

divides(*other*)

Test whether `self` divides `other`.

EXAMPLES:

```
sage: R = Zmod(6)
sage: R(2).divides(R(4))
True
sage: R(4).divides(R(2))
True
sage: R(2).divides(R(3))
False
```

```
>>> from sage.all import *
>>> R = Zmod(Integer(6))
>>> R(Integer(2)).divides(R(Integer(4)))
True
>>> R(Integer(4)).divides(R(Integer(2)))
True
>>> R(Integer(2)).divides(R(Integer(3)))
False
```

generalised_log()

Return integers $[n_1, \ldots, n_d]$ such that.

$$\prod_{i=1}^{d} x_i^{n_i} = \text{self},$$

where $x_1, \ldots, x_d$ are the generators of the unit group returned by `self.parent().unit_gens()`.

EXAMPLES:

```
sage: m = Mod(3, 1568)
sage: v = m.generalised_log(); v                                          #␣
↪needs sage.libs.pari sage.modules
[1, 3, 1]
sage: prod([Zmod(1568).unit_gens()[i] ** v[i] for i in [0..2]])           #␣
↪needs sage.libs.pari sage.modules
3
```

```
>>> from sage.all import *
>>> m = Mod(Integer(3), Integer(1568))
>>> v = m.generalised_log(); v                                          #␣
↪needs sage.libs.pari sage.modules
[1, 3, 1]
>>> prod([Zmod(Integer(1568)).unit_gens()[i] ** v[i] for i in (ellipsis_
↪range(Integer(0),Ellipsis,Integer(2)))])              # needs sage.libs.pari␣
↪sage.modules
3
```

See also

The method *log()*.

Warning

The output is given relative to the set of generators obtained by passing `algorithm='sage'` to the method *unit_gens()* of the parent (which is the default). Specifying `algorithm='pari'` usually yields a different set of generators that is incompatible with this method.

is_nilpotent()

Return `True` if `self` is nilpotent, i.e., some power of `self` is zero.

EXAMPLES:

```
sage: a = Integers(90384098234^3)
sage: factor(a.order())                                                 #␣
↪needs sage.libs.pari
2^3 * 191^3 * 236607587^3
sage: b = a(2*191)
sage: b.is_nilpotent()
False
sage: b = a(2*191*236607587)
sage: b.is_nilpotent()
True
```

```
>>> from sage.all import *
>>> a = Integers(Integer(90384098234)**Integer(3))
>>> factor(a.order())                                                   #␣
↪needs sage.libs.pari
2^3 * 191^3 * 236607587^3
>>> b = a(Integer(2)*Integer(191))
>>> b.is_nilpotent()
False
>>> b = a(Integer(2)*Integer(191)*Integer(236607587))
>>> b.is_nilpotent()
True
```

ALGORITHM: Let $m \geq \log_2(n)$, where n is the modulus. Then $x \in \mathbf{Z}/n\mathbf{Z}$ is nilpotent if and only if $x^m = 0$.

PROOF: This is clear if you reduce to the prime power case, which you can do via the Chinese Remainder Theorem.

We could alternatively factor n and check to see if the prime divisors of n all divide x. This is asymptotically slower :-).

is_one()

is_primitive_root()

Determines whether this element generates the group of units modulo n.

This is only possible if the group of units is cyclic, which occurs if n is 2, 4, a power of an odd prime or twice a power of an odd prime.

EXAMPLES:

```
sage: mod(1, 2).is_primitive_root()
True
sage: mod(3, 4).is_primitive_root()
True
sage: mod(2, 7).is_primitive_root()
False
sage: mod(3, 98).is_primitive_root()                                    #␣
↪needs sage.libs.pari
True
sage: mod(11, 1009^2).is_primitive_root()                               #␣
↪needs sage.libs.pari
True
```

```
>>> from sage.all import *
>>> mod(Integer(1), Integer(2)).is_primitive_root()
True
>>> mod(Integer(3), Integer(4)).is_primitive_root()
True
>>> mod(Integer(2), Integer(7)).is_primitive_root()
False
>>> mod(Integer(3), Integer(98)).is_primitive_root()                    ␣
↪              # needs sage.libs.pari
True
>>> mod(Integer(11), Integer(1009)**Integer(2)).is_primitive_root()     ␣
↪                        # needs sage.libs.pari
True
```

is_square()

EXAMPLES:

```
sage: Mod(3, 17).is_square()
False

sage: # needs sage.libs.pari
sage: Mod(9, 17).is_square()
True
sage: Mod(9, 17*19^2).is_square()
True
sage: Mod(-1, 17^30).is_square()
```

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```
True
sage: Mod(1/9, next_prime(2^40)).is_square()
True
sage: Mod(1/25, next_prime(2^90)).is_square()
True
```

```
>>> from sage.all import *
>>> Mod(Integer(3), Integer(17)).is_square()
False

>>> # needs sage.libs.pari
>>> Mod(Integer(9), Integer(17)).is_square()
True
>>> Mod(Integer(9), Integer(17)*Integer(19)**Integer(2)).is_square()
True
>>> Mod(-Integer(1), Integer(17)**Integer(30)).is_square()
True
>>> Mod(Integer(1)/Integer(9), next_prime(Integer(2)**Integer(40))).is_
↪square()
True
>>> Mod(Integer(1)/Integer(25), next_prime(Integer(2)**Integer(90))).is_
↪square()
True
```

ALGORITHM: Calculate the Jacobi symbol $(\texttt{self}/p)$ at each prime p dividing n. It must be 1 or 0 for each prime, and if it is 0 mod p, where $p^k||n$, then $ord_p(\texttt{self})$ must be even or greater than k.

The case $p = 2$ is handled separately.

AUTHORS:

- Robert Bradshaw

is_unit()

lift_centered()

Lift `self` to a centered congruent integer.

OUTPUT:

The unique integer i such that $-n/2 < i \leq n/2$ and $i = self \mod n$ (where n denotes the modulus).

EXAMPLES:

```
sage: Mod(0,5).lift_centered()
0
sage: Mod(1,5).lift_centered()
1
sage: Mod(2,5).lift_centered()
2
sage: Mod(3,5).lift_centered()
-2
sage: Mod(4,5).lift_centered()
-1
sage: Mod(50,100).lift_centered()
```

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```
50
sage: Mod(51,100).lift_centered()
-49
sage: Mod(-1,3^100).lift_centered()
-1
```

```
>>> from sage.all import *
>>> Mod(Integer(0),Integer(5)).lift_centered()
0
>>> Mod(Integer(1),Integer(5)).lift_centered()
1
>>> Mod(Integer(2),Integer(5)).lift_centered()
2
>>> Mod(Integer(3),Integer(5)).lift_centered()
-2
>>> Mod(Integer(4),Integer(5)).lift_centered()
-1
>>> Mod(Integer(50),Integer(100)).lift_centered()
50
>>> Mod(Integer(51),Integer(100)).lift_centered()
-49
>>> Mod(-Integer(1),Integer(3)**Integer(100)).lift_centered()
-1
```

log(*b=None*, *order=None*, *check=False*)

Compute the discrete logarithm of this element to base b, that is, return an integer x such that $b^x = a$, where a is `self`.

INPUT:

- `self` – unit modulo n
- `b` – a unit modulo n. If `b` is not given, `R.multiplicative_generator()` is used, where `R` is the parent of `self`.
- `order` – integer (unused), the order of `b`. This argument is normally unused, only there for coherence of apis with finite field elements.
- `check` – boolean (default: `False`); if set, test whether the given `order` is correct

OUTPUT:

Integer x such that $b^x = a$, if this exists; a `ValueError` otherwise.

> **Note**
>
> The algorithm first factors the modulus, then invokes Pari's pari:znlog function for each odd prime power in the factorization of the modulus. This method can be quite slow for large moduli.

EXAMPLES:

```
sage: # needs sage.libs.pari sage.modules
sage: r = Integers(125)
sage: b = r.multiplicative_generator()^3
```

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```
sage: a = b^17
sage: a.log(b)
17
sage: a.log()
51
```

```
>>> from sage.all import *
>>> # needs sage.libs.pari sage.modules
>>> r = Integers(Integer(125))
>>> b = r.multiplicative_generator()**Integer(3)
>>> a = b**Integer(17)
>>> a.log(b)
17
>>> a.log()
51
```

A bigger example:

```
sage: # needs sage.rings.finite_rings
sage: FF = FiniteField(2^32 + 61)
sage: c = FF(4294967356)
sage: x = FF(2)
sage: a = c.log(x)
sage: a
2147483678
sage: x^a
4294967356
```

```
>>> from sage.all import *
>>> # needs sage.rings.finite_rings
>>> FF = FiniteField(Integer(2)**Integer(32) + Integer(61))
>>> c = FF(Integer(4294967356))
>>> x = FF(Integer(2))
>>> a = c.log(x)
>>> a
2147483678
>>> x**a
4294967356
```

An example with a highly composite modulus:

```
sage: m = 2^99 * 77^7 * 123456789 * 13712923537615486607^2
sage: (Mod(5,m)^573581676307385495338814723792 1).log(5)                    #␣
↪needs sage.libs.pari
573581676307385495338814723792 1
```

```
>>> from sage.all import *
>>> m = Integer(2)**Integer(99) * Integer(77)**Integer(7) *␣
↪Integer(123456789) * Integer(13712923537615486607)**Integer(2)
>>> (Mod(Integer(5),m)**Integer(573581676307385495338814723792 1)).
↪log(Integer(5))                      # needs sage.libs.pari
573581676307385495338814723792 1
```

Errors are generated if the logarithm doesn't exist or the inputs are not units:

```
sage: Mod(3, 7).log(Mod(2, 7))                                          #␣
↪needs sage.libs.pari
Traceback (most recent call last):
...
ValueError: no logarithm of 3 found to base 2 modulo 7
sage: a = Mod(16, 100); b = Mod(4, 100)
sage: a.log(b)
Traceback (most recent call last):
...
ValueError: logarithm of 16 is not defined since it is not a unit modulo 100
```

```
>>> from sage.all import *
>>> Mod(Integer(3), Integer(7)).log(Mod(Integer(2), Integer(7)))          ␣
↪                                # needs sage.libs.pari
Traceback (most recent call last):
...
ValueError: no logarithm of 3 found to base 2 modulo 7
>>> a = Mod(Integer(16), Integer(100)); b = Mod(Integer(4), Integer(100))
>>> a.log(b)
Traceback (most recent call last):
...
ValueError: logarithm of 16 is not defined since it is not a unit modulo 100
```

AUTHORS:

- David Joyner and William Stein (2005-11)
- William Stein (2007-01-27): update to use PARI as requested by David Kohel.
- Simon King (2010-07-07): fix a side effect on PARI
- Lorenz Panny (2021): speedups for composite moduli

minimal_polynomial(*var='x'*)

Return the minimal polynomial of this element.

EXAMPLES:

```
sage: GF(241, 'a')(1).minimal_polynomial(var = 'z')
z + 240
```

```
>>> from sage.all import *
>>> GF(Integer(241), 'a')(Integer(1)).minimal_polynomial(var = 'z')
z + 240
```

minpoly(*var='x'*)

Return the minimal polynomial of this element.

EXAMPLES:

```
sage: GF(241, 'a')(1).minpoly()
x + 240
```

```
>>> from sage.all import *
>>> GF(Integer(241), 'a')(Integer(1)).minpoly()
x + 240
```

modulus()

EXAMPLES:

```
sage: Mod(3,17).modulus()
17
```

```
>>> from sage.all import *
>>> Mod(Integer(3),Integer(17)).modulus()
17
```

multiplicative_order()

Return the multiplicative order of `self`.

EXAMPLES:

```
sage: Mod(-1, 5).multiplicative_order()                                    #␣
↪needs sage.libs.pari
2
sage: Mod(1, 5).multiplicative_order()                                     #␣
↪needs sage.libs.pari
1
sage: Mod(0, 5).multiplicative_order()                                     #␣
↪needs sage.libs.pari
Traceback (most recent call last):
...
ArithmeticError: multiplicative order of 0 not defined
since it is not a unit modulo 5
```

```
>>> from sage.all import *
>>> Mod(-Integer(1), Integer(5)).multiplicative_order()                    ␣
↪              # needs sage.libs.pari
2
>>> Mod(Integer(1), Integer(5)).multiplicative_order()                     ␣
↪              # needs sage.libs.pari
1
>>> Mod(Integer(0), Integer(5)).multiplicative_order()                     ␣
↪              # needs sage.libs.pari
Traceback (most recent call last):
...
ArithmeticError: multiplicative order of 0 not defined
since it is not a unit modulo 5
```

norm()

Return the norm of this element, which is itself. (This is here for compatibility with higher order finite fields.)

EXAMPLES:

```
sage: k = GF(691)
sage: a = k(389)
```

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```
sage: a.norm()
389
```

```
>>> from sage.all import *
>>> k = GF(Integer(691))
>>> a = k(Integer(389))
>>> a.norm()
389
```

AUTHORS:

- Craig Citro

nth_root(*n*, *extend=False*, *all=False*, *algorithm=None*, *cunningham=False*)

Return an n-th root of `self`.

INPUT:

- `n` – integer ≥ 1
- `extend` – boolean (default: `True`); if `True`, return an n-th root in an extension ring, if necessary. Otherwise, raise a `ValueError` if the root is not in the base ring. Warning: this option is not implemented!
- `all` – boolean (default: `False`); if `True`, return all n-th roots of `self`, instead of just one
- `algorithm` – string (default: `None`); the algorithm for the prime modulus case. CRT and p-adic log techniques are used to reduce to this case. `'Johnston'` is the only currently supported option.
- `cunningham` – boolean (default: `False`); in some cases, factorization of n is computed. If cunningham is set to `True`, the factorization of n is computed using trial division for all primes in the so called Cunningham table. Refer to `sage.rings.factorint.factor_cunningham` for more information. You need to install an optional package to use this method, this can be done with the following command line: `sage -i cunningham_tables`.

OUTPUT:

If `self` has an n-th root, returns one (if `all` is `False`) or a list of all of them (if `all` is `True`). Otherwise, raises a `ValueError` (if `extend` is `False`) or a `NotImplementedError` (if `extend` is `True`).

Warning

The 'extend' option is not implemented (yet).

NOTE:

- If $n = 0$:
  - if `all=True`:
    - if `self=1`: all nonzero elements of the parent are returned in a list. Note that this could be very expensive for large parents.
    - otherwise: an empty list is returned
  - if `all=False`:
    - if `self=1`: `self` is returned
    - otherwise; a `ValueError` is raised

- If $n < 0$:
  - if `self` is invertible, the $(-n)$th root of the inverse of `self` is returned
  - otherwise a `ValueError` is raised or empty list returned.

EXAMPLES:

```
sage: K = GF(31)
sage: a = K(22)
sage: K(22).nth_root(7)
13
sage: K(25).nth_root(5)
5
sage: K(23).nth_root(3)
29

sage: # needs sage.rings.padics
sage: mod(225, 2^5*3^2).nth_root(4, all=True)
[225, 129, 33, 63, 255, 159, 9, 201, 105, 279, 183, 87, 81,
 273, 177, 207, 111, 15, 153, 57, 249, 135, 39, 231]
sage: mod(275, 2^5*7^4).nth_root(7, all=True)
[58235, 25307, 69211, 36283, 3355, 47259, 14331]
sage: mod(1,8).nth_root(2, all=True)
[1, 7, 5, 3]
sage: mod(4,8).nth_root(2, all=True)
[2, 6]
sage: mod(1,16).nth_root(4, all=True)
[1, 15, 13, 3, 9, 7, 5, 11]

sage: (mod(22,31)^200).nth_root(200)
5
sage: mod(3,6).nth_root(0, all=True)
[]
sage: mod(3,6).nth_root(0)
Traceback (most recent call last):
...
ValueError
sage: mod(1,6).nth_root(0, all=True)
[1, 2, 3, 4, 5]
```

```
>>> from sage.all import *
>>> K = GF(Integer(31))
>>> a = K(Integer(22))
>>> K(Integer(22)).nth_root(Integer(7))
13
>>> K(Integer(25)).nth_root(Integer(5))
5
>>> K(Integer(23)).nth_root(Integer(3))
29

>>> # needs sage.rings.padics
>>> mod(Integer(225), Integer(2)**Integer(5)*Integer(3)**Integer(2)).nth_
↪root(Integer(4), all=True)
[225, 129, 33, 63, 255, 159, 9, 201, 105, 279, 183, 87, 81,
```

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```
 273, 177, 207, 111, 15, 153, 57, 249, 135, 39, 231]
>>> mod(Integer(275), Integer(2)**Integer(5)*Integer(7)**Integer(4)).nth_
↪root(Integer(7), all=True)
[58235, 25307, 69211, 36283, 3355, 47259, 14331]
>>> mod(Integer(1),Integer(8)).nth_root(Integer(2), all=True)
[1, 7, 5, 3]
>>> mod(Integer(4),Integer(8)).nth_root(Integer(2), all=True)
[2, 6]
>>> mod(Integer(1),Integer(16)).nth_root(Integer(4), all=True)
[1, 15, 13, 3, 9, 7, 5, 11]

>>> (mod(Integer(22),Integer(31))**Integer(200)).nth_root(Integer(200))
5
>>> mod(Integer(3),Integer(6)).nth_root(Integer(0), all=True)
[]
>>> mod(Integer(3),Integer(6)).nth_root(Integer(0))
Traceback (most recent call last):
...
ValueError
>>> mod(Integer(1),Integer(6)).nth_root(Integer(0), all=True)
[1, 2, 3, 4, 5]
```

ALGORITHM:

The default for prime modulus is currently an algorithm described in [Joh1999].

AUTHORS:

- David Roe (2010-02-13)

polynomial(*var='x'*)

Return a constant polynomial representing this value.

EXAMPLES:

```
sage: k = GF(7)
sage: a = k.gen(); a
1
sage: a.polynomial()
1
sage: type(a.polynomial())                                                  #␣
↪needs sage.libs.flint
<class 'sage.rings.polynomial.polynomial_zmod_flint.Polynomial_zmod_flint'>
```

```
>>> from sage.all import *
>>> k = GF(Integer(7))
>>> a = k.gen(); a
1
>>> a.polynomial()
1
>>> type(a.polynomial())                                                    #␣
↪needs sage.libs.flint
<class 'sage.rings.polynomial.polynomial_zmod_flint.Polynomial_zmod_flint'>
```

rational_reconstruction()

Use rational reconstruction to try to find a lift of this element to the rational numbers.

EXAMPLES:

```
sage: R = IntegerModRing(97)
sage: a = R(2) / R(3)
sage: a
33
sage: a.rational_reconstruction()
2/3
```

```
>>> from sage.all import *
>>> R = IntegerModRing(Integer(97))
>>> a = R(Integer(2)) / R(Integer(3))
>>> a
33
>>> a.rational_reconstruction()
2/3
```

This method is also inherited by prime finite fields elements:

```
sage: k = GF(97)
sage: a = k(RationalField()('2/3'))
sage: a
33
sage: a.rational_reconstruction()
2/3
```

```
>>> from sage.all import *
>>> k = GF(Integer(97))
>>> a = k(RationalField()('2/3'))
>>> a
33
>>> a.rational_reconstruction()
2/3
```

sqrt(*extend=True*, *all=False*)

Return square root or square roots of `self` modulo n.

INPUT:

- `extend` – boolean (default: `True`); if `True`, return a square root in an extension ring, if necessary. Otherwise, raise a `ValueError` if the square root is not in the base ring.
- `all` – boolean (default: `False`); if `True`, return {all} square roots of self, instead of just one

ALGORITHM: Calculates the square roots mod p for each of the primes p dividing the order of the ring, then lifts them p-adically and uses the CRT to find a square root mod n.

See also *square_root_mod_prime_power()* and *square_root_mod_prime()* for more algorithmic details.

EXAMPLES:

```
sage: mod(-1, 17).sqrt()
4
sage: mod(5, 389).sqrt()
86
sage: mod(7, 18).sqrt()
5

sage: # needs sage.libs.pari
sage: a = mod(14, 5^60).sqrt()
sage: a*a
14
sage: mod(15, 389).sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
sage: Mod(1/9, next_prime(2^40)).sqrt()^(-2)
9
sage: Mod(1/25, next_prime(2^90)).sqrt()^(-2)
25
```

```
>>> from sage.all import *
>>> mod(-Integer(1), Integer(17)).sqrt()
4
>>> mod(Integer(5), Integer(389)).sqrt()
86
>>> mod(Integer(7), Integer(18)).sqrt()
5

>>> # needs sage.libs.pari
>>> a = mod(Integer(14), Integer(5)**Integer(60)).sqrt()
>>> a*a
14
>>> mod(Integer(15), Integer(389)).sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
>>> Mod(Integer(1)/Integer(9), next_prime(Integer(2)**Integer(40))).sqrt()**(-
↪Integer(2))
9
>>> Mod(Integer(1)/Integer(25), next_prime(Integer(2)**Integer(90))).
↪sqrt()**(-Integer(2))
25
```

```
sage: a = Mod(3, 5); a
3
sage: x = Mod(-1, 360)
sage: x.sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
sage: y = x.sqrt(); y
sqrt359
```

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```
sage: y.parent()
Univariate Quotient Polynomial Ring in sqrt359 over
 Ring of integers modulo 360 with modulus x^2 + 1
sage: y^2
359
```

```
>>> from sage.all import *
>>> a = Mod(Integer(3), Integer(5)); a
3
>>> x = Mod(-Integer(1), Integer(360))
>>> x.sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
>>> y = x.sqrt(); y
sqrt359
>>> y.parent()
Univariate Quotient Polynomial Ring in sqrt359 over
 Ring of integers modulo 360 with modulus x^2 + 1
>>> y**Integer(2)
359
```

We compute all square roots in several cases:

```
sage: R = Integers(5*2^3*3^2); R
Ring of integers modulo 360
sage: R(40).sqrt(all=True)
[20, 160, 200, 340]
sage: [x for x in R if x^2 == 40]  # Brute force verification
[20, 160, 200, 340]
sage: R(1).sqrt(all=True)
[1, 19, 71, 89, 91, 109, 161, 179, 181, 199, 251, 269, 271, 289, 341, 359]
sage: R(0).sqrt(all=True)
[0, 60, 120, 180, 240, 300]
```

```
>>> from sage.all import *
>>> R = Integers(Integer(5)*Integer(2)**Integer(3)*Integer(3)**Integer(2)); R
Ring of integers modulo 360
>>> R(Integer(40)).sqrt(all=True)
[20, 160, 200, 340]
>>> [x for x in R if x**Integer(2) == Integer(40)]  # Brute force verification
[20, 160, 200, 340]
>>> R(Integer(1)).sqrt(all=True)
[1, 19, 71, 89, 91, 109, 161, 179, 181, 199, 251, 269, 271, 289, 341, 359]
>>> R(Integer(0)).sqrt(all=True)
[0, 60, 120, 180, 240, 300]
```

```
sage: # needs sage.libs.pari
sage: R = Integers(5*13^3*37); R
Ring of integers modulo 406445
sage: v = R(-1).sqrt(all=True); v
```

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```
[78853, 111808, 160142, 193097, 213348, 246303, 294637, 327592]
sage: [x^2 for x in v]
[406444, 406444, 406444, 406444, 406444, 406444, 406444, 406444]
sage: v = R(169).sqrt(all=True); min(v), -max(v), len(v)
(13, 13, 104)
sage: all(x^2 == 169 for x in v)
True
```

```
>>> from sage.all import *
>>> # needs sage.libs.pari
>>> R = Integers(Integer(5)*Integer(13)**Integer(3)*Integer(37)); R
Ring of integers modulo 406445
>>> v = R(-Integer(1)).sqrt(all=True); v
[78853, 111808, 160142, 193097, 213348, 246303, 294637, 327592]
>>> [x**Integer(2) for x in v]
[406444, 406444, 406444, 406444, 406444, 406444, 406444, 406444]
>>> v = R(Integer(169)).sqrt(all=True); min(v), -max(v), len(v)
(13, 13, 104)
>>> all(x**Integer(2) == Integer(169) for x in v)
True
```

```
sage: # needs sage.rings.finite_rings
sage: t = FiniteField(next_prime(2^100))(4)
sage: t.sqrt(extend=False, all=True)
[2, 1267650600228229401496703205651]
sage: t = FiniteField(next_prime(2^100))(2)
sage: t.sqrt(extend=False, all=True)
[]
```

```
>>> from sage.all import *
>>> # needs sage.rings.finite_rings
>>> t = FiniteField(next_prime(Integer(2)**Integer(100)))(Integer(4))
>>> t.sqrt(extend=False, all=True)
[2, 1267650600228229401496703205651]
>>> t = FiniteField(next_prime(Integer(2)**Integer(100)))(Integer(2))
>>> t.sqrt(extend=False, all=True)
[]
```

Modulo a power of 2:

```
sage: R = Integers(2^7); R
Ring of integers modulo 128
sage: a = R(17)
sage: a.sqrt()
23
sage: a.sqrt(all=True)
[23, 41, 87, 105]
sage: [x for x in R if x^2==17]
[23, 41, 87, 105]
```

```
>>> from sage.all import *
```

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```
>>> R = Integers(Integer(2)**Integer(7)); R
Ring of integers modulo 128
>>> a = R(Integer(17))
>>> a.sqrt()
23
>>> a.sqrt(all=True)
[23, 41, 87, 105]
>>> [x for x in R if x**Integer(2)==Integer(17)]
[23, 41, 87, 105]
```

square_root(*extend=True*, *all=False*)

Return square root or square roots of `self` modulo n.

INPUT:

- `extend` – boolean (default: `True`); if `True`, return a square root in an extension ring, if necessary. Otherwise, raise a `ValueError` if the square root is not in the base ring.
- `all` – boolean (default: `False`); if `True`, return {all} square roots of self, instead of just one

ALGORITHM: Calculates the square roots mod p for each of the primes p dividing the order of the ring, then lifts them p-adically and uses the CRT to find a square root mod n.

See also *square_root_mod_prime_power()* and *square_root_mod_prime()* for more algorithmic details.

EXAMPLES:

```
sage: mod(-1, 17).sqrt()
4
sage: mod(5, 389).sqrt()
86
sage: mod(7, 18).sqrt()
5

sage: # needs sage.libs.pari
sage: a = mod(14, 5^60).sqrt()
sage: a*a
14
sage: mod(15, 389).sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
sage: Mod(1/9, next_prime(2^40)).sqrt()^(-2)
9
sage: Mod(1/25, next_prime(2^90)).sqrt()^(-2)
25
```

```
>>> from sage.all import *
>>> mod(-Integer(1), Integer(17)).sqrt()
4
>>> mod(Integer(5), Integer(389)).sqrt()
86
>>> mod(Integer(7), Integer(18)).sqrt()
5
```

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```
>>> # needs sage.libs.pari
>>> a = mod(Integer(14), Integer(5)**Integer(60)).sqrt()
>>> a*a
14
>>> mod(Integer(15), Integer(389)).sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
>>> Mod(Integer(1)/Integer(9), next_prime(Integer(2)**Integer(40))).sqrt()**(-
↪Integer(2))
9
>>> Mod(Integer(1)/Integer(25), next_prime(Integer(2)**Integer(90))).
↪sqrt()**(-Integer(2))
25
```

```
sage: a = Mod(3, 5); a
3
sage: x = Mod(-1, 360)
sage: x.sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
sage: y = x.sqrt(); y
sqrt359
sage: y.parent()
Univariate Quotient Polynomial Ring in sqrt359 over
 Ring of integers modulo 360 with modulus x^2 + 1
sage: y^2
359
```

```
>>> from sage.all import *
>>> a = Mod(Integer(3), Integer(5)); a
3
>>> x = Mod(-Integer(1), Integer(360))
>>> x.sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
>>> y = x.sqrt(); y
sqrt359
>>> y.parent()
Univariate Quotient Polynomial Ring in sqrt359 over
 Ring of integers modulo 360 with modulus x^2 + 1
>>> y**Integer(2)
359
```

We compute all square roots in several cases:

```
sage: R = Integers(5*2^3*3^2); R
Ring of integers modulo 360
sage: R(40).sqrt(all=True)
```

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```
[20, 160, 200, 340]
sage: [x for x in R if x^2 == 40]  # Brute force verification
[20, 160, 200, 340]
sage: R(1).sqrt(all=True)
[1, 19, 71, 89, 91, 109, 161, 179, 181, 199, 251, 269, 271, 289, 341, 359]
sage: R(0).sqrt(all=True)
[0, 60, 120, 180, 240, 300]
```

```
>>> from sage.all import *
>>> R = Integers(Integer(5)*Integer(2)**Integer(3)*Integer(3)**Integer(2)); R
Ring of integers modulo 360
>>> R(Integer(40)).sqrt(all=True)
[20, 160, 200, 340]
>>> [x for x in R if x**Integer(2) == Integer(40)]  # Brute force verification
[20, 160, 200, 340]
>>> R(Integer(1)).sqrt(all=True)
[1, 19, 71, 89, 91, 109, 161, 179, 181, 199, 251, 269, 271, 289, 341, 359]
>>> R(Integer(0)).sqrt(all=True)
[0, 60, 120, 180, 240, 300]
```

```
sage: # needs sage.libs.pari
sage: R = Integers(5*13^3*37); R
Ring of integers modulo 406445
sage: v = R(-1).sqrt(all=True); v
[78853, 111808, 160142, 193097, 213348, 246303, 294637, 327592]
sage: [x^2 for x in v]
[406444, 406444, 406444, 406444, 406444, 406444, 406444, 406444]
sage: v = R(169).sqrt(all=True); min(v), -max(v), len(v)
(13, 13, 104)
sage: all(x^2 == 169 for x in v)
True
```

```
>>> from sage.all import *
>>> # needs sage.libs.pari
>>> R = Integers(Integer(5)*Integer(13)**Integer(3)*Integer(37)); R
Ring of integers modulo 406445
>>> v = R(-Integer(1)).sqrt(all=True); v
[78853, 111808, 160142, 193097, 213348, 246303, 294637, 327592]
>>> [x**Integer(2) for x in v]
[406444, 406444, 406444, 406444, 406444, 406444, 406444, 406444]
>>> v = R(Integer(169)).sqrt(all=True); min(v), -max(v), len(v)
(13, 13, 104)
>>> all(x**Integer(2) == Integer(169) for x in v)
True
```

```
sage: # needs sage.rings.finite_rings
sage: t = FiniteField(next_prime(2^100))(4)
sage: t.sqrt(extend=False, all=True)
[2, 1267650600228229401496703205651]
sage: t = FiniteField(next_prime(2^100))(2)
sage: t.sqrt(extend=False, all=True)
```

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```
[]
```

```
>>> from sage.all import *
>>> # needs sage.rings.finite_rings
>>> t = FiniteField(next_prime(Integer(2)**Integer(100)))(Integer(4))
>>> t.sqrt(extend=False, all=True)
[2, 1267650600228229401496703205651]
>>> t = FiniteField(next_prime(Integer(2)**Integer(100)))(Integer(2))
>>> t.sqrt(extend=False, all=True)
[]
```

Modulo a power of 2:

```
sage: R = Integers(2^7); R
Ring of integers modulo 128
sage: a = R(17)
sage: a.sqrt()
23
sage: a.sqrt(all=True)
[23, 41, 87, 105]
sage: [x for x in R if x^2==17]
[23, 41, 87, 105]
```

```
>>> from sage.all import *
>>> R = Integers(Integer(2)**Integer(7)); R
Ring of integers modulo 128
>>> a = R(Integer(17))
>>> a.sqrt()
23
>>> a.sqrt(all=True)
[23, 41, 87, 105]
>>> [x for x in R if x**Integer(2)==Integer(17)]
[23, 41, 87, 105]
```

trace()

Return the trace of this element, which is itself. (This is here for compatibility with higher order finite fields.)

EXAMPLES:

```
sage: k = GF(691)
sage: a = k(389)
sage: a.trace()
389
```

```
>>> from sage.all import *
>>> k = GF(Integer(691))
>>> a = k(Integer(389))
>>> a.trace()
389
```

AUTHORS:

- Craig Citro

valuation(p)

The largest power r such that m is in the ideal generated by p^r or infinity if there is not a largest such power. However it is an error to take the valuation with respect to a unit.

> **Note**
>
> This is not a valuation in the mathematical sense. As shown with the examples below.

EXAMPLES:

This example shows that `(a*b).valuation(n)` is not always the same as `a.valuation(n) + b.valuation(n)`

```
sage: R = ZZ.quo(9)
sage: a = R(3)
sage: b = R(6)
sage: a.valuation(3)
1
sage: a.valuation(3) + b.valuation(3)
2
sage: (a*b).valuation(3)
+Infinity
```

```
>>> from sage.all import *
>>> R = ZZ.quo(Integer(9))
>>> a = R(Integer(3))
>>> b = R(Integer(6))
>>> a.valuation(Integer(3))
1
>>> a.valuation(Integer(3)) + b.valuation(Integer(3))
2
>>> (a*b).valuation(Integer(3))
+Infinity
```

The valuation with respect to a unit is an error

```
sage: a.valuation(4)
Traceback (most recent call last):
...
ValueError: Valuation with respect to a unit is not defined.
```

```
>>> from sage.all import *
>>> a.valuation(Integer(4))
Traceback (most recent call last):
...
ValueError: Valuation with respect to a unit is not defined.
```

class sage.rings.finite_rings.integer_mod.**IntegerMod_gmp**

Bases: *IntegerMod_abstract*

Elements of $\mathbf{Z}/n\mathbf{Z}$ for n not small enough to be operated on in word size.

AUTHORS:

- Robert Bradshaw (2006-08-24)

gcd(*other*)

Greatest common divisor.

Returns the "smallest" generator in $\mathbf{Z}/N\mathbf{Z}$ of the ideal generated by `self` and `other`.

INPUT:

- `other` – an element of the same ring as this one

EXAMPLES:

```
sage: mod(2^3*3^2*5, 3^3*2^2*17^8).gcd(mod(2^4*3*17, 3^3*2^2*17^8))
12
sage: mod(0,17^8).gcd(mod(0,17^8))
0
```

```
>>> from sage.all import *
>>> mod(Integer(2)**Integer(3)*Integer(3)**Integer(2)*Integer(5),␣
↪Integer(3)**Integer(3)*Integer(2)**Integer(2)*Integer(17)**Integer(8)).
↪gcd(mod(Integer(2)**Integer(4)*Integer(3)*Integer(17),␣
↪Integer(3)**Integer(3)*Integer(2)**Integer(2)*Integer(17)**Integer(8)))
12
>>> mod(Integer(0),Integer(17)**Integer(8)).gcd(mod(Integer(0),
↪Integer(17)**Integer(8)))
0
```

is_one()

Return `True` if this is 1, otherwise `False`.

EXAMPLES:

```
sage: mod(1,5^23).is_one()
True
sage: mod(0,5^23).is_one()
False
```

```
>>> from sage.all import *
>>> mod(Integer(1),Integer(5)**Integer(23)).is_one()
True
>>> mod(Integer(0),Integer(5)**Integer(23)).is_one()
False
```

is_unit()

Return `True` iff this element is a unit.

EXAMPLES:

```
sage: mod(13, 5^23).is_unit()
True
sage: mod(25, 5^23).is_unit()
False
```

```
>>> from sage.all import *
>>> mod(Integer(13), Integer(5)**Integer(23)).is_unit()
```

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```
True
>>> mod(Integer(25), Integer(5)**Integer(23)).is_unit()
False
```

lift()

Lift an integer modulo n to the integers.

EXAMPLES:

```
sage: a = Mod(8943, 2^70); type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>
sage: lift(a)
8943
sage: a.lift()
8943
```

```
>>> from sage.all import *
>>> a = Mod(Integer(8943), Integer(2)**Integer(70)); type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>
>>> lift(a)
8943
>>> a.lift()
8943
```

class sage.rings.finite_rings.integer_mod.**IntegerMod_hom**

Bases: Morphism

class sage.rings.finite_rings.integer_mod.**IntegerMod_int**

Bases: *IntegerMod_abstract*

Elements of $\mathbf{Z}/n\mathbf{Z}$ for n small enough to be operated on in 32 bits

AUTHORS:

- Robert Bradshaw (2006-08-24)

EXAMPLES:

```
sage: a = Mod(10,30); a
10
sage: loads(a.dumps()) == a
True
```

```
>>> from sage.all import *
>>> a = Mod(Integer(10),Integer(30)); a
10
>>> loads(a.dumps()) == a
True
```

gcd(*other*)

Greatest common divisor.

Returns the "smallest" generator in $\mathbf{Z}/N\mathbf{Z}$ of the ideal generated by `self` and `other`.

INPUT:

- `other` – an element of the same ring as this one

EXAMPLES:

```
sage: R = Zmod(60); S = Zmod(72)
sage: a = R(40).gcd(S(30)); a
2
sage: a.parent()
Ring of integers modulo 12
sage: b = R(17).gcd(60); b
1
sage: b.parent()
Ring of integers modulo 60

sage: mod(72*5, 3^3*2^2*17^2).gcd(mod(48*17, 3^3*2^2*17^2))
12
sage: mod(0,1).gcd(mod(0,1))
0
```

```
>>> from sage.all import *
>>> R = Zmod(Integer(60)); S = Zmod(Integer(72))
>>> a = R(Integer(40)).gcd(S(Integer(30))); a
2
>>> a.parent()
Ring of integers modulo 12
>>> b = R(Integer(17)).gcd(Integer(60)); b
1
>>> b.parent()
Ring of integers modulo 60

>>> mod(Integer(72)*Integer(5),␣
↪Integer(3)**Integer(3)*Integer(2)**Integer(2)*Integer(17)**Integer(2)).
↪gcd(mod(Integer(48)*Integer(17),␣
↪Integer(3)**Integer(3)*Integer(2)**Integer(2)*Integer(17)**Integer(2)))
12
>>> mod(Integer(0),Integer(1)).gcd(mod(Integer(0),Integer(1)))
0
```

is_one()

Return `True` if this is 1, otherwise `False`.

EXAMPLES:

```
sage: mod(6,5).is_one()
True
sage: mod(0,5).is_one()
False
sage: mod(1, 1).is_one()
True
sage: Zmod(1).one().is_one()
True
```

```
>>> from sage.all import *
>>> mod(Integer(6),Integer(5)).is_one()
```

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```
True
>>> mod(Integer(0),Integer(5)).is_one()
False
>>> mod(Integer(1), Integer(1)).is_one()
True
>>> Zmod(Integer(1)).one().is_one()
True
```

is_unit()

Return `True` iff this element is a unit

EXAMPLES:

```
sage: a=Mod(23,100)
sage: a.is_unit()
True
sage: a=Mod(24,100)
sage: a.is_unit()
False
```

```
>>> from sage.all import *
>>> a=Mod(Integer(23),Integer(100))
>>> a.is_unit()
True
>>> a=Mod(Integer(24),Integer(100))
>>> a.is_unit()
False
```

lift()

Lift an integer modulo n to the integers.

EXAMPLES:

```
sage: a = Mod(8943, 2^10); type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>
sage: lift(a)
751
sage: a.lift()
751
```

```
>>> from sage.all import *
>>> a = Mod(Integer(8943), Integer(2)**Integer(10)); type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>
>>> lift(a)
751
>>> a.lift()
751
```

sqrt(*extend=True*, *all=False*)

Return square root or square roots of `self` modulo n.

INPUT:

- `extend` – boolean (default: `True`); if `True`, return a square root in an extension ring, if necessary. Otherwise, raise a `ValueError` if the square root is not in the base ring.
- `all` – boolean (default: `False`); if `True`, return {all} square roots of self, instead of just one.

ALGORITHM: Calculates the square roots mod p for each of the primes p dividing the order of the ring, then lifts them p-adically and uses the CRT to find a square root mod n.

See also *square_root_mod_prime_power()* and *square_root_mod_prime()* for more algorithmic details.

EXAMPLES:

```
sage: mod(-1, 17).sqrt()
4
sage: mod(5, 389).sqrt()
86
sage: mod(7, 18).sqrt()
5

sage: # needs sage.libs.pari
sage: a = mod(14, 5^60).sqrt()
sage: a*a
14
sage: mod(15, 389).sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
sage: Mod(1/9, next_prime(2^40)).sqrt()^(-2)
9
sage: Mod(1/25, next_prime(2^90)).sqrt()^(-2)
25
```

```
>>> from sage.all import *
>>> mod(-Integer(1), Integer(17)).sqrt()
4
>>> mod(Integer(5), Integer(389)).sqrt()
86
>>> mod(Integer(7), Integer(18)).sqrt()
5

>>> # needs sage.libs.pari
>>> a = mod(Integer(14), Integer(5)**Integer(60)).sqrt()
>>> a*a
14
>>> mod(Integer(15), Integer(389)).sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
>>> Mod(Integer(1)/Integer(9), next_prime(Integer(2)**Integer(40))).sqrt()**(-
↪Integer(2))
9
>>> Mod(Integer(1)/Integer(25), next_prime(Integer(2)**Integer(90))).
↪sqrt()**(-Integer(2))
25
```

```
sage: a = Mod(3,5); a
3
sage: x = Mod(-1, 360)
sage: x.sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
sage: y = x.sqrt(); y
sqrt359
sage: y.parent()
Univariate Quotient Polynomial Ring in sqrt359
 over Ring of integers modulo 360 with modulus x^2 + 1
sage: y^2
359
```

```
>>> from sage.all import *
>>> a = Mod(Integer(3),Integer(5)); a
3
>>> x = Mod(-Integer(1), Integer(360))
>>> x.sqrt(extend=False)
Traceback (most recent call last):
...
ValueError: self must be a square
>>> y = x.sqrt(); y
sqrt359
>>> y.parent()
Univariate Quotient Polynomial Ring in sqrt359
 over Ring of integers modulo 360 with modulus x^2 + 1
>>> y**Integer(2)
359
```

We compute all square roots in several cases:

```
sage: R = Integers(5*2^3*3^2); R
Ring of integers modulo 360
sage: R(40).sqrt(all=True)
[20, 160, 200, 340]
sage: [x for x in R if x^2 == 40]  # Brute force verification
[20, 160, 200, 340]
sage: R(1).sqrt(all=True)
[1, 19, 71, 89, 91, 109, 161, 179, 181, 199, 251, 269, 271, 289, 341, 359]
sage: R(0).sqrt(all=True)
[0, 60, 120, 180, 240, 300]
sage: GF(107)(0).sqrt(all=True)
[0]
```

```
>>> from sage.all import *
>>> R = Integers(Integer(5)*Integer(2)**Integer(3)*Integer(3)**Integer(2)); R
Ring of integers modulo 360
>>> R(Integer(40)).sqrt(all=True)
[20, 160, 200, 340]
>>> [x for x in R if x**Integer(2) == Integer(40)]  # Brute force verification
```

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```
[20, 160, 200, 340]
>>> R(Integer(1)).sqrt(all=True)
[1, 19, 71, 89, 91, 109, 161, 179, 181, 199, 251, 269, 271, 289, 341, 359]
>>> R(Integer(0)).sqrt(all=True)
[0, 60, 120, 180, 240, 300]
>>> GF(Integer(107))(Integer(0)).sqrt(all=True)
[0]
```

```
sage: # needs sage.libs.pari
sage: R = Integers(5*13^3*37); R
Ring of integers modulo 406445
sage: v = R(-1).sqrt(all=True); v
[78853, 111808, 160142, 193097, 213348, 246303, 294637, 327592]
sage: [x^2 for x in v]
[406444, 406444, 406444, 406444, 406444, 406444, 406444, 406444]
sage: v = R(169).sqrt(all=True); min(v), -max(v), len(v)
(13, 13, 104)
sage: all(x^2 == 169 for x in v)
True
```

```
>>> from sage.all import *
>>> # needs sage.libs.pari
>>> R = Integers(Integer(5)*Integer(13)**Integer(3)*Integer(37)); R
Ring of integers modulo 406445
>>> v = R(-Integer(1)).sqrt(all=True); v
[78853, 111808, 160142, 193097, 213348, 246303, 294637, 327592]
>>> [x**Integer(2) for x in v]
[406444, 406444, 406444, 406444, 406444, 406444, 406444, 406444]
>>> v = R(Integer(169)).sqrt(all=True); min(v), -max(v), len(v)
(13, 13, 104)
>>> all(x**Integer(2) == Integer(169) for x in v)
True
```

Modulo a power of 2:

```
sage: R = Integers(2^7); R
Ring of integers modulo 128
sage: a = R(17)
sage: a.sqrt()
23
sage: a.sqrt(all=True)
[23, 41, 87, 105]
sage: [x for x in R if x^2==17]
[23, 41, 87, 105]
```

```
>>> from sage.all import *
>>> R = Integers(Integer(2)**Integer(7)); R
Ring of integers modulo 128
>>> a = R(Integer(17))
>>> a.sqrt()
23
```

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```
>>> a.sqrt(all=True)
[23, 41, 87, 105]
>>> [x for x in R if x**Integer(2)==Integer(17)]
[23, 41, 87, 105]
```

class sage.rings.finite_rings.integer_mod.**IntegerMod_int64**

Bases: *IntegerMod_abstract*

Elements of $\mathbf{Z}/n\mathbf{Z}$ for n small enough to be operated on in 64 bits

EXAMPLES:

```
sage: a = Mod(10,3^10); a
10
sage: type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>
sage: loads(a.dumps()) == a
True
sage: Mod(5, 2^31)
5
```

```
>>> from sage.all import *
>>> a = Mod(Integer(10),Integer(3)**Integer(10)); a
10
>>> type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>
>>> loads(a.dumps()) == a
True
>>> Mod(Integer(5), Integer(2)**Integer(31))
5
```

AUTHORS:

- Robert Bradshaw (2006-09-14)

gcd(*other*)

Greatest common divisor.

Returns the "smallest" generator in $\mathbf{Z}/N\mathbf{Z}$ of the ideal generated by `self` and `other`.

INPUT:

- `other` – an element of the same ring as this one

EXAMPLES:

```
sage: mod(2^3*3^2*5, 3^3*2^2*17^5).gcd(mod(2^4*3*17, 3^3*2^2*17^5))
12
sage: mod(0,17^5).gcd(mod(0,17^5))
0
```

```
>>> from sage.all import *
>>> mod(Integer(2)**Integer(3)*Integer(3)**Integer(2)*Integer(5),␣
↪Integer(3)**Integer(3)*Integer(2)**Integer(2)*Integer(17)**Integer(5)).
↪gcd(mod(Integer(2)**Integer(4)*Integer(3)*Integer(17),␣
```

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```
↪Integer(3)**Integer(3)*Integer(2)**Integer(2)*Integer(17)**Integer(5)))
12
>>> mod(Integer(0),Integer(17)**Integer(5)).gcd(mod(Integer(0),
↪Integer(17)**Integer(5)))
0
```

is_one()

Return `True` if this is 1, otherwise `False`.

EXAMPLES:

```
sage: (mod(-1,5^10)^2).is_one()
True
sage: mod(0,5^10).is_one()
False
```

```
>>> from sage.all import *
>>> (mod(-Integer(1),Integer(5)**Integer(10))**Integer(2)).is_one()
True
>>> mod(Integer(0),Integer(5)**Integer(10)).is_one()
False
```

is_unit()

Return `True` iff this element is a unit.

EXAMPLES:

```
sage: mod(13, 5^10).is_unit()
True
sage: mod(25, 5^10).is_unit()
False
```

```
>>> from sage.all import *
>>> mod(Integer(13), Integer(5)**Integer(10)).is_unit()
True
>>> mod(Integer(25), Integer(5)**Integer(10)).is_unit()
False
```

lift()

Lift an integer modulo n to the integers.

EXAMPLES:

```
sage: a = Mod(8943, 2^25); type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>
sage: lift(a)
8943
sage: a.lift()
8943
```

```
>>> from sage.all import *
>>> a = Mod(Integer(8943), Integer(2)**Integer(25)); type(a)
```

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```
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>
>>> lift(a)
8943
>>> a.lift()
8943
```

class sage.rings.finite_rings.integer_mod.**IntegerMod_to_Integer**

Bases: Map

Map to lift elements to Integer.

EXAMPLES:

```
sage: ZZ.convert_map_from(GF(2))
Lifting map:
  From: Finite Field of size 2
  To:   Integer Ring
```

```
>>> from sage.all import *
>>> ZZ.convert_map_from(GF(Integer(2)))
Lifting map:
  From: Finite Field of size 2
  To:   Integer Ring
```

class sage.rings.finite_rings.integer_mod.**IntegerMod_to_IntegerMod**

Bases: *IntegerMod_hom*

Very fast IntegerMod to IntegerMod homomorphism.

EXAMPLES:

```
sage: from sage.rings.finite_rings.integer_mod import IntegerMod_to_IntegerMod
sage: Rs = [Integers(3**k) for k in range(1,30,5)]
sage: [type(R(0)) for R in Rs]
[<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>]
sage: fs = [IntegerMod_to_IntegerMod(S, R)
....:       for R in Rs for S in Rs if S is not R and S.order() > R.order()]
sage: all(f(-1) == f.codomain()(-1) for f in fs)
True
sage: [f(-1) for f in fs]
[2, 2, 2, 2, 2, 728, 728, 728, 728, 177146, 177146, 177146, 43046720, 43046720,␣
↪10460353202]
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.integer_mod import IntegerMod_to_IntegerMod
>>> Rs = [Integers(Integer(3)**k) for k in range(Integer(1),Integer(30),
↪Integer(5))]
>>> [type(R(Integer(0))) for R in Rs]
```

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```
[<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>]
>>> fs = [IntegerMod_to_IntegerMod(S, R)
...       for R in Rs for S in Rs if S is not R and S.order() > R.order()]
>>> all(f(-Integer(1)) == f.codomain()(-Integer(1)) for f in fs)
True
>>> [f(-Integer(1)) for f in fs]
[2, 2, 2, 2, 2, 728, 728, 728, 728, 177146, 177146, 177146, 43046720, 43046720,␣
↪10460353202]
```

is_injective()

Return whether this morphism is injective.

EXAMPLES:

```
sage: Zmod(4).hom(Zmod(2)).is_injective()
False
```

```
>>> from sage.all import *
>>> Zmod(Integer(4)).hom(Zmod(Integer(2))).is_injective()
False
```

is_surjective()

Return whether this morphism is surjective.

EXAMPLES:

```
sage: Zmod(4).hom(Zmod(2)).is_surjective()
True
```

```
>>> from sage.all import *
>>> Zmod(Integer(4)).hom(Zmod(Integer(2))).is_surjective()
True
```

class sage.rings.finite_rings.integer_mod.**Integer_to_IntegerMod**

Bases: *IntegerMod_hom*

Fast $\mathbf{Z} \to \mathbf{Z}/n\mathbf{Z}$ morphism.

EXAMPLES:

We make sure it works for every type.

```
sage: from sage.rings.finite_rings.integer_mod import Integer_to_IntegerMod
sage: Rs = [Integers(10), Integers(10^5), Integers(10^10)]
sage: [type(R(0)) for R in Rs]
[<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>]
sage: fs = [Integer_to_IntegerMod(R) for R in Rs]
```

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```
sage: [f(-1) for f in fs]
[9, 99999, 9999999999]
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.integer_mod import Integer_to_IntegerMod
>>> Rs = [Integers(Integer(10)), Integers(Integer(10)**Integer(5)),␣
↪Integers(Integer(10)**Integer(10))]
>>> [type(R(Integer(0))) for R in Rs]
[<class 'sage.rings.finite_rings.integer_mod.IntegerMod_int'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_int64'>,
 <class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>]
>>> fs = [Integer_to_IntegerMod(R) for R in Rs]
>>> [f(-Integer(1)) for f in fs]
[9, 99999, 9999999999]
```

is_injective()

Return whether this morphism is injective.

EXAMPLES:

```
sage: ZZ.hom(Zmod(2)).is_injective()
False
```

```
>>> from sage.all import *
>>> ZZ.hom(Zmod(Integer(2))).is_injective()
False
```

is_surjective()

Return whether this morphism is surjective.

EXAMPLES:

```
sage: ZZ.hom(Zmod(2)).is_surjective()
True
```

```
>>> from sage.all import *
>>> ZZ.hom(Zmod(Integer(2))).is_surjective()
True
```

section()

sage.rings.finite_rings.integer_mod.**Mod**(*n*, *m*, *parent=None*)

Return the equivalence class of n modulo m as an element of $\mathbf{Z}/m\mathbf{Z}$.

EXAMPLES:

```
sage: x = Mod(12345678, 32098203845329048)
sage: x
12345678
sage: x^100
1017322209155072
```

```
>>> from sage.all import *
>>> x = Mod(Integer(12345678), Integer(32098203845329048))
>>> x
12345678
>>> x**Integer(100)
1017322209155072
```

You can also use the lowercase version:

```
sage: mod(12,5)
2
```

```
>>> from sage.all import *
>>> mod(Integer(12),Integer(5))
2
```

Illustrates that Issue #5971 is fixed. Consider n modulo m when $m = 0$. Then $\mathbf{Z}/0\mathbf{Z}$ is isomorphic to $\mathbf{Z}$ so n modulo 0 is equivalent to n for any integer value of n:

```
sage: Mod(10, 0)
10
sage: a = randint(-100, 100)
sage: Mod(a, 0) == a
True
```

```
>>> from sage.all import *
>>> Mod(Integer(10), Integer(0))
10
>>> a = randint(-Integer(100), Integer(100))
>>> Mod(a, Integer(0)) == a
True
```

class sage.rings.finite_rings.integer_mod.**NativeIntStruct**

Bases: object

We store the various forms of the modulus here rather than in the parent for efficiency reasons.

We may also store a cached table of all elements of a given ring in this class.

inverses

precompute_table(*parent*)

Function to compute and cache all elements of this class.

If `inverses == True`, also computes and caches the inverses of the invertible elements.

EXAMPLES:

```
sage: from sage.rings.finite_rings.integer_mod import NativeIntStruct
sage: R = IntegerModRing(10)
sage: M = NativeIntStruct(R.order())
sage: M.precompute_table(R)
sage: M.table
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
sage: M.inverses
[None, 1, None, 7, None, None, None, 3, None, 9]
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.integer_mod import NativeIntStruct
>>> R = IntegerModRing(Integer(10))
>>> M = NativeIntStruct(R.order())
>>> M.precompute_table(R)
>>> M.table
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
>>> M.inverses
[None, 1, None, 7, None, None, None, 3, None, 9]
```

This is used by the *sage.rings.finite_rings.integer_mod_ring.IntegerModRing_generic* constructor:

```
sage: from sage.rings.finite_rings.integer_mod_ring import IntegerModRing_
↪generic
sage: R = IntegerModRing_generic(39, cache=False)
sage: R(5)^-1
8
sage: R(5)^-1 is R(8)
False
sage: R = IntegerModRing_generic(39, cache=True)  # indirect doctest
sage: R(5)^-1 is R(8)
True
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.integer_mod_ring import IntegerModRing_
↪generic
>>> R = IntegerModRing_generic(Integer(39), cache=False)
>>> R(Integer(5))**-Integer(1)
8
>>> R(Integer(5))**-Integer(1) is R(Integer(8))
False
>>> R = IntegerModRing_generic(Integer(39), cache=True)  # indirect doctest
>>> R(Integer(5))**-Integer(1) is R(Integer(8))
True
```

Check that the inverse of 0 modulo 1 works, see Issue #13639:

```
sage: R = IntegerModRing_generic(1, cache=True)  # indirect doctest
sage: R(0)^-1 is R(0)
True
```

```
>>> from sage.all import *
>>> R = IntegerModRing_generic(Integer(1), cache=True)  # indirect doctest
>>> R(Integer(0))**-Integer(1) is R(Integer(0))
True
```

table

sage.rings.finite_rings.integer_mod.**is_IntegerMod**(x)

Return `True` if and only if x is an integer modulo n.

EXAMPLES:

```
sage: from sage.rings.finite_rings.integer_mod import is_IntegerMod
sage: is_IntegerMod(5)
doctest:warning...
DeprecationWarning: The function is_IntegerMod is deprecated;
use 'isinstance(..., IntegerMod_abstract)' instead.
See https://github.com/sagemath/sage/issues/38128 for details.
False
sage: is_IntegerMod(Mod(5,10))
True
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.integer_mod import is_IntegerMod
>>> is_IntegerMod(Integer(5))
doctest:warning...
DeprecationWarning: The function is_IntegerMod is deprecated;
use 'isinstance(..., IntegerMod_abstract)' instead.
See https://github.com/sagemath/sage/issues/38128 for details.
False
>>> is_IntegerMod(Mod(Integer(5),Integer(10)))
True
```

`sage.rings.finite_rings.integer_mod.`**`lucas`**(*k*, *P*, *Q=1*, *n=None*)

Return $[V_k(P,Q) \mod n, Q^{\lfloor k/2 \rfloor} \mod n]$ where V_k is the Lucas function defined by the recursive relation

$$V_k(P,Q) = PV_{k-1}(P,Q) - QV_{k-2}(P,Q)$$

with $V_0 = 2, V_1 = P$.

INPUT:

- `k` – integer; index to compute
- `P`, `Q` – integers or modular integers; initial values
- `n` – integer (optional); modulus to use if `P` is not a modular integer

REFERENCES:

- [IEEEP1363]

AUTHORS:

- Somindu Chaya Ramanna, Shashank Singh and Srinivas Vivek Venkatesh (2011-09-15, ECC2011 summer school)
- Robert Bradshaw

EXAMPLES:

```
sage: [lucas(k,4,5,11)[0] for k in range(30)]
[2, 4, 6, 4, 8, 1, 8, 5, 2, 5, 10, 4, 10, 9, 8, 9, 7, 5, 7, 3, 10, 3, 6, 9, 6, 1,␣
↪7, 1, 2, 3]

sage: lucas(20,4,5,11)
[10, 1]
```

```
>>> from sage.all import *
>>> [lucas(k,Integer(4),Integer(5),Integer(11))[Integer(0)] for k in␣
↪range(Integer(30))]
[2, 4, 6, 4, 8, 1, 8, 5, 2, 5, 10, 4, 10, 9, 8, 9, 7, 5, 7, 3, 10, 3, 6, 9, 6, 1,␣
↪7, 1, 2, 3]

>>> lucas(Integer(20),Integer(4),Integer(5),Integer(11))
[10, 1]
```

sage.rings.finite_rings.integer_mod.**lucas_q1**(*mm*, *P*)

Return $V_k(P,1)$ where V_k is the Lucas function defined by the recursive relation.

$V_k(P,Q) = PV_{k-1}(P,Q) - QV_{k-2}(P,Q)$

with $V_0 = 2, V_1(P_Q) = P$.

REFERENCES:

- [Pos1988]

AUTHORS:

- Robert Bradshaw

sage.rings.finite_rings.integer_mod.**makeNativeIntStruct**

alias of *NativeIntStruct*

sage.rings.finite_rings.integer_mod.**mod**(*n*, *m*, *parent=None*)

Return the equivalence class of n modulo m as an element of $\mathbf{Z}/m\mathbf{Z}$.

EXAMPLES:

```
sage: x = Mod(12345678, 32098203845329048)
sage: x
12345678
sage: x^100
1017322209155072
```

```
>>> from sage.all import *
>>> x = Mod(Integer(12345678), Integer(32098203845329048))
>>> x
12345678
>>> x**Integer(100)
1017322209155072
```

You can also use the lowercase version:

```
sage: mod(12,5)
2
```

```
>>> from sage.all import *
>>> mod(Integer(12),Integer(5))
2
```

Illustrates that Issue #5971 is fixed. Consider n modulo m when $m = 0$. Then $\mathbf{Z}/0\mathbf{Z}$ is isomorphic to $\mathbf{Z}$ so n modulo 0 is equivalent to n for any integer value of n:

```
sage: Mod(10, 0)
10
sage: a = randint(-100, 100)
sage: Mod(a, 0) == a
True
```

```
>>> from sage.all import *
>>> Mod(Integer(10), Integer(0))
10
>>> a = randint(-Integer(100), Integer(100))
>>> Mod(a, Integer(0)) == a
True
```

`sage.rings.finite_rings.integer_mod.`**`square_root_mod_prime`**(*a*, *p=None*)

Calculate the square root of a, where a is an integer mod p; if a is not a perfect square, this returns an (incorrect) answer without checking.

ALGORITHM: Several cases based on residue class of p mod 16.

- p mod $2 = 0$: $p = 2$ so $\sqrt{a} = a$.
- p mod $4 = 3$: $\sqrt{a} = a^{(p+1)/4}$.
- p mod $8 = 5$: $\sqrt{a} = \zeta i a$ where $\zeta = (2a)^{(p-5)/8}$, $i = \sqrt{-1}$.
- p mod $16 = 9$: Similar, work in a bi-quadratic extension of $\mathbf{F}_p$ for small p, Tonelli and Shanks for large p.
- p mod $16 = 1$: Tonelli and Shanks.

REFERENCES:

- [Mul2004]
- [Atk1992]
- [Pos1988]

AUTHORS:

- Robert Bradshaw

`sage.rings.finite_rings.integer_mod.`**`square_root_mod_prime_power`**(*a*, *p*, *e*)

Calculate the square root of a, where a is an integer mod p^e.

ALGORITHM: Compute p-adically by stripping off even powers of p to get a unit and lifting $\sqrt{unit}$ mod p via Newton's method whenever p is odd and by a variant of Hensel lifting for $p = 2$.

AUTHORS:

- Robert Bradshaw
- Lorenz Panny (2022): polynomial-time algorithm for $p = 2$

EXAMPLES:

```
sage: from sage.rings.finite_rings.integer_mod import square_root_mod_prime_power
sage: a = Mod(17,2^20)
sage: b = square_root_mod_prime_power(a,2,20)
sage: b^2 == a
True
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.integer_mod import square_root_mod_prime_power
>>> a = Mod(Integer(17),Integer(2)**Integer(20))
>>> b = square_root_mod_prime_power(a,Integer(2),Integer(20))
>>> b**Integer(2) == a
True
```

```
sage: a = Mod(72, 97^10)
sage: b = square_root_mod_prime_power(a, 97, 10)                                #␣
↪needs sage.libs.pari
sage: b^2 == a                                                                  #␣
↪needs sage.libs.pari
True
sage: mod(100, 5^7).sqrt()^2                                                    #␣
↪needs sage.libs.pari
100
```

```
>>> from sage.all import *
>>> a = Mod(Integer(72), Integer(97)**Integer(10))
>>> b = square_root_mod_prime_power(a, Integer(97), Integer(10))                 ␣
↪              # needs sage.libs.pari
>>> b**Integer(2) == a                                                           ␣
↪      # needs sage.libs.pari
True
>>> mod(Integer(100), Integer(5)**Integer(7)).sqrt()**Integer(2)                 ␣
↪                                  # needs sage.libs.pari
100
```

CHAPTER

TWO

FINITE FIELDS

2.1 Finite fields

Sage supports arithmetic in finite prime and extension fields. Several implementation for prime fields are implemented natively in Sage for several sizes of primes p. These implementations are

- `sage.rings.finite_rings.integer_mod.IntegerMod_int`,
- `sage.rings.finite_rings.integer_mod.IntegerMod_int64`, and
- `sage.rings.finite_rings.integer_mod.IntegerMod_gmp`.

Small extension fields of cardinality $< 2^{16}$ are implemented using tables of Zech logs via the Givaro C++ library (`sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro`). While this representation is very fast it is limited to finite fields of small cardinality. Larger finite extension fields of order $q >= 2^{16}$ are internally represented as polynomials over smaller finite prime fields. If the characteristic of such a field is 2 then NTL is used internally to represent the field (`sage.rings.finite_rings.finite_field_ntl_gf2e.FiniteField_ntl_gf2e`). In all other case the PARI C library is used (`sage.rings.finite_rings.finite_field_pari_ffelt.FiniteField_pari_ffelt`).

However, this distinction is internal only and the user usually does not have to worry about it because consistency across all implementations is aimed for. In all extension field implementations the user may either specify a minimal polynomial or leave the choice to Sage.

For small finite fields the default choice are Conway polynomials.

The Conway polynomial C_n is the lexicographically first monic irreducible, primitive polynomial of degree n over $GF(p)$ with the property that for a root α of C_n we have that $\beta = \alpha^{(p^n-1)/(p^m-1)}$ is a root of C_m for all m dividing n. Sage contains a database of Conway polynomials which also can be queried independently of finite field construction.

A pseudo-Conway polynomial satisfies all of the conditions required of a Conway polynomial except the condition that it is lexicographically first. They are therefore not unique. If no variable name is specified for an extension field, Sage will fit the finite field into a compatible lattice of field extensions defined by pseudo-Conway polynomials. This lattice is stored in an *`AlgebraicClosureFiniteField`* object; different algebraic closure objects can be created by using a different `prefix` keyword to the finite field constructor.

Note that the computation of pseudo-Conway polynomials is expensive when the degree is large and highly composite. If a variable name is specified then a random polynomial is used instead, which will be much faster to find.

While Sage supports basic arithmetic in finite fields some more advanced features for computing with finite fields are still not implemented. For instance, Sage does not calculate embeddings of finite fields yet.

EXAMPLES:

```
sage: k = GF(5); type(k)
<class 'sage.rings.finite_rings.finite_field_prime_modn.FiniteField_prime_modn_with_
↪category'>
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)); type(k)
<class 'sage.rings.finite_rings.finite_field_prime_modn.FiniteField_prime_modn_with_
↪category'>
```

```
sage: k = GF(5^2,'c'); type(k)                                                  #␣
↪needs sage.libs.linbox
<class 'sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro_with_category'>
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(2),'c'); type(k)                                 ␣
↪                # needs sage.libs.linbox
<class 'sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro_with_category'>
```

One can also give the cardinality $q = p^n$ as the tuple (p, n):

```
sage: k = GF((5, 2),'c'); k
Finite Field in c of size 5^2
```

```
>>> from sage.all import *
>>> k = GF((Integer(5), Integer(2)),'c'); k
Finite Field in c of size 5^2
```

```
sage: k = GF(2^16,'c'); type(k)                                                 #␣
↪needs sage.libs.ntl
<class 'sage.rings.finite_rings.finite_field_ntl_gf2e.FiniteField_ntl_gf2e_with_
↪category'>
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(16),'c'); type(k)                                ␣
↪                # needs sage.libs.ntl
<class 'sage.rings.finite_rings.finite_field_ntl_gf2e.FiniteField_ntl_gf2e_with_
↪category'>
```

```
sage: k = GF((3, 16),'c'); type(k)
<class 'sage.rings.finite_rings.finite_field_pari_ffelt.FiniteField_pari_ffelt_with_
↪category'>
```

```
>>> from sage.all import *
>>> k = GF((Integer(3), Integer(16)),'c'); type(k)
<class 'sage.rings.finite_rings.finite_field_pari_ffelt.FiniteField_pari_ffelt_with_
↪category'>
```

Finite Fields support iteration, starting with 0.

```
sage: k = GF(9, 'a')
sage: for i,x in enumerate(k):  print("{} {}".format(i, x))
0 0
1 a
2 a + 1
3 2*a + 1
```

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```
4 2
5 2*a
6 2*a + 2
7 a + 2
8 1
sage: for a in GF(5):
....:     print(a)
0
1
2
3
4
```

```
>>> from sage.all import *
>>> k = GF(Integer(9), 'a')
>>> for i,x in enumerate(k):  print("{} {}".format(i, x))
0 0
1 a
2 a + 1
3 2*a + 1
4 2
5 2*a
6 2*a + 2
7 a + 2
8 1
>>> for a in GF(Integer(5)):
...     print(a)
0
1
2
3
4
```

We output the base rings of several finite fields.

```
sage: k = GF(3); type(k)
<class 'sage.rings.finite_rings.finite_field_prime_modn.FiniteField_prime_modn_with_
↪category'>
sage: k.base_ring()
Finite Field of size 3
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)); type(k)
<class 'sage.rings.finite_rings.finite_field_prime_modn.FiniteField_prime_modn_with_
↪category'>
>>> k.base_ring()
Finite Field of size 3
```

```
sage: # needs sage.libs.linbox
sage: k = GF(9,'alpha'); type(k)
<class 'sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro_with_category'>
```

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```
sage: k.base_ring()
Finite Field of size 3
```

```
>>> from sage.all import *
>>> # needs sage.libs.linbox
>>> k = GF(Integer(9),'alpha'); type(k)
<class 'sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro_with_category'>
>>> k.base_ring()
Finite Field of size 3
```

```
sage: k = GF((3, 40),'b'); type(k)
<class 'sage.rings.finite_rings.finite_field_pari_ffelt.FiniteField_pari_ffelt_with_
↪category'>
sage: k.base_ring()
Finite Field of size 3
```

```
>>> from sage.all import *
>>> k = GF((Integer(3), Integer(40)),'b'); type(k)
<class 'sage.rings.finite_rings.finite_field_pari_ffelt.FiniteField_pari_ffelt_with_
↪category'>
>>> k.base_ring()
Finite Field of size 3
```

Further examples:

```
sage: GF(2).is_field()
True
sage: GF(next_prime(10^20)).is_field()
True
sage: GF(19^20,'a').is_field()
True
sage: GF(8,'a').is_field()
True
```

```
>>> from sage.all import *
>>> GF(Integer(2)).is_field()
True
>>> GF(next_prime(Integer(10)**Integer(20))).is_field()
True
>>> GF(Integer(19)**Integer(20),'a').is_field()
True
>>> GF(Integer(8),'a').is_field()
True
```

AUTHORS:

- William Stein: initial version
- Robert Bradshaw: prime field implementation
- Martin Albrecht: Givaro and ntl.GF2E implementations

class sage.rings.finite_rings.finite_field_constructor.**FiniteFieldFactory**(**args*, ***kwds*)

Bases: UniqueFactory

Return the globally unique finite field of given order with generator labeled by the given name and possibly with given modulus.

INPUT:

- `order` – a prime power
- `name` – string, optional. Note that there can be a substantial speed penalty (in creating extension fields) when omitting the variable name, since doing so triggers the computation of pseudo-Conway polynomials in order to define a coherent lattice of extensions of the prime field. The speed penalty grows with the size of extension degree and with the number of factors of the extension degree.
- `modulus` – (optional) either a defining polynomial for the field, or a string specifying an algorithm to use to generate such a polynomial. If `modulus` is a string, it is passed to `irreducible_element()` as the parameter `algorithm`; see there for the permissible values of this parameter. In particular, you can specify `modulus="primitive"` to get a primitive polynomial. You may not specify a modulus if you do not specify a variable name.
- `impl` – (optional) a string specifying the implementation of the finite field. Possible values are:
  - `'modn'` – ring of integers modulo p (only for prime fields)
  - `'givaro'` – Givaro, which uses Zech logs (only for fields of at most 65521 elements)
  - `'ntl'` – NTL using GF2X (only in characteristic 2)
  - `'pari'` or `'pari_ffelt'` – PARI's `FFELT` type (only for extension fields)
- `elem_cache` – (default: order < 500) cache all elements to avoid creation time; ignored unless `impl='givaro'`
- `repr` – (default: `'poly'`) ignored unless `impl='givaro'`; controls the way elements are printed to the user:
  - 'log': repr is `log_repr()`
  - 'int': repr is `int_repr()`
  - 'poly': repr is `poly_repr()`
- `check_irreducible` – verify that the polynomial modulus is irreducible
- `proof` – boolean (default: `True`); if `True`, use provable primality test; otherwise only use pseudoprimality test

ALIAS: You can also use `GF` instead of `FiniteField` – they are identical.

EXAMPLES:

```
sage: k.<a> = FiniteField(9); k
Finite Field in a of size 3^2
sage: parent(a)
Finite Field in a of size 3^2
sage: charpoly(a, 'y')
y^2 + 2*y + 2
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(9), names=('a',)); (a,) = k._first_ngens(1); k
Finite Field in a of size 3^2
>>> parent(a)
Finite Field in a of size 3^2
```

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```
>>> charpoly(a, 'y')
y^2 + 2*y + 2
```

We illustrate the proof flag. The following example would hang for a very long time if we didn't use `proof=False`.

> **Note**
>
> Magma only supports `proof=False` for making finite fields, so falsely appears to be faster than Sage – see Issue #10975.

```
sage: k = FiniteField(10^1000 + 453, proof=False)
sage: k = FiniteField((10^1000 + 453)^2, 'a', proof=False)      # long time --␣
↪about 5 seconds
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(10)**Integer(1000) + Integer(453), proof=False)
>>> k = FiniteField((Integer(10)**Integer(1000) + Integer(453))**Integer(2), 'a',␣
↪proof=False)      # long time -- about 5 seconds
```

```
sage: F.<x> = GF(5)[]
sage: K.<a> = GF(5**5, name='a', modulus=x^5 - x +1 )
sage: f = K.modulus(); f
x^5 + 4*x + 1
sage: type(f)
 <class 'sage.rings.polynomial.polynomial_zmod_flint.Polynomial_zmod_flint'>
```

```
>>> from sage.all import *
>>> F = GF(Integer(5))['x']; (x,) = F._first_ngens(1)
>>> K = GF(Integer(5)**Integer(5), name='a', modulus=x**Integer(5) - x␣
↪+Integer(1) , names=('a',)); (a,) = K._first_ngens(1)
>>> f = K.modulus(); f
x^5 + 4*x + 1
>>> type(f)
 <class 'sage.rings.polynomial.polynomial_zmod_flint.Polynomial_zmod_flint'>
```

By default, the given generator is not guaranteed to be primitive (a generator of the multiplicative group), use `modulus="primitive"` if you need this:

```
sage: K.<a> = GF(5^45)
sage: a.multiplicative_order()
7105427357601001858711242675781
sage: a.is_square()
True
sage: K.<b> = GF(5^45, modulus='primitive')
sage: b.multiplicative_order()
28421709430404007434844970703124
```

```
>>> from sage.all import *
>>> K = GF(Integer(5)**Integer(45), names=('a',)); (a,) = K._first_ngens(1)
>>> a.multiplicative_order()
```

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```
710542735760100185871124267578 1
>>> a.is_square()
True
>>> K = GF(Integer(5)**Integer(45), modulus='primitive', names=('b',)); (b,) = K._
↪first_ngens(1)
>>> b.multiplicative_order()
28421709430404007434844970703124
```

The modulus must be irreducible:

```
sage: K.<a> = GF(5**5, name='a', modulus=x^5 - x)
Traceback (most recent call last):
...
ValueError: finite field modulus must be irreducible but it is not
```

```
>>> from sage.all import *
>>> K = GF(Integer(5)**Integer(5), name='a', modulus=x**Integer(5) - x, names=('a
↪',)); (a,) = K._first_ngens(1)
Traceback (most recent call last):
...
ValueError: finite field modulus must be irreducible but it is not
```

You can't accidentally fool the constructor into thinking the modulus is irreducible when it is not, since it actually tests irreducibility modulo p. Also, the modulus has to be of the right degree (this is always checked):

```
sage: F.<x> = QQ[]
sage: factor(x^5 + 2)
x^5 + 2
sage: K.<a> = GF(5^5, modulus=x^5 + 2)
Traceback (most recent call last):
...
ValueError: finite field modulus must be irreducible but it is not
sage: K.<a> = GF(5^5, modulus=x^3 + 3*x + 3, check_irreducible=False)
Traceback (most recent call last):
...
ValueError: the degree of the modulus does not equal the degree of the field
```

```
>>> from sage.all import *
>>> F = QQ['x']; (x,) = F._first_ngens(1)
>>> factor(x**Integer(5) + Integer(2))
x^5 + 2
>>> K = GF(Integer(5)**Integer(5), modulus=x**Integer(5) + Integer(2), names=('a',
↪)); (a,) = K._first_ngens(1)
Traceback (most recent call last):
...
ValueError: finite field modulus must be irreducible but it is not
>>> K = GF(Integer(5)**Integer(5), modulus=x**Integer(3) + Integer(3)*x +␣
↪Integer(3), check_irreducible=False, names=('a',)); (a,) = K._first_ngens(1)
Traceback (most recent call last):
...
ValueError: the degree of the modulus does not equal the degree of the field
```

Any type which can be converted to the polynomial ring $GF(p)[x]$ is accepted as modulus:

```
sage: K.<a> = GF(13^3, modulus=[1,0,0,2])
sage: K.<a> = GF(13^10, modulus=pari("ffinit(13,10)"))
sage: var('x')
x
sage: K.<a> = GF(13^2, modulus=x^2 - 2)
sage: K.<a> = GF(13^2, modulus=sin(x))
Traceback (most recent call last):
...
TypeError: self must be a numeric expression
```

```
>>> from sage.all import *
>>> K = GF(Integer(13)**Integer(3), modulus=[Integer(1),Integer(0),Integer(0),
↪Integer(2)], names=('a',)); (a,) = K._first_ngens(1)
>>> K = GF(Integer(13)**Integer(10), modulus=pari("ffinit(13,10)"), names=('a',));
↪ (a,) = K._first_ngens(1)
>>> var('x')
x
>>> K = GF(Integer(13)**Integer(2), modulus=x**Integer(2) - Integer(2), names=('a
↪',)); (a,) = K._first_ngens(1)
>>> K = GF(Integer(13)**Integer(2), modulus=sin(x), names=('a',)); (a,) = K._
↪first_ngens(1)
Traceback (most recent call last):
...
TypeError: self must be a numeric expression
```

If you wish to live dangerously, you can tell the constructor not to test irreducibility using `check_irreducible=False`, but this can easily lead to crashes and hangs – so do not do it unless you know that the modulus really is irreducible!

```
sage: K.<a> = GF(5**2, name='a', modulus=x^2 + 2, check_irreducible=False)
```

```
>>> from sage.all import *
>>> K = GF(Integer(5)**Integer(2), name='a', modulus=x**Integer(2) + Integer(2),␣
↪check_irreducible=False, names=('a',)); (a,) = K._first_ngens(1)
```

Even for prime fields, you can specify a modulus. This will not change how Sage computes in this field, but it will change the result of the `modulus()` and `gen()` methods:

```
sage: k.<a> = GF(5, modulus='primitive')
sage: k.modulus()
x + 3
sage: a
2
```

```
>>> from sage.all import *
>>> k = GF(Integer(5), modulus='primitive', names=('a',)); (a,) = k._first_
↪ngens(1)
>>> k.modulus()
x + 3
>>> a
2
```

The order of a finite field must be a prime power:

```
sage: GF(1)
Traceback (most recent call last):
...
ValueError: the order of a finite field must be at least 2
sage: GF(100)
Traceback (most recent call last):
...
ValueError: the order of a finite field must be a prime power
```

```
>>> from sage.all import *
>>> GF(Integer(1))
Traceback (most recent call last):
...
ValueError: the order of a finite field must be at least 2
>>> GF(Integer(100))
Traceback (most recent call last):
...
ValueError: the order of a finite field must be a prime power
```

Finite fields with explicit random modulus are not cached:

```
sage: k.<a> = GF(5**10, modulus='random')
sage: n.<a> = GF(5**10, modulus='random')
sage: while k.modulus() == n.modulus():
....:     n.<a> = GF(5**10, modulus='random')
sage: n is k
False
sage: GF(5**10, 'a') is GF(5**10, 'a')
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(10), modulus='random', names=('a',)); (a,) = k._
↪first_ngens(1)
>>> n = GF(Integer(5)**Integer(10), modulus='random', names=('a',)); (a,) = n._
↪first_ngens(1)
>>> while k.modulus() == n.modulus():
...     n = GF(Integer(5)**Integer(10), modulus='random', names=('a',)); (a,) = n.
↪_first_ngens(1)
>>> n is k
False
>>> GF(Integer(5)**Integer(10), 'a') is GF(Integer(5)**Integer(10), 'a')
True
```

We check that various ways of creating the same finite field yield the same object, which is cached:

```
sage: K = GF(7, 'a')
sage: L = GF(7, 'b')
sage: K is L            # name is ignored for prime fields
True
sage: K is GF(7, modulus=K.modulus())
True
sage: K = GF(4,'a'); K.modulus()
x^2 + x + 1
```

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```
sage: L = GF(4,'a', K.modulus())
sage: K is L
True
sage: M = GF(4,'a', K.modulus().change_variable_name('y'))
sage: K is M
True
```

```
>>> from sage.all import *
>>> K = GF(Integer(7), 'a')
>>> L = GF(Integer(7), 'b')
>>> K is L            # name is ignored for prime fields
True
>>> K is GF(Integer(7), modulus=K.modulus())
True
>>> K = GF(Integer(4),'a'); K.modulus()
x^2 + x + 1
>>> L = GF(Integer(4),'a', K.modulus())
>>> K is L
True
>>> M = GF(Integer(4),'a', K.modulus().change_variable_name('y'))
>>> K is M
True
```

You may print finite field elements as integers. This currently only works if the order of field is $< 2^{16}$, though:

```
sage: k.<a> = GF(2^8, repr='int')
sage: a
2
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8), repr='int', names=('a',)); (a,) = k._first_
↪ngens(1)
>>> a
2
```

The following demonstrate coercions for finite fields using Conway polynomials:

```
sage: k = GF(5^2); a = k.gen()
sage: l = GF(5^5); b = l.gen()
sage: a + b
3*z10^5 + z10^4 + z10^2 + 3*z10 + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(2)); a = k.gen()
>>> l = GF(Integer(5)**Integer(5)); b = l.gen()
>>> a + b
3*z10^5 + z10^4 + z10^2 + 3*z10 + 1
```

Note that embeddings are compatible in lattices of such finite fields:

```
sage: m = GF(5^3); c = m.gen()
sage: (a+b)+c == a+(b+c)
```

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```
True
sage: (a*b)*c == a*(b*c)
True
sage: from sage.categories.pushout import pushout
sage: n = pushout(k, l)
sage: o = pushout(l, m)
sage: q = pushout(n, o)
sage: q(o(b)) == q(n(b))
True
```

```
>>> from sage.all import *
>>> m = GF(Integer(5)**Integer(3)); c = m.gen()
>>> (a+b)+c == a+(b+c)
True
>>> (a*b)*c == a*(b*c)
True
>>> from sage.categories.pushout import pushout
>>> n = pushout(k, l)
>>> o = pushout(l, m)
>>> q = pushout(n, o)
>>> q(o(b)) == q(n(b))
True
```

Another check that embeddings are defined properly:

```
sage: k = GF(3**10)
sage: l = GF(3**20)
sage: l(k.gen()**10) == l(k.gen())**10
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(10))
>>> l = GF(Integer(3)**Integer(20))
>>> l(k.gen()**Integer(10)) == l(k.gen())**Integer(10)
True
```

Using pseudo-Conway polynomials is slow for highly composite extension degrees:

```
sage: k = GF(3^120)  # long time (about 3 seconds)
sage: GF(3^40).gen().minimal_polynomial()(k.gen()^((3^120-1)/(3^40-1)))  # long␣
↪time (because of previous line)
0
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(120))  # long time (about 3 seconds)
>>> GF(Integer(3)**Integer(40)).gen().minimal_polynomial()(k.
↪gen()**((Integer(3)**Integer(120)-Integer(1))/(Integer(3)**Integer(40)-
↪Integer(1))))  # long time (because of previous line)
0
```

Before Issue #17569, the boolean keyword argument `conway` was required when creating finite fields without a variable name. This keyword argument is now removed (Issue #21433). You can still pass in `prefix` as an

argument, which has the effect of changing the variable name of the algebraic closure:

```
sage: K = GF(3^10, prefix='w'); L = GF(3^10); K is L
False
sage: K.variable_name(), L.variable_name()
('w10', 'z10')
sage: list(K.polynomial()) == list(L.polynomial())
True
```

```
>>> from sage.all import *
>>> K = GF(Integer(3)**Integer(10), prefix='w'); L = GF(Integer(3)**Integer(10));␣
↪K is L
False
>>> K.variable_name(), L.variable_name()
('w10', 'z10')
>>> list(K.polynomial()) == list(L.polynomial())
True
```

create_key_and_extra_args(*order*, *name=None*, *modulus=None*, *names=None*, *impl=None*, *proof=None*, *check_prime=True*, *check_irreducible=True*, *prefix=None*, *repr=None*, *elem_cache=None*, ***kwds*)

EXAMPLES:

```
sage: GF.create_key_and_extra_args(9, 'a')                                    #␣
↪needs sage.libs.linbox
((9, ('a',), x^2 + 2*x + 2, 'givaro', 3, 2, True, None, 'poly', True, True,␣
↪True), {})
```

```
>>> from sage.all import *
>>> GF.create_key_and_extra_args(Integer(9), 'a')                             ␣
↪     # needs sage.libs.linbox
((9, ('a',), x^2 + 2*x + 2, 'givaro', 3, 2, True, None, 'poly', True, True,␣
↪True), {})
```

The order q can also be given as a pair (p, n):

```
sage: GF.create_key_and_extra_args((3, 2), 'a')                               #␣
↪needs sage.libs.linbox
((9, ('a',), x^2 + 2*x + 2, 'givaro', 3, 2, True, None, 'poly', True, True,␣
↪True), {})
```

```
>>> from sage.all import *
>>> GF.create_key_and_extra_args((Integer(3), Integer(2)), 'a')               ␣
↪               # needs sage.libs.linbox
((9, ('a',), x^2 + 2*x + 2, 'givaro', 3, 2, True, None, 'poly', True, True,␣
↪True), {})
```

We do not take invalid keyword arguments and raise a value error to better ensure uniqueness:

```
sage: GF.create_key_and_extra_args(9, 'a', foo='value')
Traceback (most recent call last):
...
```

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```
TypeError: ...create_key_and_extra_args() got an unexpected keyword argument
↪'foo'
```

```
>>> from sage.all import *
>>> GF.create_key_and_extra_args(Integer(9), 'a', foo='value')
Traceback (most recent call last):
...
TypeError: ...create_key_and_extra_args() got an unexpected keyword argument
↪'foo'
```

Moreover, `repr` and `elem_cache` are ignored when not using givaro:

```
sage: GF.create_key_and_extra_args(16, 'a', impl='ntl', repr='poly')        #␣
↪needs sage.libs.ntl
((16, ('a',), x^4 + x + 1, 'ntl', 2, 4, True, None, None, None, True, True),
↪{})
sage: GF.create_key_and_extra_args(16, 'a', impl='ntl', elem_cache=False)   #␣
↪needs sage.libs.ntl
((16, ('a',), x^4 + x + 1, 'ntl', 2, 4, True, None, None, None, True, True),
↪{})
sage: GF(16, impl='ntl') is GF(16, impl='ntl', repr='foo')                  #␣
↪needs sage.libs.ntl
True
```

```
>>> from sage.all import *
>>> GF.create_key_and_extra_args(Integer(16), 'a', impl='ntl', repr='poly')  ␣
↪     # needs sage.libs.ntl
((16, ('a',), x^4 + x + 1, 'ntl', 2, 4, True, None, None, None, True, True),
↪{})
>>> GF.create_key_and_extra_args(Integer(16), 'a', impl='ntl', elem_
↪cache=False)   # needs sage.libs.ntl
((16, ('a',), x^4 + x + 1, 'ntl', 2, 4, True, None, None, None, True, True),
↪{})
>>> GF(Integer(16), impl='ntl') is GF(Integer(16), impl='ntl', repr='foo')   ␣
↪              # needs sage.libs.ntl
True
```

We handle extra arguments for the givaro finite field and create unique objects for their defaults:

```
sage: GF(25, impl='givaro') is GF(25, impl='givaro', repr='poly')           #␣
↪needs sage.libs.linbox
True
sage: GF(25, impl='givaro') is GF(25, impl='givaro', elem_cache=True)       #␣
↪needs sage.libs.linbox
True
sage: GF(625, impl='givaro') is GF(625, impl='givaro', elem_cache=False)    #␣
↪needs sage.libs.linbox
True
```

```
>>> from sage.all import *
>>> GF(Integer(25), impl='givaro') is GF(Integer(25), impl='givaro', repr=
↪'poly')           # needs sage.libs.linbox
```

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```
True
>>> GF(Integer(25), impl='givaro') is GF(Integer(25), impl='givaro', elem_
↪cache=True)        # needs sage.libs.linbox
True
>>> GF(Integer(625), impl='givaro') is GF(Integer(625), impl='givaro', elem_
↪cache=False)     # needs sage.libs.linbox
True
```

We explicitly take `structure`, `implementation` and `prec` attributes for compatibility with `AlgebraicExtensionFunctor` but we ignore them as they are not used, see Issue #21433:

```
sage: GF.create_key_and_extra_args(9, 'a', structure=None)                    #␣
↪needs sage.libs.linbox
((9, ('a',), x^2 + 2*x + 2, 'givaro', 3, 2, True, None, 'poly', True, True,␣
↪True), {})
```

```
>>> from sage.all import *
>>> GF.create_key_and_extra_args(Integer(9), 'a', structure=None)             ␣
↪     # needs sage.libs.linbox
((9, ('a',), x^2 + 2*x + 2, 'givaro', 3, 2, True, None, 'poly', True, True,␣
↪True), {})
```

create_object(*version*, *key*, ***kwds*)

EXAMPLES:

```
sage: K = GF(19) # indirect doctest
sage: TestSuite(K).run()
```

```
>>> from sage.all import *
>>> K = GF(Integer(19)) # indirect doctest
>>> TestSuite(K).run()
```

We try to create finite fields with various implementations:

```
sage: k = GF(2, impl='modn')
sage: k = GF(2, impl='givaro')                                                #␣
↪needs sage.libs.linbox
sage: k = GF(2, impl='ntl')                                                   #␣
↪needs sage.libs.ntl
sage: k = GF(2, impl='pari')
Traceback (most recent call last):
...
ValueError: the degree must be at least 2
sage: k = GF(2, impl='supercalifragilisticexpialidocious')
Traceback (most recent call last):
...
ValueError: no such finite field implementation:
↪'supercalifragilisticexpialidocious'
sage: k.<a> = GF(2^15, impl='modn')
Traceback (most recent call last):
...
ValueError: the 'modn' implementation requires a prime order
```

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```
sage: k.<a> = GF(2^15, impl='givaro')                                    #␣
↪needs sage.libs.linbox
sage: k.<a> = GF(2^15, impl='ntl')                                       #␣
↪needs sage.libs.ntl
sage: k.<a> = GF(2^15, impl='pari')
sage: k.<a> = GF(3^60, impl='modn')
Traceback (most recent call last):
...
ValueError: the 'modn' implementation requires a prime order
sage: k.<a> = GF(3^60, impl='givaro')                                    #␣
↪needs sage.libs.linbox
Traceback (most recent call last):
...
ValueError: q must be < 2^16
sage: k.<a> = GF(3^60, impl='ntl')                                       #␣
↪needs sage.libs.ntl
Traceback (most recent call last):
...
ValueError: q must be a 2-power
sage: k.<a> = GF(3^60, impl='pari')
```

```
>>> from sage.all import *
>>> k = GF(Integer(2), impl='modn')
>>> k = GF(Integer(2), impl='givaro')                                    ␣
↪     # needs sage.libs.linbox
>>> k = GF(Integer(2), impl='ntl')                                       ␣
↪     # needs sage.libs.ntl
>>> k = GF(Integer(2), impl='pari')
Traceback (most recent call last):
...
ValueError: the degree must be at least 2
>>> k = GF(Integer(2), impl='supercalifragilisticexpialidocious')
Traceback (most recent call last):
...
ValueError: no such finite field implementation:
↪'supercalifragilisticexpialidocious'
>>> k = GF(Integer(2)**Integer(15), impl='modn', names=('a',)); (a,) = k._
↪first_ngens(1)
Traceback (most recent call last):
...
ValueError: the 'modn' implementation requires a prime order
>>> k = GF(Integer(2)**Integer(15), impl='givaro', names=('a',)); (a,) = k._
↪first_ngens(1)# needs sage.libs.linbox
>>> k = GF(Integer(2)**Integer(15), impl='ntl', names=('a',)); (a,) = k._
↪first_ngens(1)# needs sage.libs.ntl
>>> k = GF(Integer(2)**Integer(15), impl='pari', names=('a',)); (a,) = k._
↪first_ngens(1)
>>> k = GF(Integer(3)**Integer(60), impl='modn', names=('a',)); (a,) = k._
↪first_ngens(1)
Traceback (most recent call last):
...
ValueError: the 'modn' implementation requires a prime order
```

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```
>>> k = GF(Integer(3)**Integer(60), impl='givaro', names=('a',)); (a,) = k._
↪first_ngens(1)# needs sage.libs.linbox
Traceback (most recent call last):
...
ValueError: q must be < 2^16
>>> k = GF(Integer(3)**Integer(60), impl='ntl', names=('a',)); (a,) = k._
↪first_ngens(1)# needs sage.libs.ntl
Traceback (most recent call last):
...
ValueError: q must be a 2-power
>>> k = GF(Integer(3)**Integer(60), impl='pari', names=('a',)); (a,) = k._
↪first_ngens(1)
```

`sage.rings.finite_rings.finite_field_constructor.`**`is_PrimeFiniteField`**(*x*)

Return `True` if `x` is a prime finite field.

This function is deprecated.

EXAMPLES:

```
sage: from sage.rings.finite_rings.finite_field_constructor import is_
↪PrimeFiniteField
sage: is_PrimeFiniteField(QQ)
doctest:...: DeprecationWarning: the function is_PrimeFiniteField is deprecated;␣
↪use isinstance(x, sage.rings.finite_rings.finite_field_base.FiniteField) and x.
↪is_prime_field() instead
See https://github.com/sagemath/sage/issues/32664 for details.
False
sage: is_PrimeFiniteField(GF(7))
True
sage: is_PrimeFiniteField(GF(7^2, 'a'))
False
sage: is_PrimeFiniteField(GF(next_prime(10^90, proof=False)))
True
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.finite_field_constructor import is_
↪PrimeFiniteField
>>> is_PrimeFiniteField(QQ)
doctest:...: DeprecationWarning: the function is_PrimeFiniteField is deprecated;␣
↪use isinstance(x, sage.rings.finite_rings.finite_field_base.FiniteField) and x.
↪is_prime_field() instead
See https://github.com/sagemath/sage/issues/32664 for details.
False
>>> is_PrimeFiniteField(GF(Integer(7)))
True
>>> is_PrimeFiniteField(GF(Integer(7)**Integer(2), 'a'))
False
>>> is_PrimeFiniteField(GF(next_prime(Integer(10)**Integer(90), proof=False)))
True
```

2.2 Base class for finite fields

AUTHORS:

- Adrien Brochard, David Roe, Jeroen Demeyer, Julian Rueth, Niles Johnson, Peter Bruin, Travis Scrimshaw, Xavier Caruso: initial version

class sage.rings.finite_rings.finite_field_base.**FiniteField**

Bases: Field

Abstract base class for finite fields.

algebraic_closure(*name='z'*, ***kwds*)

Return an algebraic closure of `self`.

INPUT:

- `name` – string (default: `'z'`); prefix to use for variable names of subfields
- `implementation` – string (optional); specifies how to construct the algebraic closure. The only value supported at the moment is `'pseudo_conway'`. For more details, see *algebraic_closure_finite_field*.

OUTPUT:

An algebraic closure of `self`. Note that mathematically speaking, this is only unique up to *non-unique* isomorphism. To obtain canonically defined algebraic closures, one needs an algorithm that also provides a canonical isomorphism between any two algebraic closures constructed using the algorithm.

This non-uniqueness problem can in principle be solved by using *Conway polynomials*; see for example Wikipedia article Conway_polynomial_(finite_fields). These have the drawback that computing them takes a long time. Therefore Sage implements a variant called *pseudo-Conway polynomials*, which are easier to compute but do not determine an algebraic closure up to unique isomorphism.

The output of this method is cached, so that within the same Sage session, calling it multiple times will return the same algebraic closure (i.e. the same Sage object). Despite this, the non-uniqueness of the current implementation means that coercion and pickling cannot work as one might expect. See below for an example.

EXAMPLES:

```
sage: F = GF(5).algebraic_closure()
sage: F
Algebraic closure of Finite Field of size 5
sage: F.gen(3)
z3
```

```
>>> from sage.all import *
>>> F = GF(Integer(5)).algebraic_closure()
>>> F
Algebraic closure of Finite Field of size 5
>>> F.gen(Integer(3))
z3
```

For finite fields, the algebraic closure is always (isomorphic to) the algebraic closure of the prime field:

```
sage: GF(5^2).algebraic_closure() == F
True
```

```
>>> from sage.all import *
>>> GF(Integer(5)**Integer(2)).algebraic_closure() == F
True
```

The default name is 'z' but you can change it through the option `name`:

```
sage: Ft = GF(5).algebraic_closure('t')
sage: Ft.gen(3)
t3
```

```
>>> from sage.all import *
>>> Ft = GF(Integer(5)).algebraic_closure('t')
>>> Ft.gen(Integer(3))
t3
```

Because Sage currently only implements algebraic closures using a non-unique definition (see above), it is currently impossible to implement pickling in such a way that a pickled and unpickled element compares equal to the original:

```
sage: F = GF(7).algebraic_closure()
sage: x = F.gen(2)
sage: loads(dumps(x)) == x
False
```

```
>>> from sage.all import *
>>> F = GF(Integer(7)).algebraic_closure()
>>> x = F.gen(Integer(2))
>>> loads(dumps(x)) == x
False
```

Note

For non-prime finite fields, this method currently simply returns the algebraic closure of the prime field. This may or may not change in the future when extension towers are supported properly.

an_embedding(K)

Return some embedding of this field into another field K, and raise a `ValueError` if none exists.

See also

`sage.rings.ring.Field.an_embedding()`

EXAMPLES:

```
sage: GF(4,'a').an_embedding(GF(2).algebraic_closure())
Ring morphism:
  From: Finite Field in a of size 2^2
  To:   Algebraic closure of Finite Field of size 2
  Defn: a |--> ...
```

```
>>> from sage.all import *
>>> GF(Integer(4),'a').an_embedding(GF(Integer(2)).algebraic_closure())
Ring morphism:
  From: Finite Field in a of size 2^2
  To:   Algebraic closure of Finite Field of size 2
  Defn: a |--> ...
```

cardinality()

Return the cardinality of `self`.

Same as *order()*.

EXAMPLES:

```
sage: GF(997).cardinality()
997
```

```
>>> from sage.all import *
>>> GF(Integer(997)).cardinality()
997
```

construction()

Return the construction of this finite field, as a `ConstructionFunctor` and the base field.

EXAMPLES:

```
sage: v = GF(3^3).construction(); v
(AlgebraicExtensionFunctor, Finite Field of size 3)
sage: v[0].polys[0]
3
sage: v = GF(2^1000, 'a').construction(); v[0].polys[0]
a^1000 + a^5 + a^4 + a^3 + 1
```

```
>>> from sage.all import *
>>> v = GF(Integer(3)**Integer(3)).construction(); v
(AlgebraicExtensionFunctor, Finite Field of size 3)
>>> v[Integer(0)].polys[Integer(0)]
3
>>> v = GF(Integer(2)**Integer(1000), 'a').construction(); v[Integer(0)].
↪polys[Integer(0)]
a^1000 + a^5 + a^4 + a^3 + 1
```

The implementation is taken into account, by Issue #15223:

```
sage: k = FiniteField(9, 'a', impl='pari_ffelt')
sage: F, R = k.construction()
sage: F(R) is k
True
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(9), 'a', impl='pari_ffelt')
>>> F, R = k.construction()
>>> F(R) is k
True
```

dual_basis(*basis=None*, *check=True*)

Return the dual basis of `basis`, or the dual basis of the power basis if no basis is supplied.

If $e = \{e_0, e_1, ..., e_{n-1}\}$ is a basis of $\mathbf{F}_{p^n}$ as a vector space over $\mathbf{F}_p$, then the dual basis of e, $d = \{d_0, d_1, ..., d_{n-1}\}$, is the unique basis such that $\mathrm{Tr}(e_i d_j) = \delta_{i,j}, 0 \leq i, j \leq n-1$, where Tr is the trace function.

INPUT:

- `basis` – (default: `None`); a basis of the finite field `self`, $\mathbf{F}_{p^n}$, as a vector space over the base field $\mathbf{F}_p$. Uses the power basis $\{x^i : 0 \leq i \leq n-1\}$ as input if no basis is supplied, where x is the generator of `self`.
- `check` – (default: `True`); verifies that `basis` is a valid basis of `self`

ALGORITHM:

The algorithm used to calculate the dual basis comes from pages 110–111 of [McE1987].

Let $e = \{e_0, e_1, ..., e_{n-1}\}$ be a basis of $\mathbf{F}_{p^n}$ as a vector space over $\mathbf{F}_p$ and $d = \{d_0, d_1, ..., d_{n-1}\}$ be the dual basis of e. Since e is a basis, we can rewrite any $d_c, 0 \leq c \leq n-1$, as $d_c = \beta_0 e_0 + \beta_1 e_1 + ... + \beta_{n-1} e_{n-1}$, for some $\beta_0, \beta_1, ..., \beta_{n-1} \in \mathbf{F}_p$. Using properties of the trace function, we can rewrite the n equations of the form $\mathrm{Tr}(e_i d_c) = \delta_{i,c}$ and express the result as the matrix vector product: $A[\beta_0, \beta_1, ..., \beta_{n-1}] = i_c$, where the i, j-th element of A is $\mathrm{Tr}(\mathrm{e_i e_j})$ and i_c is the i-th column of the $n \times n$ identity matrix. Since A is an invertible matrix, $[\beta_0, \beta_1, ..., \beta_{n-1}] = A^{-1} i_c$, from which we can easily calculate d_c.

EXAMPLES:

```
sage: F.<a> = GF(2^4)
sage: F.dual_basis(basis=None, check=False)                                #␣
↪needs sage.modules
[a^3 + 1, a^2, a, 1]
```

```
>>> from sage.all import *
>>> F = GF(Integer(2)**Integer(4), names=('a',)); (a,) = F._first_ngens(1)
>>> F.dual_basis(basis=None, check=False)                                  #␣
↪needs sage.modules
[a^3 + 1, a^2, a, 1]
```

We can test that the dual basis returned satisfies the defining property of a dual basis: $\mathrm{Tr}(e_i d_j) = \delta_{i,j}, 0 \leq i, j \leq n-1$

```
sage: # needs sage.modules
sage: F.<a> = GF(7^4)
sage: e = [4*a^3, 2*a^3 + a^2 + 3*a + 5,
....:      3*a^3 + 5*a^2 + 4*a + 2, 2*a^3 + 2*a^2 + 2]
sage: d = F.dual_basis(e, check=True); d
[3*a^3 + 4*a^2 + 6*a + 2, a^3 + 6*a + 5,
3*a^3 + 6*a^2 + 2*a + 5, 5*a^2 + 4*a + 3]
sage: vals = [[(x * y).trace() for x in e] for y in d]
sage: matrix(vals) == matrix.identity(4)
True
```

```
>>> from sage.all import *
>>> # needs sage.modules
>>> F = GF(Integer(7)**Integer(4), names=('a',)); (a,) = F._first_ngens(1)
>>> e = [Integer(4)*a**Integer(3), Integer(2)*a**Integer(3) + a**Integer(2) +␣
```

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```
↪Integer(3)*a + Integer(5),
...       Integer(3)*a**Integer(3) + Integer(5)*a**Integer(2) + Integer(4)*a +␣
↪Integer(2), Integer(2)*a**Integer(3) + Integer(2)*a**Integer(2) +␣
↪Integer(2)]
>>> d = F.dual_basis(e, check=True); d
[3*a^3 + 4*a^2 + 6*a + 2, a^3 + 6*a + 5,
3*a^3 + 6*a^2 + 2*a + 5, 5*a^2 + 4*a + 3]
>>> vals = [[(x * y).trace() for x in e] for y in d]
>>> matrix(vals) == matrix.identity(Integer(4))
True
```

We can test that if d is the dual basis of e, then e is the dual basis of d:

```
sage: # needs sage.modules
sage: F.<a> = GF(7^8)
sage: e = [a^0, a^1, a^2, a^3, a^4, a^5, a^6, a^7]
sage: d = F.dual_basis(e, check=False); d
[6*a^6 + 4*a^5 + 4*a^4 + a^3 + 6*a^2 + 3,
 6*a^7 + 4*a^6 + 4*a^5 + 2*a^4 + a^2,
 4*a^6 + 5*a^5 + 5*a^4 + 4*a^3 + 5*a^2 + a + 6,
 5*a^7 + a^6 + a^4 + 4*a^3 + 4*a^2 + 1,
 2*a^7 + 5*a^6 + a^5 + a^3 + 5*a^2 + 2*a + 4,
 a^7 + 2*a^6 + 5*a^5 + a^4 + 5*a^2 + 4*a + 4,
 a^7 + a^6 + 2*a^5 + 5*a^4 + a^3 + 4*a^2 + 4*a + 6,
 5*a^7 + a^6 + a^5 + 2*a^4 + 5*a^3 + 6*a]
sage: F.dual_basis(d)
[1, a, a^2, a^3, a^4, a^5, a^6, a^7]
```

```
>>> from sage.all import *
>>> # needs sage.modules
>>> F = GF(Integer(7)**Integer(8), names=('a',)); (a,) = F._first_ngens(1)
>>> e = [a**Integer(0), a**Integer(1), a**Integer(2), a**Integer(3),␣
↪a**Integer(4), a**Integer(5), a**Integer(6), a**Integer(7)]
>>> d = F.dual_basis(e, check=False); d
[6*a^6 + 4*a^5 + 4*a^4 + a^3 + 6*a^2 + 3,
 6*a^7 + 4*a^6 + 4*a^5 + 2*a^4 + a^2,
 4*a^6 + 5*a^5 + 5*a^4 + 4*a^3 + 5*a^2 + a + 6,
 5*a^7 + a^6 + a^4 + 4*a^3 + 4*a^2 + 1,
 2*a^7 + 5*a^6 + a^5 + a^3 + 5*a^2 + 2*a + 4,
 a^7 + 2*a^6 + 5*a^5 + a^4 + 5*a^2 + 4*a + 4,
 a^7 + a^6 + 2*a^5 + 5*a^4 + a^3 + 4*a^2 + 4*a + 6,
 5*a^7 + a^6 + a^5 + 2*a^4 + 5*a^3 + 6*a]
>>> F.dual_basis(d)
[1, a, a^2, a^3, a^4, a^5, a^6, a^7]
```

We cannot calculate the dual basis if `basis` is not a valid basis.

```
sage: F.<a> = GF(2^3)
sage: F.dual_basis([a], check=True)                                    #␣
↪needs sage.modules
Traceback (most recent call last):
...
```

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```
ValueError: basis length should be 3, not 1

sage: F.dual_basis([a^0, a, a^0 + a], check=True)                          #␣
↪needs sage.modules
Traceback (most recent call last):
...
ValueError: value of 'basis' keyword is not a basis
```

```
>>> from sage.all import *
>>> F = GF(Integer(2)**Integer(3), names=('a',)); (a,) = F._first_ngens(1)
>>> F.dual_basis([a], check=True)                                          #␣
↪needs sage.modules
Traceback (most recent call last):
...
ValueError: basis length should be 3, not 1

>>> F.dual_basis([a**Integer(0), a, a**Integer(0) + a], check=True)        ␣
↪                 # needs sage.modules
Traceback (most recent call last):
...
ValueError: value of 'basis' keyword is not a basis
```

AUTHOR:

- Thomas Gagne (2015-06-16)

embeddings(K)

Return a list of all embeddings of this field in another field K.

EXAMPLES:

```
sage: GF(2).embeddings(GF(4))
[Ring morphism:
   From: Finite Field of size 2
   To:   Finite Field in z2 of size 2^2
   Defn: 1 |--> 1]
sage: GF(4).embeddings(GF(2).algebraic_closure())
[Ring morphism:
   From: Finite Field in z2 of size 2^2
   To:   Algebraic closure of Finite Field of size 2
   Defn: z2 |--> z2,
 Ring morphism:
   From: Finite Field in z2 of size 2^2
   To:   Algebraic closure of Finite Field of size 2
   Defn: z2 |--> z2 + 1]
```

```
>>> from sage.all import *
>>> GF(Integer(2)).embeddings(GF(Integer(4)))
[Ring morphism:
   From: Finite Field of size 2
   To:   Finite Field in z2 of size 2^2
   Defn: 1 |--> 1]
>>> GF(Integer(4)).embeddings(GF(Integer(2)).algebraic_closure())
```

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```
[Ring morphism:
   From: Finite Field in z2 of size 2^2
   To:   Algebraic closure of Finite Field of size 2
   Defn: z2 |--> z2,
 Ring morphism:
   From: Finite Field in z2 of size 2^2
   To:   Algebraic closure of Finite Field of size 2
   Defn: z2 |--> z2 + 1]
```

extension(*modulus*, *name=None*, *names=None*, *map=False*, *embedding=None*, *latex_name=None*, *latex_names=None*, ***kwds*)

Return an extension of this finite field.

INPUT:

- `modulus` – a polynomial with coefficients in `self`, or an integer
- `name` or `names` – string; the name of the generator in the new extension
- `latex_name` or `latex_names` – string; latex name of the generator in the new extension
- `map` – boolean (default: `False`); if `False`, return just the extension E. If `True`, return a pair (E, f), where f is an embedding of `self` into E.
- `embedding` – currently not used; for compatibility with other `AlgebraicExtensionFunctor` calls
- `**kwds` – further keywords, passed to the finite field constructor.

OUTPUT:

An extension of the given modulus, or pseudo-Conway of the given degree if `modulus` is an integer.

EXAMPLES:

```
sage: k = GF(2)
sage: R.<x> = k[]
sage: k.extension(x^1000 + x^5 + x^4 + x^3 + 1, 'a')
Finite Field in a of size 2^1000
sage: k = GF(3^4)
sage: R.<x> = k[]
sage: k.extension(3)
Finite Field in z12 of size 3^12
sage: K = k.extension(2, 'a')
sage: k.is_subring(K)
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(2))
>>> R = k['x']; (x,) = R._first_ngens(1)
>>> k.extension(x**Integer(1000) + x**Integer(5) + x**Integer(4) +␣
↪x**Integer(3) + Integer(1), 'a')
Finite Field in a of size 2^1000
>>> k = GF(Integer(3)**Integer(4))
>>> R = k['x']; (x,) = R._first_ngens(1)
>>> k.extension(Integer(3))
Finite Field in z12 of size 3^12
>>> K = k.extension(Integer(2), 'a')
```

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```
>>> k.is_subring(K)
True
```

An example using the `map` argument:

```
sage: F = GF(5)
sage: E, f = F.extension(2, 'b', map=True)
sage: E
Finite Field in b of size 5^2
sage: f
Ring morphism:
  From: Finite Field of size 5
  To:   Finite Field in b of size 5^2
  Defn: 1 |--> 1
sage: f.parent()
Set of field embeddings
 from Finite Field of size 5
   to Finite Field in b of size 5^2
```

```
>>> from sage.all import *
>>> F = GF(Integer(5))
>>> E, f = F.extension(Integer(2), 'b', map=True)
>>> E
Finite Field in b of size 5^2
>>> f
Ring morphism:
  From: Finite Field of size 5
  To:   Finite Field in b of size 5^2
  Defn: 1 |--> 1
>>> f.parent()
Set of field embeddings
 from Finite Field of size 5
   to Finite Field in b of size 5^2
```

Extensions of non-prime finite fields by polynomials are not yet supported: we fall back to generic code:

```
sage: k.extension(x^5 + x^2 + x - 1)
Univariate Quotient Polynomial Ring in x over Finite Field in z4 of size 3^4
 with modulus x^5 + x^2 + x + 2
```

```
>>> from sage.all import *
>>> k.extension(x**Integer(5) + x**Integer(2) + x - Integer(1))
Univariate Quotient Polynomial Ring in x over Finite Field in z4 of size 3^4
 with modulus x^5 + x^2 + x + 2
```

factored_order()

Return the factored order of this field. For compatibility with *integer_mod_ring*.

EXAMPLES:

```
sage: GF(7^2,'a').factored_order()
7^2
```

```
>>> from sage.all import *
>>> GF(Integer(7)**Integer(2),'a').factored_order()
7^2
```

factored_unit_order()

Return the factorization of `self.order()-1`, as a 1-tuple.

The format is for compatibility with *integer_mod_ring*.

EXAMPLES:

```
sage: GF(7^2,'a').factored_unit_order()
(2^4 * 3,)
```

```
>>> from sage.all import *
>>> GF(Integer(7)**Integer(2),'a').factored_unit_order()
(2^4 * 3,)
```

free_module(*base=None*, *basis=None*, *map=True*)

Return the vector space over the subfield isomorphic to this finite field as a vector space, along with the isomorphisms.

INPUT:

- `base` – a subfield of or a morphism into this finite field. If not given, the prime subfield is assumed. A subfield means a finite field with coercion to this finite field.
- `basis` – a basis of the finite field as a vector space over the subfield. If not given, one is chosen automatically.
- `map` – boolean (default: `True`); if `True`, isomorphisms from and to the vector space are also returned

The `basis` maps to the standard basis of the vector space by the isomorphisms.

OUTPUT: if `map` is `False`,

- vector space over the subfield or the domain of the morphism, isomorphic to this finite field.

and if `map` is `True`, then also

- an isomorphism from the vector space to the finite field.
- the inverse isomorphism to the vector space from the finite field.

EXAMPLES:

```
sage: GF(27,'a').vector_space(map=False)                                    #␣
↪needs sage.modules
Vector space of dimension 3 over Finite Field of size 3

sage: # needs sage.modules
sage: F = GF(8)
sage: E = GF(64)
sage: V, from_V, to_V = E.vector_space(F, map=True)
sage: V
Vector space of dimension 2 over Finite Field in z3 of size 2^3
sage: to_V(E.gen())
(0, 1)
sage: all(from_V(to_V(e)) == e for e in E)
```

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```
True
sage: all(to_V(e1 + e2) == to_V(e1) + to_V(e2) for e1 in E for e2 in E)
True
sage: all(to_V(c * e) == c * to_V(e) for e in E for c in F)
True

sage: # needs sage.modules
sage: basis = [E.gen(), E.gen() + 1]
sage: W, from_W, to_W = E.vector_space(F, basis, map=True)
sage: all(from_W(to_W(e)) == e for e in E)
True
sage: all(to_W(c * e) == c * to_W(e) for e in E for c in F)
True
sage: all(to_W(e1 + e2) == to_W(e1) + to_W(e2) for e1 in E for e2 in E)  #␣
↪long time
True
sage: to_W(basis[0]); to_W(basis[1])
(1, 0)
(0, 1)

sage: # needs sage.modules
sage: x = polygen(ZZ)
sage: F = GF(9, 't', modulus=x^2 + x - 1)
sage: E = GF(81)
sage: h = Hom(F,E).an_element()
sage: V, from_V, to_V = E.vector_space(h, map=True)
sage: V
Vector space of dimension 2 over Finite Field in t of size 3^2
sage: V.base_ring() is F
True
sage: all(from_V(to_V(e)) == e for e in E)
True
sage: all(to_V(e1 + e2) == to_V(e1) + to_V(e2) for e1 in E for e2 in E)
True
sage: all(to_V(h(c) * e) == c * to_V(e) for e in E for c in F)
True
```

```
>>> from sage.all import *
>>> GF(Integer(27),'a').vector_space(map=False)                          ␣
↪     # needs sage.modules
Vector space of dimension 3 over Finite Field of size 3

>>> # needs sage.modules
>>> F = GF(Integer(8))
>>> E = GF(Integer(64))
>>> V, from_V, to_V = E.vector_space(F, map=True)
>>> V
Vector space of dimension 2 over Finite Field in z3 of size 2^3
>>> to_V(E.gen())
(0, 1)
>>> all(from_V(to_V(e)) == e for e in E)
True
```

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```
>>> all(to_V(e1 + e2) == to_V(e1) + to_V(e2) for e1 in E for e2 in E)
True
>>> all(to_V(c * e) == c * to_V(e) for e in E for c in F)
True

>>> # needs sage.modules
>>> basis = [E.gen(), E.gen() + Integer(1)]
>>> W, from_W, to_W = E.vector_space(F, basis, map=True)
>>> all(from_W(to_W(e)) == e for e in E)
True
>>> all(to_W(c * e) == c * to_W(e) for e in E for c in F)
True
>>> all(to_W(e1 + e2) == to_W(e1) + to_W(e2) for e1 in E for e2 in E)  # long␣
↪time
True
>>> to_W(basis[Integer(0)]); to_W(basis[Integer(1)])
(1, 0)
(0, 1)

>>> # needs sage.modules
>>> x = polygen(ZZ)
>>> F = GF(Integer(9), 't', modulus=x**Integer(2) + x - Integer(1))
>>> E = GF(Integer(81))
>>> h = Hom(F,E).an_element()
>>> V, from_V, to_V = E.vector_space(h, map=True)
>>> V
Vector space of dimension 2 over Finite Field in t of size 3^2
>>> V.base_ring() is F
True
>>> all(from_V(to_V(e)) == e for e in E)
True
>>> all(to_V(e1 + e2) == to_V(e1) + to_V(e2) for e1 in E for e2 in E)
True
>>> all(to_V(h(c) * e) == c * to_V(e) for e in E for c in F)
True
```

frobenius_endomorphism(*n=1*)

INPUT:

- n – integer (default: 1)

OUTPUT:

The n-th power of the absolute arithmetic Frobenius endomorphism on this finite field.

EXAMPLES:

```
sage: k.<t> = GF(3^5)
sage: Frob = k.frobenius_endomorphism(); Frob
Frobenius endomorphism t |--> t^3 on Finite Field in t of size 3^5

sage: a = k.random_element()                                          #␣
↪needs sage.modules
sage: Frob(a) == a^3                                                  #␣
```

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```
↪needs sage.modules
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(5), names=('t',)); (t,) = k._first_ngens(1)
>>> Frob = k.frobenius_endomorphism(); Frob
Frobenius endomorphism t |--> t^3 on Finite Field in t of size 3^5

>>> a = k.random_element()                                                  #␣
↪needs sage.modules
>>> Frob(a) == a**Integer(3)                                                ␣
↪       # needs sage.modules
True
```

We can specify a power:

```
sage: k.frobenius_endomorphism(2)
Frobenius endomorphism t |--> t^(3^2) on Finite Field in t of size 3^5
```

```
>>> from sage.all import *
>>> k.frobenius_endomorphism(Integer(2))
Frobenius endomorphism t |--> t^(3^2) on Finite Field in t of size 3^5
```

The result is simplified if possible:

```
sage: k.frobenius_endomorphism(6)
Frobenius endomorphism t |--> t^3 on Finite Field in t of size 3^5
sage: k.frobenius_endomorphism(5)
Identity endomorphism of Finite Field in t of size 3^5
```

```
>>> from sage.all import *
>>> k.frobenius_endomorphism(Integer(6))
Frobenius endomorphism t |--> t^3 on Finite Field in t of size 3^5
>>> k.frobenius_endomorphism(Integer(5))
Identity endomorphism of Finite Field in t of size 3^5
```

Comparisons work:

```
sage: k.frobenius_endomorphism(6) == Frob
True
sage: from sage.categories.morphism import IdentityMorphism
sage: k.frobenius_endomorphism(5) == IdentityMorphism(k)
True
```

```
>>> from sage.all import *
>>> k.frobenius_endomorphism(Integer(6)) == Frob
True
>>> from sage.categories.morphism import IdentityMorphism
>>> k.frobenius_endomorphism(Integer(5)) == IdentityMorphism(k)
True
```

AUTHOR:

- Xavier Caruso (2012-06-29)

from_bytes(*input_bytes*, *byteorder='big'*)

Return the integer represented by the given array of bytes.

Internally relies on the python `int.from_bytes()` method.

INPUT:

- `input_bytes` – a bytes-like object or iterable producing bytes
- `byteorder` – string (default: `'big'`); determines the byte order of `input_bytes` (can only be `'big'` or `'little'`)

EXAMPLES:

```
sage: input_bytes = b"some_bytes"
sage: F = GF(2**127 - 1)
sage: F.from_bytes(input_bytes)
5451276169337902908307 07
sage: a = F.from_bytes(input_bytes, byteorder='little'); a
5449436595289963090041 47
sage: type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>
```

```
>>> from sage.all import *
>>> input_bytes = b"some_bytes"
>>> F = GF(Integer(2)**Integer(127) - Integer(1))
>>> F.from_bytes(input_bytes)
5451276169337902908307 07
>>> a = F.from_bytes(input_bytes, byteorder='little'); a
5449436595289963090041 47
>>> type(a)
<class 'sage.rings.finite_rings.integer_mod.IntegerMod_gmp'>
```

```
sage: input_bytes = b"some_bytes"
sage: F_ext = GF(65537**5)
sage: F_ext.from_bytes(input_bytes)
29549*z5^4 + 40876*z5^3 + 52171*z5^2 + 13604*z5 + 20843
sage: F_ext.from_bytes(input_bytes, byteorder='little')
29539*z5^4 + 42728*z5^3 + 47440*z5^2 + 12423*z5 + 27473
```

```
>>> from sage.all import *
>>> input_bytes = b"some_bytes"
>>> F_ext = GF(Integer(65537)**Integer(5))
>>> F_ext.from_bytes(input_bytes)
29549*z5^4 + 40876*z5^3 + 52171*z5^2 + 13604*z5 + 20843
>>> F_ext.from_bytes(input_bytes, byteorder='little')
29539*z5^4 + 42728*z5^3 + 47440*z5^2 + 12423*z5 + 27473
```

from_integer(*n*, *reverse=False*)

Return the finite field element obtained by reinterpreting the base-p expansion of n as a polynomial and evaluating it at the generator of this finite field.

If `reverse` is set to `True` (default: `False`), the list of digits is reversed prior to evaluation.

Inverse of *`sage.rings.finite_rings.element_base.FinitePolyExtElement.to_integer()`*.

INPUT:

- n – integer between 0 and the cardinality of this field minus 1

EXAMPLES:

```
sage: p = 4091
sage: F = GF(p^4, 'a')
sage: n = 100*p^3 + 37*p^2 + 12*p + 6
sage: F.from_integer(n)
100*a^3 + 37*a^2 + 12*a + 6
sage: F.from_integer(n) in F
True
sage: F.from_integer(n, reverse=True)
6*a^3 + 12*a^2 + 37*a + 100
```

```
>>> from sage.all import *
>>> p = Integer(4091)
>>> F = GF(p**Integer(4), 'a')
>>> n = Integer(100)*p**Integer(3) + Integer(37)*p**Integer(2) +␣
↪Integer(12)*p + Integer(6)
>>> F.from_integer(n)
100*a^3 + 37*a^2 + 12*a + 6
>>> F.from_integer(n) in F
True
>>> F.from_integer(n, reverse=True)
6*a^3 + 12*a^2 + 37*a + 100
```

galois_group()

Return the Galois group of this finite field, a cyclic group generated by Frobenius.

EXAMPLES:

```
sage: # needs sage.groups
sage: G = GF(3^6).galois_group(); G
Galois group C6 of GF(3^6)
sage: F = G.gen()
sage: F^2
Frob^2
sage: F^6
1
```

```
>>> from sage.all import *
>>> # needs sage.groups
>>> G = GF(Integer(3)**Integer(6)).galois_group(); G
Galois group C6 of GF(3^6)
>>> F = G.gen()
>>> F**Integer(2)
Frob^2
>>> F**Integer(6)
1
```

gen()

Return a generator of this field (over its prime field). As this is an abstract base class, this just raises a NotImplementedError.

EXAMPLES:

```
sage: K = GF(17)
sage: sage.rings.finite_rings.finite_field_base.FiniteField.gen(K)
Traceback (most recent call last):
...
NotImplementedError
```

```
>>> from sage.all import *
>>> K = GF(Integer(17))
>>> sage.rings.finite_rings.finite_field_base.FiniteField.gen(K)
Traceback (most recent call last):
...
NotImplementedError
```

is_conway()

Return `True` if `self` is defined by a Conway polynomial.

EXAMPLES:

```
sage: GF(5^3, 'a').is_conway()
True
sage: GF(5^3, 'a', modulus='adleman-lenstra').is_conway()
False
sage: GF(next_prime(2^16, 2), 'a').is_conway()
False
```

```
>>> from sage.all import *
>>> GF(Integer(5)**Integer(3), 'a').is_conway()
True
>>> GF(Integer(5)**Integer(3), 'a', modulus='adleman-lenstra').is_conway()
False
>>> GF(next_prime(Integer(2)**Integer(16), Integer(2)), 'a').is_conway()
False
```

is_field(*proof=True*)

Return whether or not the finite field is a field, i.e., always returns `True`.

EXAMPLES:

```
sage: k.<a> = FiniteField(3^4)
sage: k.is_field()
True
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(3)**Integer(4), names=('a',)); (a,) = k._first_
↪ngens(1)
>>> k.is_field()
True
```

is_perfect()

Return whether this field is perfect, i.e., every element has a p-th root. Always returns `True` since finite fields are perfect.

EXAMPLES:

```
sage: GF(2).is_perfect()
True
```

```
>>> from sage.all import *
>>> GF(Integer(2)).is_perfect()
True
```

is_prime_field()

Return `True` if `self` is a prime field, i.e., has degree 1.

EXAMPLES:

```
sage: GF(3^7, 'a').is_prime_field()
False
sage: GF(3, 'a').is_prime_field()
True
```

```
>>> from sage.all import *
>>> GF(Integer(3)**Integer(7), 'a').is_prime_field()
False
>>> GF(Integer(3), 'a').is_prime_field()
True
```

modulus()

Return the minimal polynomial of the generator of `self` over the prime finite field.

The minimal polynomial of an element a in a field is the unique monic irreducible polynomial of smallest degree with coefficients in the base field that has a as a root. In finite field extensions, $\mathbf{F}_{p^n}$, the base field is $\mathbf{F}_p$.

OUTPUT: a monic polynomial over $\mathbf{F}_p$ in the variable x

EXAMPLES:

```
sage: F.<a> = GF(7^2); F
Finite Field in a of size 7^2
sage: F.polynomial_ring()
Univariate Polynomial Ring in a over Finite Field of size 7
sage: f = F.modulus(); f
x^2 + 6*x + 3
sage: f(a)
0
```

```
>>> from sage.all import *
>>> F = GF(Integer(7)**Integer(2), names=('a',)); (a,) = F._first_ngens(1); F
Finite Field in a of size 7^2
>>> F.polynomial_ring()
Univariate Polynomial Ring in a over Finite Field of size 7
>>> f = F.modulus(); f
x^2 + 6*x + 3
>>> f(a)
0
```

Although f is irreducible over the base field, we can double-check whether or not f factors in F as follows.

The command `F['x'](f)` coerces f as a polynomial with coefficients in F. (Instead of a polynomial with coefficients over the base field.)

```
sage: f.factor()
x^2 + 6*x + 3
sage: F['x'](f).factor()
(x + a + 6) * (x + 6*a)
```

```
>>> from sage.all import *
>>> f.factor()
x^2 + 6*x + 3
>>> F['x'](f).factor()
(x + a + 6) * (x + 6*a)
```

Here is an example with a degree 3 extension:

```
sage: G.<b> = GF(7^3); G
Finite Field in b of size 7^3
sage: g = G.modulus(); g
x^3 + 6*x^2 + 4
sage: g.degree(); G.degree()
3
3
```

```
>>> from sage.all import *
>>> G = GF(Integer(7)**Integer(3), names=('b',)); (b,) = G._first_ngens(1); G
Finite Field in b of size 7^3
>>> g = G.modulus(); g
x^3 + 6*x^2 + 4
>>> g.degree(); G.degree()
3
3
```

For prime fields, this returns $x - 1$ unless a custom modulus was given when constructing this field:

```
sage: k = GF(199)
sage: k.modulus()
x + 198
sage: var('x')
x
sage: k = GF(199, modulus=x+1)
sage: k.modulus()
x + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(199))
>>> k.modulus()
x + 198
>>> var('x')
x
>>> k = GF(Integer(199), modulus=x+Integer(1))
>>> k.modulus()
x + 1
```

The given modulus is always made monic:

```
sage: k.<a> = GF(7^2, modulus=2*x^2 - 3, impl='pari_ffelt')
sage: k.modulus()
x^2 + 2
```

```
>>> from sage.all import *
>>> k = GF(Integer(7)**Integer(2), modulus=Integer(2)*x**Integer(2) -␣
↪Integer(3), impl='pari_ffelt', names=('a',)); (a,) = k._first_ngens(1)
>>> k.modulus()
x^2 + 2
```

multiplicative_generator()

Return a primitive element of this finite field, i.e. a generator of the multiplicative group.

You can use *multiplicative_generator()* or *primitive_element()*, these mean the same thing.

> **Warning**
>
> This generator might change from one version of Sage to another.

EXAMPLES:

```
sage: k = GF(997)
sage: k.multiplicative_generator()
7
sage: k.<a> = GF(11^3)
sage: k.primitive_element()
a
sage: k.<b> = GF(19^32)
sage: k.multiplicative_generator()
b + 4
```

```
>>> from sage.all import *
>>> k = GF(Integer(997))
>>> k.multiplicative_generator()
7
>>> k = GF(Integer(11)**Integer(3), names=('a',)); (a,) = k._first_ngens(1)
>>> k.primitive_element()
a
>>> k = GF(Integer(19)**Integer(32), names=('b',)); (b,) = k._first_ngens(1)
>>> k.multiplicative_generator()
b + 4
```

ngens()

The number of generators of the finite field. Always 1.

EXAMPLES:

```
sage: k = FiniteField(3^4, 'b')
sage: k.ngens()
1
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(3)**Integer(4), 'b')
>>> k.ngens()
1
```

order()

Return the order of this finite field.

EXAMPLES:

```
sage: GF(997).order()
997
```

```
>>> from sage.all import *
>>> GF(Integer(997)).order()
997
```

polynomial(*name=None*)

Return the minimal polynomial of the generator of `self` over the prime finite field.

INPUT:

- `name` – a variable name to use for the polynomial. By default, use the name given when constructing this field.

OUTPUT: a monic polynomial over $\mathbf{F}_p$ in the variable `name`

See also

Except for the `name` argument, this is identical to the *`modulus()`* method.

EXAMPLES:

```
sage: k.<a> = FiniteField(9)
sage: k.polynomial('x')
x^2 + 2*x + 2
sage: k.polynomial()
a^2 + 2*a + 2

sage: F = FiniteField(9, 'a', impl='pari_ffelt')
sage: F.polynomial()
a^2 + 2*a + 2

sage: F = FiniteField(7^20, 'a', impl='pari_ffelt')
sage: f = F.polynomial(); f
a^20 + a^12 + 6*a^11 + 2*a^10 + 5*a^9 + 2*a^8 + 3*a^7 + a^6 + 3*a^5 + 3*a^3 +␣
↪a + 3
sage: f(F.gen())
0

sage: # needs sage.libs.ntl
sage: k.<a> = GF(2^20, impl='ntl')
sage: k.polynomial()
```

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```
a^20 + a^10 + a^9 + a^7 + a^6 + a^5 + a^4 + a + 1
sage: k.polynomial('FOO')
FOO^20 + FOO^10 + FOO^9 + FOO^7 + FOO^6 + FOO^5 + FOO^4 + FOO + 1
sage: a^20
a^10 + a^9 + a^7 + a^6 + a^5 + a^4 + a + 1
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(9), names=('a',)); (a,) = k._first_ngens(1)
>>> k.polynomial('x')
x^2 + 2*x + 2
>>> k.polynomial()
a^2 + 2*a + 2

>>> F = FiniteField(Integer(9), 'a', impl='pari_ffelt')
>>> F.polynomial()
a^2 + 2*a + 2

>>> F = FiniteField(Integer(7)**Integer(20), 'a', impl='pari_ffelt')
>>> f = F.polynomial(); f
a^20 + a^12 + 6*a^11 + 2*a^10 + 5*a^9 + 2*a^8 + 3*a^7 + a^6 + 3*a^5 + 3*a^3 +␣
↪a + 3
>>> f(F.gen())
0

>>> # needs sage.libs.ntl
>>> k = GF(Integer(2)**Integer(20), impl='ntl', names=('a',)); (a,) = k._
↪first_ngens(1)
>>> k.polynomial()
a^20 + a^10 + a^9 + a^7 + a^6 + a^5 + a^4 + a + 1
>>> k.polynomial('FOO')
FOO^20 + FOO^10 + FOO^9 + FOO^7 + FOO^6 + FOO^5 + FOO^4 + FOO + 1
>>> a**Integer(20)
a^10 + a^9 + a^7 + a^6 + a^5 + a^4 + a + 1
```

polynomial_ring(*variable_name=None*)

Return the polynomial ring over the prime subfield in the same variable as this finite field.

EXAMPLES:

```
sage: k.<alpha> = FiniteField(3^4)
sage: k.polynomial_ring()
Univariate Polynomial Ring in alpha over Finite Field of size 3
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(3)**Integer(4), names=('alpha',)); (alpha,) = k._
↪first_ngens(1)
>>> k.polynomial_ring()
Univariate Polynomial Ring in alpha over Finite Field of size 3
```

primitive_element()

Return a primitive element of this finite field, i.e. a generator of the multiplicative group.

You can use *multiplicative_generator()* or *primitive_element()*, these mean the same thing.

> **Warning**
>
> This generator might change from one version of Sage to another.

EXAMPLES:

```
sage: k = GF(997)
sage: k.multiplicative_generator()
7
sage: k.<a> = GF(11^3)
sage: k.primitive_element()
a
sage: k.<b> = GF(19^32)
sage: k.multiplicative_generator()
b + 4
```

```
>>> from sage.all import *
>>> k = GF(Integer(997))
>>> k.multiplicative_generator()
7
>>> k = GF(Integer(11)**Integer(3), names=('a',)); (a,) = k._first_ngens(1)
>>> k.primitive_element()
a
>>> k = GF(Integer(19)**Integer(32), names=('b',)); (b,) = k._first_ngens(1)
>>> k.multiplicative_generator()
b + 4
```

random_element(**args*, ***kwds*)

A random element of the finite field. Passes arguments to `random_element()` function of underlying vector space.

EXAMPLES:

```
sage: k = GF(19^4, 'a')
sage: k.random_element().parent() is k                                  #␣
↪needs sage.modules
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(19)**Integer(4), 'a')
>>> k.random_element().parent() is k                                    #␣
↪needs sage.modules
True
```

Passes extra positional or keyword arguments through:

```
sage: k.random_element(prob=0)                                          #␣
↪needs sage.modules
0
```

```
>>> from sage.all import *
>>> k.random_element(prob=Integer(0))                                   ␣
```

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```
↪      # needs sage.modules
0
```

some_elements()

Return a collection of elements of this finite field for use in unit testing.

EXAMPLES:

```
sage: k = GF(2^8,'a')
sage: k.some_elements()  # random output                                    #␣
↪needs sage.modules
[a^4 + a^3 + 1, a^6 + a^4 + a^3, a^5 + a^4 + a, a^2 + a]
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8),'a')
>>> k.some_elements()  # random output                                    #␣
↪needs sage.modules
[a^4 + a^3 + 1, a^6 + a^4 + a^3, a^5 + a^4 + a, a^2 + a]
```

subfield(*degree*, *name=None*, *map=False*)

Return the subfield of the field of `degree`.

The inclusion maps between these subfields will always commute, but they are only added as coercion maps if the following condition holds for the generator g of the field, where d is the degree of this field over the prime field:

The element $g^{(p^d-1)/(p^n-1)}$ generates the subfield of degree n for all divisors n of d.

INPUT:

- `degree` – integer; degree of the subfield
- `name` – string; name of the generator of the subfield
- `map` – boolean (default: `False`); whether to also return the inclusion map

EXAMPLES:

```
sage: k = GF(2^21)
sage: k.subfield(3)
Finite Field in z3 of size 2^3
sage: k.subfield(7, 'a')
Finite Field in a of size 2^7
sage: k.coerce_map_from(_)
Ring morphism:
  From: Finite Field in a of size 2^7
  To:   Finite Field in z21 of size 2^21
  Defn: a |--> z21^20 + z21^19 + z21^17 + z21^15 + z21^14 + z21^6 + z21^4 +␣
↪z21^3 + z21
sage: k.subfield(8)
Traceback (most recent call last):
...
ValueError: no subfield of order 2^8
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(21))
>>> k.subfield(Integer(3))
Finite Field in z3 of size 2^3
>>> k.subfield(Integer(7), 'a')
Finite Field in a of size 2^7
>>> k.coerce_map_from(_)
Ring morphism:
  From: Finite Field in a of size 2^7
  To:   Finite Field in z21 of size 2^21
  Defn: a |--> z21^20 + z21^19 + z21^17 + z21^15 + z21^14 + z21^6 + z21^4 +␣
↪z21^3 + z21
>>> k.subfield(Integer(8))
Traceback (most recent call last):
...
ValueError: no subfield of order 2^8
```

subfields(*degree=0*, *name=None*)

Return all subfields of `self` of the given `degree`, or all possible degrees if `degree` is 0.

The subfields are returned as absolute fields together with an embedding into `self`.

INPUT:

- `degree` – integer (default: 0)
- `name` – string; a dictionary or `None`:
  - If `degree` is nonzero, then `name` must be a string (or `None`, if this is a pseudo-Conway extension), and will be the variable name of the returned field.
  - If `degree` is zero, the dictionary should have keys the divisors of the degree of this field, with the desired variable name for the field of that degree as an entry.
  - As a shortcut, you can provide a string and the degree of each subfield will be appended for the variable name of that subfield.
  - If `None`, uses the prefix of this field.

OUTPUT:

A list of pairs `(K, e)`, where `K` ranges over the subfields of this field and `e` gives an embedding of `K` into `self`.

EXAMPLES:

```
sage: k = GF(2^21)
sage: k.subfields()
[(Finite Field of size 2,
  Ring morphism:
      From: Finite Field of size 2
      To:   Finite Field in z21 of size 2^21
      Defn: 1 |--> 1),
 (Finite Field in z3 of size 2^3,
  Ring morphism:
      From: Finite Field in z3 of size 2^3
      To:   Finite Field in z21 of size 2^21
      Defn: z3 |--> z21^20 + z21^19 + z21^17 + z21^15 + z21^11
```

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```
                    + z21^9 + z21^8 + z21^6 + z21^2),
 (Finite Field in z7 of size 2^7,
  Ring morphism:
      From: Finite Field in z7 of size 2^7
      To:   Finite Field in z21 of size 2^21
      Defn: z7 |--> z21^20 + z21^19 + z21^17 + z21^15 + z21^14
                    + z21^6 + z21^4 + z21^3 + z21),
 (Finite Field in z21 of size 2^21,
  Identity endomorphism of Finite Field in z21 of size 2^21)]
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(21))
>>> k.subfields()
[(Finite Field of size 2,
  Ring morphism:
      From: Finite Field of size 2
      To:   Finite Field in z21 of size 2^21
      Defn: 1 |--> 1),
 (Finite Field in z3 of size 2^3,
  Ring morphism:
      From: Finite Field in z3 of size 2^3
      To:   Finite Field in z21 of size 2^21
      Defn: z3 |--> z21^20 + z21^19 + z21^17 + z21^15 + z21^11
                    + z21^9 + z21^8 + z21^6 + z21^2),
 (Finite Field in z7 of size 2^7,
  Ring morphism:
      From: Finite Field in z7 of size 2^7
      To:   Finite Field in z21 of size 2^21
      Defn: z7 |--> z21^20 + z21^19 + z21^17 + z21^15 + z21^14
                    + z21^6 + z21^4 + z21^3 + z21),
 (Finite Field in z21 of size 2^21,
  Identity endomorphism of Finite Field in z21 of size 2^21)]
```

unit_group_exponent()

The exponent of the unit group of the finite field. For a finite field, this is always the order minus 1.

EXAMPLES:

```
sage: k = GF(2^10, 'a')
sage: k.order()
1024
sage: k.unit_group_exponent()
1023
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(10), 'a')
>>> k.order()
1024
>>> k.unit_group_exponent()
1023
```

zeta(*n=None*)

Return an element of multiplicative order `n` in this finite field. If there is no such element, raise `ValueError`.

Warning

In general, this returns an arbitrary element of the correct order. There are no compatibility guarantees: `F.zeta(9)^3` may not be equal to `F.zeta(3)`.

EXAMPLES:

```
sage: k = GF(7)
sage: k.zeta()
3
sage: k.zeta().multiplicative_order()
6
sage: k.zeta(3)
2
sage: k.zeta(3).multiplicative_order()
3
sage: k = GF(49, 'a')
sage: k.zeta().multiplicative_order()
48
sage: k.zeta(6)
3
sage: k.zeta(5)
Traceback (most recent call last):
...
ValueError: no 5th root of unity in Finite Field in a of size 7^2
```

```
>>> from sage.all import *
>>> k = GF(Integer(7))
>>> k.zeta()
3
>>> k.zeta().multiplicative_order()
6
>>> k.zeta(Integer(3))
2
>>> k.zeta(Integer(3)).multiplicative_order()
3
>>> k = GF(Integer(49), 'a')
>>> k.zeta().multiplicative_order()
48
>>> k.zeta(Integer(6))
3
>>> k.zeta(Integer(5))
Traceback (most recent call last):
...
ValueError: no 5th root of unity in Finite Field in a of size 7^2
```

Even more examples:

```
sage: GF(9,'a').zeta_order()
8
sage: GF(9,'a').zeta()
a
```

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```
sage: GF(9,'a').zeta(4)
a + 1
sage: GF(9,'a').zeta()^2
a + 1
```

```
>>> from sage.all import *
>>> GF(Integer(9),'a').zeta_order()
8
>>> GF(Integer(9),'a').zeta()
a
>>> GF(Integer(9),'a').zeta(Integer(4))
a + 1
>>> GF(Integer(9),'a').zeta()**Integer(2)
a + 1
```

This works even in very large finite fields, provided that `n` can be factored (see Issue #25203):

```
sage: k.<a> = GF(2^2000)
sage: p = 887794514874294500114604143902514703409869050359101317733635669441651
↪7527310181938001
sage: z = k.zeta(p)
sage: z
a^1999 + a^1996 + a^1995 + a^1994 + ... + a^7 + a^5 + a^4 + 1
sage: z ^ p
1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(2000), names=('a',)); (a,) = k._first_ngens(1)
>>> p = Integer(887794514874294500114604143902514703409869050359101317733635669
↪4416517527310181938001)
>>> z = k.zeta(p)
>>> z
a^1999 + a^1996 + a^1995 + a^1994 + ... + a^7 + a^5 + a^4 + 1
>>> z ** p
1
```

zeta_order()

Return the order of the distinguished root of unity in `self`.

EXAMPLES:

```
sage: GF(9,'a').zeta_order()
8
sage: GF(9,'a').zeta()
a
sage: GF(9,'a').zeta().multiplicative_order()
8
```

```
>>> from sage.all import *
>>> GF(Integer(9),'a').zeta_order()
8
>>> GF(Integer(9),'a').zeta()
```

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```
a
>>> GF(Integer(9),'a').zeta().multiplicative_order()
8
```

sage.rings.finite_rings.finite_field_base.**is_FiniteField**(*R*)

Return whether the implementation of R has the interface provided by the standard finite field implementation.

This function is deprecated.

EXAMPLES:

```
sage: from sage.rings.finite_rings.finite_field_base import is_FiniteField
sage: is_FiniteField(GF(9,'a'))
doctest:...: DeprecationWarning: the function is_FiniteField is deprecated; use␣
↪isinstance(x, sage.rings.finite_rings.finite_field_base.FiniteField) instead
See https://github.com/sagemath/sage/issues/32664 for details.
True
sage: is_FiniteField(GF(next_prime(10^10)))
True
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.finite_field_base import is_FiniteField
>>> is_FiniteField(GF(Integer(9),'a'))
doctest:...: DeprecationWarning: the function is_FiniteField is deprecated; use␣
↪isinstance(x, sage.rings.finite_rings.finite_field_base.FiniteField) instead
See https://github.com/sagemath/sage/issues/32664 for details.
True
>>> is_FiniteField(GF(next_prime(Integer(10)**Integer(10))))
True
```

Note that the integers modulo n are not backed by the finite field type:

```
sage: is_FiniteField(Integers(7))
False
```

```
>>> from sage.all import *
>>> is_FiniteField(Integers(Integer(7)))
False
```

sage.rings.finite_rings.finite_field_base.**unpickle_FiniteField_ext**(*_type*, *order*, *variable_name*, *modulus*, *kwargs*)

Used to unpickle extensions of finite fields. Now superseded (hence no doctest), but kept around for backward compatibility.

sage.rings.finite_rings.finite_field_base.**unpickle_FiniteField_prm**(*_type*, *order*, *variable_name*, *kwargs*)

Used to unpickle finite prime fields. Now superseded (hence no doctest), but kept around for backward compatibility.

2.3 Base class for finite field elements

AUTHORS:

- David Roe (2010-01-14): factored out of sage.structure.element
- Sebastian Oehms (2018-07-19): added `conjugate()` (see Issue #26761)

class sage.rings.finite_rings.element_base.**Cache_base**

Bases: `SageObject`

fetch_int(*number*)

Given an integer less than p^n with base 2 representation $a_0 + a_1 \cdot 2 + \cdots + a_k 2^k$, this returns $a_0 + a_1 x + \cdots + a_k x^k$, where x is the generator of this finite field.

EXAMPLES:

```
sage: k.<a> = GF(2^48)
sage: k._cache.fetch_int(2^33 + 2 + 1)                                    #␣
↪needs sage.libs.ntl
a^33 + a + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(48), names=('a',)); (a,) = k._first_ngens(1)
>>> k._cache.fetch_int(Integer(2)**Integer(33) + Integer(2) + Integer(1))    ␣
↪                                  # needs sage.libs.ntl
a^33 + a + 1
```

class sage.rings.finite_rings.element_base.**FinitePolyExtElement**

Bases: *FiniteRingElement*

Elements represented as polynomials modulo a given ideal.

additive_order()

Return the additive order of this finite field element.

EXAMPLES:

```
sage: k.<a> = FiniteField(2^12, 'a')
sage: b = a^3 + a + 1
sage: b.additive_order()
2
sage: k(0).additive_order()
1
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(2)**Integer(12), 'a', names=('a',)); (a,) = k._
↪first_ngens(1)
>>> b = a**Integer(3) + a + Integer(1)
>>> b.additive_order()
2
>>> k(Integer(0)).additive_order()
1
```

charpoly(*var='x'*, *algorithm='pari'*)

Return the characteristic polynomial of `self` as a polynomial with given variable.

INPUT:

- `var` – string (default: `'x'`)
- `algorithm` – string (default: `'pari'`):
  - `'pari'` – use pari's charpoly
  - `'matrix'` – return the charpoly computed from the matrix of left multiplication by `self`

The result is not cached.

EXAMPLES:

```
sage: from sage.rings.finite_rings.element_base import FinitePolyExtElement
sage: k.<a> = FiniteField(19^2)
sage: parent(a)
Finite Field in a of size 19^2
sage: b = a**20
sage: p = FinitePolyExtElement.charpoly(b, "x", algorithm='pari')
sage: q = FinitePolyExtElement.charpoly(b, "x", algorithm='matrix')          #␣
↪needs sage.modules
sage: q == p                                                                 #␣
↪needs sage.modules
True
sage: p
x^2 + 15*x + 4
sage: factor(p)
(x + 17)^2
sage: b.minpoly('x')
x + 17
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.element_base import FinitePolyExtElement
>>> k = FiniteField(Integer(19)**Integer(2), names=('a',)); (a,) = k._first_
↪ngens(1)
>>> parent(a)
Finite Field in a of size 19^2
>>> b = a**Integer(20)
>>> p = FinitePolyExtElement.charpoly(b, "x", algorithm='pari')
>>> q = FinitePolyExtElement.charpoly(b, "x", algorithm='matrix')          #␣
↪needs sage.modules
>>> q == p                                                                 #␣
↪needs sage.modules
True
>>> p
x^2 + 15*x + 4
>>> factor(p)
(x + 17)^2
>>> b.minpoly('x')
x + 17
```

conjugate()

This methods returns the result of the Frobenius morphism in the case where the field is a quadratic extension, say $GF(q^2)$, where $q = p^k$ is a prime power and p the characteristic of the field.

OUTPUT:

Instance of this class representing the image under the Frobenius morphisms.

EXAMPLES:

```
sage: F.<a> = GF(16)
sage: b = a.conjugate(); b
a + 1
sage: a == b.conjugate()
True

sage: F.<a> = GF(27)
sage: a.conjugate()
Traceback (most recent call last):
...
TypeError: cardinality of the field must be a square number
```

```
>>> from sage.all import *
>>> F = GF(Integer(16), names=('a',)); (a,) = F._first_ngens(1)
>>> b = a.conjugate(); b
a + 1
>>> a == b.conjugate()
True

>>> F = GF(Integer(27), names=('a',)); (a,) = F._first_ngens(1)
>>> a.conjugate()
Traceback (most recent call last):
...
TypeError: cardinality of the field must be a square number
```

frobenius(*k=1*)

Return the (p^k)-th power of self, where p is the characteristic of the field.

INPUT:

- `k` – integer (default: 1, must fit in C int type)

Note that if k is negative, then this computes the appropriate root.

EXAMPLES:

```
sage: F.<a> = GF(29^2)
sage: z = a^2 + 5*a + 1
sage: z.pth_power()
19*a + 20
sage: z.pth_power(10)
10*a + 28
sage: z.pth_power(-10) == z
True
sage: F.<b> = GF(2^12)
sage: y = b^3 + b + 1
sage: y == (y.pth_power(-3))^(2^3)
True
sage: y.pth_power(2)
b^7 + b^6 + b^5 + b^4 + b^3 + b
```

```
>>> from sage.all import *
>>> F = GF(Integer(29)**Integer(2), names=('a',)); (a,) = F._first_ngens(1)
>>> z = a**Integer(2) + Integer(5)*a + Integer(1)
>>> z.pth_power()
19*a + 20
>>> z.pth_power(Integer(10))
10*a + 28
>>> z.pth_power(-Integer(10)) == z
True
>>> F = GF(Integer(2)**Integer(12), names=('b',)); (b,) = F._first_ngens(1)
>>> y = b**Integer(3) + b + Integer(1)
>>> y == (y.pth_power(-Integer(3)))**(Integer(2)**Integer(3))
True
>>> y.pth_power(Integer(2))
b^7 + b^6 + b^5 + b^4 + b^3 + b
```

is_square()

Return `True` if and only if this element is a perfect square.

EXAMPLES:

```
sage: k.<a> = FiniteField(9, impl='givaro', modulus='primitive')          #␣
↪needs sage.libs.linbox
sage: a.is_square()                                                       #␣
↪needs sage.libs.linbox
False
sage: (a**2).is_square()                                                  #␣
↪needs sage.libs.linbox
True
sage: k.<a> = FiniteField(4, impl='ntl', modulus='primitive')             #␣
↪needs sage.libs.ntl
sage: (a**2).is_square()                                                  #␣
↪needs sage.libs.ntl
True
sage: k.<a> = FiniteField(17^5, impl='pari_ffelt', modulus='primitive')   #␣
↪needs sage.libs.pari
sage: a.is_square()                                                       #␣
↪needs sage.libs.pari
False
sage: (a**2).is_square()                                                  #␣
↪needs sage.libs.pari
True
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(9), impl='givaro', modulus='primitive', names=('a
↪',)); (a,) = k._first_ngens(1)# needs sage.libs.linbox
>>> a.is_square()                                                         #␣
↪needs sage.libs.linbox
False
>>> (a**Integer(2)).is_square()                                           ␣
↪     # needs sage.libs.linbox
True
>>> k = FiniteField(Integer(4), impl='ntl', modulus='primitive', names=('a',
```

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```
↪)); (a,) = k._first_ngens(1)# needs sage.libs.ntl
>>> (a**Integer(2)).is_square()                                          ␣
↪     # needs sage.libs.ntl
True
>>> k = FiniteField(Integer(17)**Integer(5), impl='pari_ffelt', modulus=
↪'primitive', names=('a',)); (a,) = k._first_ngens(1)# needs sage.libs.pari
>>> a.is_square()                                                        #␣
↪needs sage.libs.pari
False
>>> (a**Integer(2)).is_square()                                          ␣
↪     # needs sage.libs.pari
True
```

```
sage: k(0).is_square()                                                   #␣
↪needs sage.libs.linbox
True
```

```
>>> from sage.all import *
>>> k(Integer(0)).is_square()                                            ␣
↪     # needs sage.libs.linbox
True
```

list()

Return the list of coefficients (in little-endian) of this finite field element when written as a polynomial in the generator.

Equivalent to calling `list()` on this element.

EXAMPLES:

```
sage: x = polygen(GF(71))
sage: F.<u> = GF(71^7, modulus=x^7 + x + 1)
sage: a = 3 + u + 3*u^2 + 3*u^3 + 7*u^4
sage: a.list()
[3, 1, 3, 3, 7, 0, 0]
sage: a.list() == list(a) == [a[i] for i in range(F.degree())]
True
```

```
>>> from sage.all import *
>>> x = polygen(GF(Integer(71)))
>>> F = GF(Integer(71)**Integer(7), modulus=x**Integer(7) + x + Integer(1),␣
↪names=('u',)); (u,) = F._first_ngens(1)
>>> a = Integer(3) + u + Integer(3)*u**Integer(2) + Integer(3)*u**Integer(3)␣
↪+ Integer(7)*u**Integer(4)
>>> a.list()
[3, 1, 3, 3, 7, 0, 0]
>>> a.list() == list(a) == [a[i] for i in range(F.degree())]
True
```

The coefficients returned are those of a fully reduced representative of the finite field element:

```
sage: b = u^777
sage: b.list()
[9, 69, 4, 27, 40, 10, 56]
sage: (u.polynomial()^777).list()
[0, 0, 0, 0, ..., 0, 1]
```

```
>>> from sage.all import *
>>> b = u**Integer(777)
>>> b.list()
[9, 69, 4, 27, 40, 10, 56]
>>> (u.polynomial()**Integer(777)).list()
[0, 0, 0, 0, ..., 0, 1]
```

matrix(*reverse=False*)

Return the matrix of left multiplication by the element on the power basis $1, x, x^2, \ldots, x^{d-1}$ for the field extension.

Thus the emph{columns} of this matrix give the images of each of the x^i.

INPUT:

- `reverse` – if `True`, act on vectors in reversed order

EXAMPLES:

```
sage: # needs sage.modules
sage: k.<a> = GF(2^4)
sage: b = k.random_element()
sage: vector(a*b) == a.matrix() * vector(b)
True
sage: (a*b)._vector_(reverse=True) == a.matrix(reverse=True) * b._vector_
↪(reverse=True)
True
```

```
>>> from sage.all import *
>>> # needs sage.modules
>>> k = GF(Integer(2)**Integer(4), names=('a',)); (a,) = k._first_ngens(1)
>>> b = k.random_element()
>>> vector(a*b) == a.matrix() * vector(b)
True
>>> (a*b)._vector_(reverse=True) == a.matrix(reverse=True) * b._vector_
↪(reverse=True)
True
```

minimal_polynomial(*var='x'*)

Return the minimal polynomial of this element (over the corresponding prime subfield).

EXAMPLES:

```
sage: k.<a> = FiniteField(3^4)
sage: parent(a)
Finite Field in a of size 3^4
sage: b=a**20;p=charpoly(b,"y");p
y^4 + 2*y^2 + 1
```

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```
sage: factor(p)
(y^2 + 1)^2
sage: b.minimal_polynomial('y')
y^2 + 1
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(3)**Integer(4), names=('a',)); (a,) = k._first_
↪ngens(1)
>>> parent(a)
Finite Field in a of size 3^4
>>> b=a**Integer(20);p=charpoly(b,"y");p
y^4 + 2*y^2 + 1
>>> factor(p)
(y^2 + 1)^2
>>> b.minimal_polynomial('y')
y^2 + 1
```

minpoly(*var='x'*, *algorithm='pari'*)

Return the minimal polynomial of this element (over the corresponding prime subfield).

INPUT:

- `var` – string (default: `'x'`)
- `algorithm` – string (default: `'pari'`):
  - `'pari'` – use pari's minpoly
  - `'matrix'` – return the minpoly computed from the matrix of left multiplication by self

EXAMPLES:

```
sage: from sage.rings.finite_rings.element_base import FinitePolyExtElement
sage: k.<a> = FiniteField(19^2)
sage: parent(a)
Finite Field in a of size 19^2
sage: b=a**20
sage: p=FinitePolyExtElement.minpoly(b,"x", algorithm='pari')
sage: q=FinitePolyExtElement.minpoly(b,"x", algorithm='matrix')
sage: q == p
True
sage: p
x + 17
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.element_base import FinitePolyExtElement
>>> k = FiniteField(Integer(19)**Integer(2), names=('a',)); (a,) = k._first_
↪ngens(1)
>>> parent(a)
Finite Field in a of size 19^2
>>> b=a**Integer(20)
>>> p=FinitePolyExtElement.minpoly(b,"x", algorithm='pari')
>>> q=FinitePolyExtElement.minpoly(b,"x", algorithm='matrix')
>>> q == p
```

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```
True
>>> p
x + 17
```

multiplicative_order()

Return the multiplicative order of this field element.

EXAMPLES:

```
sage: S.<a> = GF(5^3); S
Finite Field in a of size 5^3
sage: a.multiplicative_order()
124
sage: (a^8).multiplicative_order()
31
sage: S(0).multiplicative_order()
Traceback (most recent call last):
...
ArithmeticError: Multiplicative order of 0 not defined.
```

```
>>> from sage.all import *
>>> S = GF(Integer(5)**Integer(3), names=('a',)); (a,) = S._first_ngens(1); S
Finite Field in a of size 5^3
>>> a.multiplicative_order()
124
>>> (a**Integer(8)).multiplicative_order()
31
>>> S(Integer(0)).multiplicative_order()
Traceback (most recent call last):
...
ArithmeticError: Multiplicative order of 0 not defined.
```

norm()

Return the norm of `self` down to the prime subfield.

This is the product of the Galois conjugates of `self`.

EXAMPLES:

```
sage: S.<b> = GF(5^2); S
Finite Field in b of size 5^2
sage: b.norm()
2
sage: b.charpoly('t')
t^2 + 4*t + 2
```

```
>>> from sage.all import *
>>> S = GF(Integer(5)**Integer(2), names=('b',)); (b,) = S._first_ngens(1); S
Finite Field in b of size 5^2
>>> b.norm()
2
>>> b.charpoly('t')
t^2 + 4*t + 2
```

Next we consider a cubic extension:

```
sage: S.<a> = GF(5^3); S
Finite Field in a of size 5^3
sage: a.norm()
2
sage: a.charpoly('t')
t^3 + 3*t + 3
sage: a * a^5 * (a^25)
2
```

```
>>> from sage.all import *
>>> S = GF(Integer(5)**Integer(3), names=('a',)); (a,) = S._first_ngens(1); S
Finite Field in a of size 5^3
>>> a.norm()
2
>>> a.charpoly('t')
t^3 + 3*t + 3
>>> a * a**Integer(5) * (a**Integer(25))
2
```

nth_root(*n*, *extend=False*, *all=False*, *algorithm=None*, *cunningham=False*)

Return an n-th root of `self`.

INPUT:

- `n` – integer ≥ 1
- `extend` – boolean (default: `False`); if `True`, return an n-th root in an extension ring, if necessary. Otherwise, raise a `ValueError` if the root is not in the base ring. Warning: this option is not implemented!
- `all` – boolean (default: `False`); if `True`, return all n-th roots of `self`, instead of just one
- `algorithm` – string (default: `None`); `'Johnston'` is the only currently supported option. For IntegerMod elements, the problem is reduced to the prime modulus case using CRT and p-adic logs, and then this algorithm used.

OUTPUT:

If `self` has an n-th root, returns one (if `all` is `False`) or a list of all of them (if `all` is `True`). Otherwise, raises a `ValueError` (if `extend` is `False`) or a `NotImplementedError` (if `extend` is `True`).

Warning

The `extend` option is not implemented (yet).

EXAMPLES:

```
sage: K = GF(31)
sage: a = K(22)
sage: K(22).nth_root(7)
13
sage: K(25).nth_root(5)
5
sage: K(23).nth_root(3)
```

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```
29

sage: K.<a> = GF(625)
sage: (3*a^2+a+1).nth_root(13)**13
3*a^2 + a + 1

sage: k.<a> = GF(29^2)
sage: b = a^2 + 5*a + 1
sage: b.nth_root(11)
3*a + 20
sage: b.nth_root(5)
Traceback (most recent call last):
...
ValueError: no nth root
sage: b.nth_root(5, all = True)
[]
sage: b.nth_root(3, all = True)
[14*a + 18, 10*a + 13, 5*a + 27]

sage: k.<a> = GF(29^5)
sage: b = a^2 + 5*a + 1
sage: b.nth_root(5)
19*a^4 + 2*a^3 + 2*a^2 + 16*a + 3
sage: b.nth_root(7)
Traceback (most recent call last):
...
ValueError: no nth root
sage: b.nth_root(4, all=True)
[]
```

```
>>> from sage.all import *
>>> K = GF(Integer(31))
>>> a = K(Integer(22))
>>> K(Integer(22)).nth_root(Integer(7))
13
>>> K(Integer(25)).nth_root(Integer(5))
5
>>> K(Integer(23)).nth_root(Integer(3))
29

>>> K = GF(Integer(625), names=('a',)); (a,) = K._first_ngens(1)
>>> (Integer(3)*a**Integer(2)+a+Integer(1)).nth_root(Integer(13))**Integer(13)
3*a^2 + a + 1

>>> k = GF(Integer(29)**Integer(2), names=('a',)); (a,) = k._first_ngens(1)
>>> b = a**Integer(2) + Integer(5)*a + Integer(1)
>>> b.nth_root(Integer(11))
3*a + 20
>>> b.nth_root(Integer(5))
Traceback (most recent call last):
...
ValueError: no nth root
```

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```
>>> b.nth_root(Integer(5), all = True)
[]
>>> b.nth_root(Integer(3), all = True)
[14*a + 18, 10*a + 13, 5*a + 27]

>>> k = GF(Integer(29)**Integer(5), names=('a',)); (a,) = k._first_ngens(1)
>>> b = a**Integer(2) + Integer(5)*a + Integer(1)
>>> b.nth_root(Integer(5))
19*a^4 + 2*a^3 + 2*a^2 + 16*a + 3
>>> b.nth_root(Integer(7))
Traceback (most recent call last):
...
ValueError: no nth root
>>> b.nth_root(Integer(4), all=True)
[]
```

ALGORITHM:

The default is currently an algorithm described in [Joh1999].

AUTHOR:

- David Roe (2010-02-13)

pth_power(*k=1*)

Return the (p^k)-th power of self, where p is the characteristic of the field.

INPUT:

- `k` – integer (default: 1, must fit in C int type)

Note that if k is negative, then this computes the appropriate root.

EXAMPLES:

```
sage: F.<a> = GF(29^2)
sage: z = a^2 + 5*a + 1
sage: z.pth_power()
19*a + 20
sage: z.pth_power(10)
10*a + 28
sage: z.pth_power(-10) == z
True
sage: F.<b> = GF(2^12)
sage: y = b^3 + b + 1
sage: y == (y.pth_power(-3))^(2^3)
True
sage: y.pth_power(2)
b^7 + b^6 + b^5 + b^4 + b^3 + b
```

```
>>> from sage.all import *
>>> F = GF(Integer(29)**Integer(2), names=('a',)); (a,) = F._first_ngens(1)
>>> z = a**Integer(2) + Integer(5)*a + Integer(1)
>>> z.pth_power()
19*a + 20
>>> z.pth_power(Integer(10))
```

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```
10*a + 28
>>> z.pth_power(-Integer(10)) == z
True
>>> F = GF(Integer(2)**Integer(12), names=('b',)); (b,) = F._first_ngens(1)
>>> y = b**Integer(3) + b + Integer(1)
>>> y == (y.pth_power(-Integer(3)))**(Integer(2)**Integer(3))
True
>>> y.pth_power(Integer(2))
b^7 + b^6 + b^5 + b^4 + b^3 + b
```

pth_root(*k=1*)

Return the (p^k)-th root of self, where p is the characteristic of the field.

INPUT:

- k – integer (default: 1, must fit in C int type)

Note that if k is negative, then this computes the appropriate power.

EXAMPLES:

```
sage: F.<b> = GF(2^12)
sage: y = b^3 + b + 1
sage: y == (y.pth_root(3))^(2^3)
True
sage: y.pth_root(2)
b^11 + b^10 + b^9 + b^7 + b^5 + b^4 + b^2 + b
```

```
>>> from sage.all import *
>>> F = GF(Integer(2)**Integer(12), names=('b',)); (b,) = F._first_ngens(1)
>>> y = b**Integer(3) + b + Integer(1)
>>> y == (y.pth_root(Integer(3)))**(Integer(2)**Integer(3))
True
>>> y.pth_root(Integer(2))
b^11 + b^10 + b^9 + b^7 + b^5 + b^4 + b^2 + b
```

sqrt(*extend=False*, *all=False*)

See *square_root()*.

EXAMPLES:

```
sage: k.<a> = GF(3^17)
sage: (a^3 - a - 1).sqrt()
a^16 + 2*a^15 + a^13 + 2*a^12 + a^10 + 2*a^9 + 2*a^8 + a^7 + a^6 + 2*a^5 + a^
↪4 + 2*a^2 + 2*a + 2
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(17), names=('a',)); (a,) = k._first_ngens(1)
>>> (a**Integer(3) - a - Integer(1)).sqrt()
a^16 + 2*a^15 + a^13 + 2*a^12 + a^10 + 2*a^9 + 2*a^8 + a^7 + a^6 + 2*a^5 + a^
↪4 + 2*a^2 + 2*a + 2
```

square_root(*extend=False*, *all=False*)

The square root function.

INPUT:

- `extend` – boolean (default: `True`); if `True`, return a square root in an extension ring, if necessary. Otherwise, raise a `ValueError` if the root is not in the base ring.

> **Warning**
>
> This option is not implemented!

- `all` – boolean (default: `False`); if `True`, return all square roots of `self`, instead of just one

> **Warning**
>
> The `'extend'` option is not implemented (yet).

EXAMPLES:

```
sage: F = FiniteField(7^2, 'a')
sage: F(2).square_root()
4
sage: F(3).square_root()
2*a + 6
sage: F(3).square_root()**2
3
sage: F(4).square_root()
2
sage: K = FiniteField(7^3, 'alpha', impl='pari_ffelt')
sage: K(3).square_root()
Traceback (most recent call last):
...
ValueError: must be a perfect square.
```

```
>>> from sage.all import *
>>> F = FiniteField(Integer(7)**Integer(2), 'a')
>>> F(Integer(2)).square_root()
4
>>> F(Integer(3)).square_root()
2*a + 6
>>> F(Integer(3)).square_root()**Integer(2)
3
>>> F(Integer(4)).square_root()
2
>>> K = FiniteField(Integer(7)**Integer(3), 'alpha', impl='pari_ffelt')
>>> K(Integer(3)).square_root()
Traceback (most recent call last):
...
ValueError: must be a perfect square.
```

`to_bytes`(*byteorder='big'*)

Return an array of bytes representing an integer.

Internally relies on the python `int.to_bytes()` method. Length of byte array is determined from the field's order.

INPUT:

- `byteorder` – string (default: `'big'`); determines the byte order of the output; can only be `'big'` or `'little'`

EXAMPLES:

```
sage: F.<z5> = GF(3^5)
sage: a = z5^4 + 2*z5^3 + 1
sage: a.to_bytes()
b'\x88'
```

```
>>> from sage.all import *
>>> F = GF(Integer(3)**Integer(5), names=('z5',)); (z5,) = F._first_ngens(1)
>>> a = z5**Integer(4) + Integer(2)*z5**Integer(3) + Integer(1)
>>> a.to_bytes()
b'\x88'
```

```
sage: F.<z3> = GF(163^3)
sage: a = 136*z3^2 + 10*z3 + 125
sage: a.to_bytes()
b'7)\xa3'
```

```
>>> from sage.all import *
>>> F = GF(Integer(163)**Integer(3), names=('z3',)); (z3,) = F._first_ngens(1)
>>> a = Integer(136)*z3**Integer(2) + Integer(10)*z3 + Integer(125)
>>> a.to_bytes()
b'7)\xa3'
```

to_integer(*reverse=False*)

Return an integer representation of this finite field element obtained by lifting its representative polynomial to $\mathbf{Z}$ and evaluating it at the characteristic p.

If `reverse` is set to `True` (default: `False`), the list of coefficients is reversed prior to evaluation.

Inverse of *sage.rings.finite_rings.finite_field_base.FiniteField.from_integer()*.

EXAMPLES:

```
sage: F.<t> = GF(7^5)
sage: F(5).to_integer()
5
sage: t.to_integer()
7
sage: (t^2).to_integer()
49
sage: (t^2+1).to_integer()
50
sage: (t^2+t+1).to_integer()
57
```

```
>>> from sage.all import *
>>> F = GF(Integer(7)**Integer(5), names=('t',)); (t,) = F._first_ngens(1)
>>> F(Integer(5)).to_integer()
5
```

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```
>>> t.to_integer()
7
>>> (t**Integer(2)).to_integer()
49
>>> (t**Integer(2)+Integer(1)).to_integer()
50
>>> (t**Integer(2)+t+Integer(1)).to_integer()
57
```

```
sage: F.<t> = GF(2^8)
sage: u = F.from_integer(0xd1)
sage: bin(u.to_integer(False))
'0b11010001'
sage: bin(u.to_integer(True))
'0b10001011'
```

```
>>> from sage.all import *
>>> F = GF(Integer(2)**Integer(8), names=('t',)); (t,) = F._first_ngens(1)
>>> u = F.from_integer(Integer(0xd1))
>>> bin(u.to_integer(False))
'0b11010001'
>>> bin(u.to_integer(True))
'0b10001011'
```

trace()

Return the trace of this element, which is the sum of the Galois conjugates.

EXAMPLES:

```
sage: S.<a> = GF(5^3); S
Finite Field in a of size 5^3
sage: a.trace()
0
sage: a.charpoly('t')
t^3 + 3*t + 3
sage: a + a^5 + a^25
0
sage: z = a^2 + a + 1
sage: z.trace()
2
sage: z.charpoly('t')
t^3 + 3*t^2 + 2*t + 2
sage: z + z^5 + z^25
2
```

```
>>> from sage.all import *
>>> S = GF(Integer(5)**Integer(3), names=('a',)); (a,) = S._first_ngens(1); S
Finite Field in a of size 5^3
>>> a.trace()
0
>>> a.charpoly('t')
t^3 + 3*t + 3
```

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```
>>> a + a**Integer(5) + a**Integer(25)
0
>>> z = a**Integer(2) + a + Integer(1)
>>> z.trace()
2
>>> z.charpoly('t')
t^3 + 3*t^2 + 2*t + 2
>>> z + z**Integer(5) + z**Integer(25)
2
```

class sage.rings.finite_rings.element_base.**FiniteRingElement**

Bases: CommutativeRingElement

to_bytes(*byteorder='big'*)

Return an array of bytes representing an integer.

Internally relies on the python `int.to_bytes()` method. Length of byte array is determined from the field's order.

INPUT:

- `byteorder` – string (default: `'big'`); determines the byte order of `input_bytes`; can only be `'big'` or `'little'`

EXAMPLES:

```
sage: F = GF(65537)
sage: a = F(8726)
sage: a.to_bytes()
b'\x00"\x16'
sage: a.to_bytes(byteorder='little')
b'\x16"\x00'
```

```
>>> from sage.all import *
>>> F = GF(Integer(65537))
>>> a = F(Integer(8726))
>>> a.to_bytes()
b'\x00"\x16'
>>> a.to_bytes(byteorder='little')
b'\x16"\x00'
```

sage.rings.finite_rings.element_base.**is_FiniteFieldElement**(*x*)

Return `True` if `x` is a finite field element.

This function is deprecated.

EXAMPLES:

```
sage: from sage.rings.finite_rings.element_base import is_FiniteFieldElement
sage: is_FiniteFieldElement(1)
doctest:...: DeprecationWarning: the function is_FiniteFieldElement is deprecated;
↪ use isinstance(x, sage.structure.element.FieldElement) and x.parent().is_
↪finite() instead
See https://github.com/sagemath/sage/issues/32664 for details.
False
```

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```
sage: is_FiniteFieldElement(IntegerRing())
False
sage: is_FiniteFieldElement(GF(5)(2))
True
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.element_base import is_FiniteFieldElement
>>> is_FiniteFieldElement(Integer(1))
doctest:...: DeprecationWarning: the function is_FiniteFieldElement is deprecated;
↪ use isinstance(x, sage.structure.element.FieldElement) and x.parent().is_
↪finite() instead
See https://github.com/sagemath/sage/issues/32664 for details.
False
>>> is_FiniteFieldElement(IntegerRing())
False
>>> is_FiniteFieldElement(GF(Integer(5))(Integer(2)))
True
```

2.4 Homset for finite fields

This is the set of all field homomorphisms between two finite fields.

EXAMPLES:

```
sage: R.<t> = ZZ[]
sage: E.<a> = GF(25, modulus = t^2 - 2)
sage: F.<b> = GF(625)
sage: H = Hom(E, F)
sage: f = H([4*b^3 + 4*b^2 + 4*b]); f
Ring morphism:
  From: Finite Field in a of size 5^2
  To:   Finite Field in b of size 5^4
  Defn: a |--> 4*b^3 + 4*b^2 + 4*b
sage: f(2)
2
sage: f(a)
4*b^3 + 4*b^2 + 4*b
sage: len(H)
2
sage: [phi(2*a)^2 for phi in Hom(E, F)]
[3, 3]
```

```
>>> from sage.all import *
>>> R = ZZ['t']; (t,) = R._first_ngens(1)
>>> E = GF(Integer(25), modulus = t**Integer(2) - Integer(2), names=('a',)); (a,) = E.
↪_first_ngens(1)
>>> F = GF(Integer(625), names=('b',)); (b,) = F._first_ngens(1)
>>> H = Hom(E, F)
>>> f = H([Integer(4)*b**Integer(3) + Integer(4)*b**Integer(2) + Integer(4)*b]); f
Ring morphism:
  From: Finite Field in a of size 5^2
```

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```
  To:   Finite Field in b of size 5^4
  Defn: a |--> 4*b^3 + 4*b^2 + 4*b
>>> f(Integer(2))
2
>>> f(a)
4*b^3 + 4*b^2 + 4*b
>>> len(H)
2
>>> [phi(Integer(2)*a)**Integer(2) for phi in Hom(E, F)]
[3, 3]
```

We can also create endomorphisms:

```
sage: End(E)
Automorphism group of Finite Field in a of size 5^2
sage: End(GF(7))[0]
Ring endomorphism of Finite Field of size 7
  Defn: 1 |--> 1
sage: H = Hom(GF(7), GF(49, 'c'))
sage: H[0](2)
2
```

```
>>> from sage.all import *
>>> End(E)
Automorphism group of Finite Field in a of size 5^2
>>> End(GF(Integer(7)))[Integer(0)]
Ring endomorphism of Finite Field of size 7
  Defn: 1 |--> 1
>>> H = Hom(GF(Integer(7)), GF(Integer(49), 'c'))
>>> H[Integer(0)](Integer(2))
2
```

class sage.rings.finite_rings.homset.**FiniteFieldHomset**(*R*, *S*, *category=None*)

Bases: RingHomset_generic

Set of homomorphisms with domain a given finite field.

index(*item*)

Return the index of `self`.

EXAMPLES:

```
sage: K.<z> = GF(1024)
sage: g = End(K)[3]
sage: End(K).index(g) == 3
True
```

```
>>> from sage.all import *
>>> K = GF(Integer(1024), names=('z',)); (z,) = K._first_ngens(1)
>>> g = End(K)[Integer(3)]
>>> End(K).index(g) == Integer(3)
True
```

is_aut()

Check if `self` is an automorphism.

EXAMPLES:

```
sage: Hom(GF(4, 'a'), GF(16, 'b')).is_aut()
False
sage: Hom(GF(4, 'a'), GF(4, 'c')).is_aut()
False
sage: Hom(GF(4, 'a'), GF(4, 'a')).is_aut()
True
```

```
>>> from sage.all import *
>>> Hom(GF(Integer(4), 'a'), GF(Integer(16), 'b')).is_aut()
False
>>> Hom(GF(Integer(4), 'a'), GF(Integer(4), 'c')).is_aut()
False
>>> Hom(GF(Integer(4), 'a'), GF(Integer(4), 'a')).is_aut()
True
```

list()

Return a list of all the elements in this set of field homomorphisms.

EXAMPLES:

```
sage: K.<a> = GF(25)
sage: End(K)
Automorphism group of Finite Field in a of size 5^2
sage: list(End(K))
[Ring endomorphism of Finite Field in a of size 5^2
  Defn: a |--> 4*a + 1,
 Ring endomorphism of Finite Field in a of size 5^2
  Defn: a |--> a]
sage: L.<z> = GF(7^6)
sage: [g for g in End(L) if (g^3)(z) == z]
[Ring endomorphism of Finite Field in z of size 7^6
  Defn: z |--> z,
 Ring endomorphism of Finite Field in z of size 7^6
  Defn: z |--> 5*z^4 + 5*z^3 + 4*z^2 + 3*z + 1,
 Ring endomorphism of Finite Field in z of size 7^6
  Defn: z |--> 3*z^5 + 5*z^4 + 5*z^2 + 2*z + 3]
```

```
>>> from sage.all import *
>>> K = GF(Integer(25), names=('a',)); (a,) = K._first_ngens(1)
>>> End(K)
Automorphism group of Finite Field in a of size 5^2
>>> list(End(K))
[Ring endomorphism of Finite Field in a of size 5^2
  Defn: a |--> 4*a + 1,
 Ring endomorphism of Finite Field in a of size 5^2
  Defn: a |--> a]
>>> L = GF(Integer(7)**Integer(6), names=('z',)); (z,) = L._first_ngens(1)
>>> [g for g in End(L) if (g**Integer(3))(z) == z]
[Ring endomorphism of Finite Field in z of size 7^6
```

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```
 Defn: z |--> z,
Ring endomorphism of Finite Field in z of size 7^6
 Defn: z |--> 5*z^4 + 5*z^3 + 4*z^2 + 3*z + 1,
Ring endomorphism of Finite Field in z of size 7^6
 Defn: z |--> 3*z^5 + 5*z^4 + 5*z^2 + 2*z + 3]
```

Between isomorphic fields with different moduli:

```
sage: k1 = GF(1009)
sage: k2 = GF(1009, modulus='primitive')
sage: Hom(k1, k2).list()
[Ring morphism:
   From: Finite Field of size 1009
   To:   Finite Field of size 1009
   Defn: 1 |--> 1]
sage: Hom(k2, k1).list()
[Ring morphism:
   From: Finite Field of size 1009
   To:   Finite Field of size 1009
   Defn: 11 |--> 11]

sage: k1.<a> = GF(1009^2, modulus='first_lexicographic')
sage: k2.<b> = GF(1009^2, modulus='conway')
sage: Hom(k1, k2).list()
[Ring morphism:
   From: Finite Field in a of size 1009^2
   To:   Finite Field in b of size 1009^2
   Defn: a |--> 290*b + 864,
 Ring morphism:
   From: Finite Field in a of size 1009^2
   To:   Finite Field in b of size 1009^2
   Defn: a |--> 719*b + 145]
```

```
>>> from sage.all import *
>>> k1 = GF(Integer(1009))
>>> k2 = GF(Integer(1009), modulus='primitive')
>>> Hom(k1, k2).list()
[Ring morphism:
   From: Finite Field of size 1009
   To:   Finite Field of size 1009
   Defn: 1 |--> 1]
>>> Hom(k2, k1).list()
[Ring morphism:
   From: Finite Field of size 1009
   To:   Finite Field of size 1009
   Defn: 11 |--> 11]

>>> k1 = GF(Integer(1009)**Integer(2), modulus='first_lexicographic', names=(
↪'a',)); (a,) = k1._first_ngens(1)
>>> k2 = GF(Integer(1009)**Integer(2), modulus='conway', names=('b',)); (b,)␣
↪= k2._first_ngens(1)
>>> Hom(k1, k2).list()
```

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```
[Ring morphism:
   From: Finite Field in a of size 1009^2
   To:   Finite Field in b of size 1009^2
   Defn: a |--> 290*b + 864,
 Ring morphism:
   From: Finite Field in a of size 1009^2
   To:   Finite Field in b of size 1009^2
   Defn: a |--> 719*b + 145]
```

order()

Return the order of this set of field homomorphisms.

EXAMPLES:

```
sage: K.<a> = GF(125)
sage: End(K)
Automorphism group of Finite Field in a of size 5^3
sage: End(K).order()
3
sage: L.<b> = GF(25)
sage: Hom(L, K).order() == Hom(K, L).order() == 0
True
```

```
>>> from sage.all import *
>>> K = GF(Integer(125), names=('a',)); (a,) = K._first_ngens(1)
>>> End(K)
Automorphism group of Finite Field in a of size 5^3
>>> End(K).order()
3
>>> L = GF(Integer(25), names=('b',)); (b,) = L._first_ngens(1)
>>> Hom(L, K).order() == Hom(K, L).order() == Integer(0)
True
```

2.5 Finite field morphisms

This file provides several classes implementing:

- embeddings between finite fields
- Frobenius isomorphism on finite fields

EXAMPLES:

```
sage: from sage.rings.finite_rings.hom_finite_field import FiniteFieldHomomorphism_
↪generic
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.hom_finite_field import FiniteFieldHomomorphism_
↪generic
```

Construction of an embedding:

```
sage: k.<t> = GF(3^7)
sage: K.<T> = GF(3^21)
sage: f = FiniteFieldHomomorphism_generic(Hom(k, K)); f # random
Ring morphism:
  From: Finite Field in t of size 3^7
  To:   Finite Field in T of size 3^21
  Defn: t |--> T^20 + 2*T^18 + T^16 + 2*T^13 + T^9 + 2*T^8 + T^7 + T^6 + T^5 + T^3 +␣
↪2*T^2 + T

sage: f(t) # random
T^20 + 2*T^18 + T^16 + 2*T^13 + T^9 + 2*T^8 + T^7 + T^6 + T^5 + T^3 + 2*T^2 + T
sage: f(t) == f.im_gens()[0]
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(7), names=('t',)); (t,) = k._first_ngens(1)
>>> K = GF(Integer(3)**Integer(21), names=('T',)); (T,) = K._first_ngens(1)
>>> f = FiniteFieldHomomorphism_generic(Hom(k, K)); f # random
Ring morphism:
  From: Finite Field in t of size 3^7
  To:   Finite Field in T of size 3^21
  Defn: t |--> T^20 + 2*T^18 + T^16 + 2*T^13 + T^9 + 2*T^8 + T^7 + T^6 + T^5 + T^3 +␣
↪2*T^2 + T

>>> f(t) # random
T^20 + 2*T^18 + T^16 + 2*T^13 + T^9 + 2*T^8 + T^7 + T^6 + T^5 + T^3 + 2*T^2 + T
>>> f(t) == f.im_gens()[Integer(0)]
True
```

The map f has a method `section` which returns a partially defined map which is the inverse of f on the image of f:

```
sage: g = f.section(); g # random
Section of Ring morphism:
  From: Finite Field in t of size 3^7
  To:   Finite Field in T of size 3^21
  Defn: t |--> T^20 + 2*T^18 + T^16 + 2*T^13 + T^9 + 2*T^8 + T^7 + T^6 + T^5 + T^3 +␣
↪2*T^2 + T
sage: a = k.random_element()
sage: g(f(a)) == a
True
sage: g(T)
Traceback (most recent call last):
...
ValueError: T is not in the image of Ring morphism:
  From: Finite Field in t of size 3^7
  To:   Finite Field in T of size 3^21
  Defn: ...
```

```
>>> from sage.all import *
>>> g = f.section(); g # random
Section of Ring morphism:
  From: Finite Field in t of size 3^7
```

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```
  To:   Finite Field in T of size 3^21
  Defn: t |--> T^20 + 2*T^18 + T^16 + 2*T^13 + T^9 + 2*T^8 + T^7 + T^6 + T^5 + T^3 +␣
↪2*T^2 + T
>>> a = k.random_element()
>>> g(f(a)) == a
True
>>> g(T)
Traceback (most recent call last):
...
ValueError: T is not in the image of Ring morphism:
  From: Finite Field in t of size 3^7
  To:   Finite Field in T of size 3^21
  Defn: ...
```

There is no embedding of $GF(5^6)$ into $GF(5^11)$:

```
sage: k.<t> = GF(5^6)
sage: K.<T> = GF(5^11)
sage: FiniteFieldHomomorphism_generic(Hom(k, K))
Traceback (most recent call last):
...
ValueError: No embedding of Finite Field in t of size 5^6 into Finite Field in T of␣
↪size 5^11
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(6), names=('t',)); (t,) = k._first_ngens(1)
>>> K = GF(Integer(5)**Integer(11), names=('T',)); (T,) = K._first_ngens(1)
>>> FiniteFieldHomomorphism_generic(Hom(k, K))
Traceback (most recent call last):
...
ValueError: No embedding of Finite Field in t of size 5^6 into Finite Field in T of␣
↪size 5^11
```

Construction of Frobenius endomorphisms:

```
sage: k.<t> = GF(7^14)
sage: Frob = k.frobenius_endomorphism(); Frob
Frobenius endomorphism t |--> t^7 on Finite Field in t of size 7^14
sage: Frob(t)
t^7
```

```
>>> from sage.all import *
>>> k = GF(Integer(7)**Integer(14), names=('t',)); (t,) = k._first_ngens(1)
>>> Frob = k.frobenius_endomorphism(); Frob
Frobenius endomorphism t |--> t^7 on Finite Field in t of size 7^14
>>> Frob(t)
t^7
```

Some basic arithmetics is supported:

```
sage: Frob^2
Frobenius endomorphism t |--> t^(7^2) on Finite Field in t of size 7^14
```

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```
sage: f = k.frobenius_endomorphism(7); f
Frobenius endomorphism t |--> t^(7^7) on Finite Field in t of size 7^14
sage: f*Frob
Frobenius endomorphism t |--> t^(7^8) on Finite Field in t of size 7^14

sage: Frob.order()
14
sage: f.order()
2
```

```
>>> from sage.all import *
>>> Frob**Integer(2)
Frobenius endomorphism t |--> t^(7^2) on Finite Field in t of size 7^14
>>> f = k.frobenius_endomorphism(Integer(7)); f
Frobenius endomorphism t |--> t^(7^7) on Finite Field in t of size 7^14
>>> f*Frob
Frobenius endomorphism t |--> t^(7^8) on Finite Field in t of size 7^14

>>> Frob.order()
14
>>> f.order()
2
```

Note that simplifications are made automatically:

```
sage: Frob^16
Frobenius endomorphism t |--> t^(7^2) on Finite Field in t of size 7^14
sage: Frob^28
Identity endomorphism of Finite Field in t of size 7^14
```

```
>>> from sage.all import *
>>> Frob**Integer(16)
Frobenius endomorphism t |--> t^(7^2) on Finite Field in t of size 7^14
>>> Frob**Integer(28)
Identity endomorphism of Finite Field in t of size 7^14
```

And that comparisons work:

```
sage: Frob == Frob^15
True
sage: Frob^14 == Hom(k, k).identity()
True
```

```
>>> from sage.all import *
>>> Frob == Frob**Integer(15)
True
>>> Frob**Integer(14) == Hom(k, k).identity()
True
```

AUTHOR:

- Xavier Caruso (2012-06-29)

class sage.rings.finite_rings.hom_finite_field.**FiniteFieldHomomorphism_generic**

Bases: RingHomomorphism_im_gens

A class implementing embeddings between finite fields.

is_injective()

Return True since a embedding between finite fields is always injective.

EXAMPLES:

```
sage: from sage.rings.finite_rings.hom_finite_field import␣
↪FiniteFieldHomomorphism_generic
sage: k.<t> = GF(3^3)
sage: K.<T> = GF(3^9)
sage: f = FiniteFieldHomomorphism_generic(Hom(k, K))
sage: f.is_injective()
True
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.hom_finite_field import␣
↪FiniteFieldHomomorphism_generic
>>> k = GF(Integer(3)**Integer(3), names=('t',)); (t,) = k._first_ngens(1)
>>> K = GF(Integer(3)**Integer(9), names=('T',)); (T,) = K._first_ngens(1)
>>> f = FiniteFieldHomomorphism_generic(Hom(k, K))
>>> f.is_injective()
True
```

is_surjective()

Return True if this embedding is surjective (and hence an isomorphism.

EXAMPLES:

```
sage: from sage.rings.finite_rings.hom_finite_field import␣
↪FiniteFieldHomomorphism_generic
sage: k.<t> = GF(3^3)
sage: K.<T> = GF(3^9)
sage: f = FiniteFieldHomomorphism_generic(Hom(k, K))
sage: f.is_surjective()
False
sage: g = FiniteFieldHomomorphism_generic(Hom(k, k))
sage: g.is_surjective()
True
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.hom_finite_field import␣
↪FiniteFieldHomomorphism_generic
>>> k = GF(Integer(3)**Integer(3), names=('t',)); (t,) = k._first_ngens(1)
>>> K = GF(Integer(3)**Integer(9), names=('T',)); (T,) = K._first_ngens(1)
>>> f = FiniteFieldHomomorphism_generic(Hom(k, K))
>>> f.is_surjective()
False
>>> g = FiniteFieldHomomorphism_generic(Hom(k, k))
>>> g.is_surjective()
True
```

section()

Return the `inverse` of this embedding.

It is a partially defined map whose domain is the codomain of the embedding, but which is only defined on the image of the embedding.

EXAMPLES:

```
sage: from sage.rings.finite_rings.hom_finite_field import␣
↪FiniteFieldHomomorphism_generic
sage: k.<t> = GF(3^7)
sage: K.<T> = GF(3^21)
sage: f = FiniteFieldHomomorphism_generic(Hom(k, K))
sage: g = f.section(); g # random
Section of Ring morphism:
  From: Finite Field in t of size 3^7
  To:   Finite Field in T of size 3^21
  Defn: t |--> T^20 + 2*T^18 + T^16 + 2*T^13 + T^9 + 2*T^8 + T^7 + T^6 + T^5␣
↪+ T^3 + 2*T^2 + T
sage: a = k.random_element()
sage: b = k.random_element()
sage: g(f(a) + f(b)) == a + b
True
sage: g(T)
Traceback (most recent call last):
...
ValueError: T is not in the image of Ring morphism:
  From: Finite Field in t of size 3^7
  To:   Finite Field in T of size 3^21
  Defn: ...
```

```
>>> from sage.all import *
>>> from sage.rings.finite_rings.hom_finite_field import␣
↪FiniteFieldHomomorphism_generic
>>> k = GF(Integer(3)**Integer(7), names=('t',)); (t,) = k._first_ngens(1)
>>> K = GF(Integer(3)**Integer(21), names=('T',)); (T,) = K._first_ngens(1)
>>> f = FiniteFieldHomomorphism_generic(Hom(k, K))
>>> g = f.section(); g # random
Section of Ring morphism:
  From: Finite Field in t of size 3^7
  To:   Finite Field in T of size 3^21
  Defn: t |--> T^20 + 2*T^18 + T^16 + 2*T^13 + T^9 + 2*T^8 + T^7 + T^6 + T^5␣
↪+ T^3 + 2*T^2 + T
>>> a = k.random_element()
>>> b = k.random_element()
>>> g(f(a) + f(b)) == a + b
True
>>> g(T)
Traceback (most recent call last):
...
ValueError: T is not in the image of Ring morphism:
  From: Finite Field in t of size 3^7
  To:   Finite Field in T of size 3^21
  Defn: ...
```

class sage.rings.finite_rings.hom_finite_field.**FrobeniusEndomorphism_finite_field**

Bases: FrobeniusEndomorphism_generic

A class implementing Frobenius endomorphisms on finite fields.

fixed_field()

Return the fixed field of `self`.

OUTPUT:

- a tuple (K, e), where K is the subfield of the domain consisting of elements fixed by `self` and e is an embedding of K into the domain.

> **Note**
>
> The name of the variable used for the subfield (if it is not a prime subfield) is suffixed by `_fixed`.

EXAMPLES:

```
sage: k.<t> = GF(5^6)
sage: f = k.frobenius_endomorphism(2)
sage: kfixed, embed = f.fixed_field()
sage: kfixed
Finite Field in t_fixed of size 5^2
sage: embed # random
Ring morphism:
  From: Finite Field in t_fixed of size 5^2
  To:   Finite Field in t of size 5^6
  Defn: t_fixed |--> 4*t^5 + 2*t^4 + 4*t^2 + t

sage: tfixed = kfixed.gen()
sage: embed(tfixed) # random
4*t^5 + 2*t^4 + 4*t^2 + t
sage: embed(tfixed) == embed.im_gens()[0]
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(6), names=('t',)); (t,) = k._first_ngens(1)
>>> f = k.frobenius_endomorphism(Integer(2))
>>> kfixed, embed = f.fixed_field()
>>> kfixed
Finite Field in t_fixed of size 5^2
>>> embed # random
Ring morphism:
  From: Finite Field in t_fixed of size 5^2
  To:   Finite Field in t of size 5^6
  Defn: t_fixed |--> 4*t^5 + 2*t^4 + 4*t^2 + t

>>> tfixed = kfixed.gen()
>>> embed(tfixed) # random
4*t^5 + 2*t^4 + 4*t^2 + t
>>> embed(tfixed) == embed.im_gens()[Integer(0)]
True
```

inverse()

Return the inverse of this Frobenius endomorphism.

EXAMPLES:

```
sage: k.<a> = GF(7^11)
sage: f = k.frobenius_endomorphism(5)
sage: (f.inverse() * f).is_identity()
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(7)**Integer(11), names=('a',)); (a,) = k._first_ngens(1)
>>> f = k.frobenius_endomorphism(Integer(5))
>>> (f.inverse() * f).is_identity()
True
```

is_identity()

Return `True` if this morphism is the identity morphism.

EXAMPLES:

```
sage: k.<t> = GF(5^3)
sage: Frob = k.frobenius_endomorphism()
sage: Frob.is_identity()
False
sage: (Frob^3).is_identity()
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(3), names=('t',)); (t,) = k._first_ngens(1)
>>> Frob = k.frobenius_endomorphism()
>>> Frob.is_identity()
False
>>> (Frob**Integer(3)).is_identity()
True
```

is_injective()

Return `True` since any power of the Frobenius endomorphism over a finite field is always injective.

EXAMPLES:

```
sage: k.<t> = GF(5^3)
sage: Frob = k.frobenius_endomorphism()
sage: Frob.is_injective()
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(3), names=('t',)); (t,) = k._first_ngens(1)
>>> Frob = k.frobenius_endomorphism()
>>> Frob.is_injective()
True
```

is_surjective()

Return `True` since any power of the Frobenius endomorphism over a finite field is always surjective.

EXAMPLES:

```
sage: k.<t> = GF(5^3)
sage: Frob = k.frobenius_endomorphism()
sage: Frob.is_surjective()
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(3), names=('t',)); (t,) = k._first_ngens(1)
>>> Frob = k.frobenius_endomorphism()
>>> Frob.is_surjective()
True
```

order()

Return the order of this endomorphism.

EXAMPLES:

```
sage: k.<t> = GF(5^12)
sage: Frob = k.frobenius_endomorphism()
sage: Frob.order()
12
sage: (Frob^2).order()
6
sage: (Frob^9).order()
4
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(12), names=('t',)); (t,) = k._first_ngens(1)
>>> Frob = k.frobenius_endomorphism()
>>> Frob.order()
12
>>> (Frob**Integer(2)).order()
6
>>> (Frob**Integer(9)).order()
4
```

power()

Return an integer n such that this endomorphism is the n-th power of the absolute (arithmetic) Frobenius.

EXAMPLES:

```
sage: k.<t> = GF(5^12)
sage: Frob = k.frobenius_endomorphism()
sage: Frob.power()
1
sage: (Frob^9).power()
9
sage: (Frob^13).power()
1
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(12), names=('t',)); (t,) = k._first_ngens(1)
```

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```
>>> Frob = k.frobenius_endomorphism()
>>> Frob.power()
1
>>> (Frob**Integer(9)).power()
9
>>> (Frob**Integer(13)).power()
1
```

class sage.rings.finite_rings.hom_finite_field.**SectionFiniteFieldHomomorphism_generic**

Bases: Section

A class implementing sections of embeddings between finite fields.

CHAPTER

THREE

PRIME FIELDS

3.1 Finite prime fields

AUTHORS:

- William Stein: initial version
- Martin Albrecht (2008-01): refactoring

class sage.rings.finite_rings.finite_field_prime_modn.**FiniteField_prime_modn**(*p*, *check=True*, *modulus=None*)

Bases: *FiniteField*, *IntegerModRing_generic*

Finite field of order p where p is prime.

EXAMPLES:

```
sage: FiniteField(3)
Finite Field of size 3

sage: FiniteField(next_prime(1000))                                    #␣
↪needs sage.rings.finite_rings
Finite Field of size 1009
```

```
>>> from sage.all import *
>>> FiniteField(Integer(3))
Finite Field of size 3

>>> FiniteField(next_prime(Integer(1000)))                              ␣
↪     # needs sage.rings.finite_rings
Finite Field of size 1009
```

characteristic()

Return the characteristic of code{self}.

EXAMPLES:

```
sage: k = GF(7)
sage: k.characteristic()
7
```

```
>>> from sage.all import *
>>> k = GF(Integer(7))
>>> k.characteristic()
7
```

construction()

Return the construction of this finite field (for use by `sage.categories.pushout`).

EXAMPLES:

```
sage: GF(3).construction()
(QuotientFunctor, Integer Ring)
```

```
>>> from sage.all import *
>>> GF(Integer(3)).construction()
(QuotientFunctor, Integer Ring)
```

degree()

Return the degree of `self` over its prime field.

This always returns 1.

EXAMPLES:

```
sage: FiniteField(3).degree()
1
```

```
>>> from sage.all import *
>>> FiniteField(Integer(3)).degree()
1
```

gen(*n=0*)

Return a generator of `self` over its prime field, which is a root of `self.modulus()`.

Unless a custom modulus was given when constructing this prime field, this returns 1.

INPUT:

- `n` – must be 0

OUTPUT:

An element a of `self` such that `self.modulus()(a) == 0`.

> **Warning**
>
> This generator is not guaranteed to be a generator for the multiplicative group. To obtain the latter, use `multiplicative_generator()` or use the `modulus="primitive"` option when constructing the field.

EXAMPLES:

```
sage: k = GF(13)
sage: k.gen()
1
```

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```
sage: # needs sage.rings.finite_rings
sage: k = GF(1009, modulus='primitive')
sage: k.gen()  # this gives a primitive element
11
sage: k.gen(1)
Traceback (most recent call last):
...
IndexError: only one generator
```

```
>>> from sage.all import *
>>> k = GF(Integer(13))
>>> k.gen()
1

>>> # needs sage.rings.finite_rings
>>> k = GF(Integer(1009), modulus='primitive')
>>> k.gen()  # this gives a primitive element
11
>>> k.gen(Integer(1))
Traceback (most recent call last):
...
IndexError: only one generator
```

gens()

Return a tuple containing the generator of `self`.

> **Warning**
>
> The generator is not guaranteed to be a generator for the multiplicative group. To obtain the latter, use `multiplicative_generator()` or use the `modulus="primitive"` option when constructing the field.

EXAMPLES:

```
sage: k = GF(1009, modulus='primitive')
sage: k.gens()
(11,)

sage: k = GF(1009)
sage: k.gens()
(1,)
```

```
>>> from sage.all import *
>>> k = GF(Integer(1009), modulus='primitive')
>>> k.gens()
(11,)

>>> k = GF(Integer(1009))
>>> k.gens()
(1,)
```

is_prime_field()

Return `True` since this is a prime field.

EXAMPLES:

```
sage: k.<a> = GF(3)
sage: k.is_prime_field()
True

sage: # needs sage.rings.finite_rings
sage: k.<a> = GF(3^2)
sage: k.is_prime_field()
False
```

```
>>> from sage.all import *
>>> k = GF(Integer(3), names=('a',)); (a,) = k._first_ngens(1)
>>> k.is_prime_field()
True

>>> # needs sage.rings.finite_rings
>>> k = GF(Integer(3)**Integer(2), names=('a',)); (a,) = k._first_ngens(1)
>>> k.is_prime_field()
False
```

order()

Return the order of this finite field.

EXAMPLES:

```
sage: k = GF(5)
sage: k.order()
5
```

```
>>> from sage.all import *
>>> k = GF(Integer(5))
>>> k.order()
5
```

polynomial(*name=None*)

Return the polynomial `name`.

EXAMPLES:

```
sage: k.<a> = GF(3)
sage: k.polynomial()
x
```

```
>>> from sage.all import *
>>> k = GF(Integer(3), names=('a',)); (a,) = k._first_ngens(1)
>>> k.polynomial()
x
```

3.2 Finite field morphisms for prime fields

Special implementation for prime finite field of:

- embeddings of such field into general finite fields
- Frobenius endomorphisms (= identity with our assumptions)

See also

sage.rings.finite_rings.hom_finite_field

AUTHOR:

- Xavier Caruso (2012-06-29)

class sage.rings.finite_rings.hom_prime_finite_field.**FiniteFieldHomomorphism_prime**

Bases: *FiniteFieldHomomorphism_generic*

A class implementing embeddings of prime finite fields into general finite fields.

class sage.rings.finite_rings.hom_prime_finite_field.**FrobeniusEndomorphism_prime**

Bases: *FrobeniusEndomorphism_finite_field*

A class implementing Frobenius endomorphism on prime finite fields (i.e. identity map :-).

fixed_field()

Return the fixed field of `self`.

OUTPUT:

- a tuple (K, e), where K is the subfield of the domain consisting of elements fixed by `self` and e is an embedding of K into the domain.

Note

Since here the domain is a prime field, the subfield is the same prime field and the embedding is necessarily the identity map.

EXAMPLES:

```
sage: k.<t> = GF(5)
sage: f = k.frobenius_endomorphism(2); f
Identity endomorphism of Finite Field of size 5
sage: kfixed, embed = f.fixed_field()

sage: kfixed == k
True
sage: [ embed(x) == x for x in kfixed ]
[True, True, True, True, True]
```

```
>>> from sage.all import *
>>> k = GF(Integer(5), names=('t',)); (t,) = k._first_ngens(1)
>>> f = k.frobenius_endomorphism(Integer(2)); f
Identity endomorphism of Finite Field of size 5
```

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```
>>> kfixed, embed = f.fixed_field()

>>> kfixed == k
True
>>> [ embed(x) == x for x in kfixed ]
[True, True, True, True, True]
```

class sage.rings.finite_rings.hom_prime_finite_field.**SectionFiniteFieldHomomorphism_prime**

Bases: *SectionFiniteFieldHomomorphism_generic*

CHAPTER

FOUR

FINITE FIELDS USING PARI

4.1 Finite fields implemented via PARI's FFELT type

AUTHORS:

- Peter Bruin (June 2013): initial version, based on finite_field_ext_pari.py by William Stein et al.

class sage.rings.finite_rings.finite_field_pari_ffelt.**FiniteField_pari_ffelt**(*p*, *modulus*, *name=None*)

Bases: *FiniteField*

Finite fields whose cardinality is a prime power (not a prime), implemented using PARI's `FFELT` type.

INPUT:

- `p` – prime number
- `modulus` – an irreducible polynomial of degree at least 2 over the field of p elements
- `name` – string: name of the distinguished generator (default: variable name of `modulus`)

OUTPUT:

A finite field of order $q = p^n$, generated by a distinguished element with minimal polynomial `modulus`. Elements are represented as polynomials in `name` of degree less than n.

> **Note**
>
> Direct construction of *FiniteField_pari_ffelt* objects requires specifying a characteristic and a modulus. To construct a finite field by specifying a cardinality and an algorithm for finding an irreducible polynomial, use the `FiniteField` constructor with `impl='pari_ffelt'`.

EXAMPLES:

Some computations with a finite field of order 9:

```
sage: k = FiniteField(9, 'a', impl='pari_ffelt')
sage: k
Finite Field in a of size 3^2
sage: k.is_field()
True
sage: k.characteristic()
3
sage: a = k.gen()
sage: a
```

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```
a
sage: a.parent()
Finite Field in a of size 3^2
sage: a.charpoly('x')
x^2 + 2*x + 2
sage: [a^i for i in range(8)]
[1, a, a + 1, 2*a + 1, 2, 2*a, 2*a + 2, a + 2]
sage: TestSuite(k).run()
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(9), 'a', impl='pari_ffelt')
>>> k
Finite Field in a of size 3^2
>>> k.is_field()
True
>>> k.characteristic()
3
>>> a = k.gen()
>>> a
a
>>> a.parent()
Finite Field in a of size 3^2
>>> a.charpoly('x')
x^2 + 2*x + 2
>>> [a**i for i in range(Integer(8))]
[1, a, a + 1, 2*a + 1, 2, 2*a, 2*a + 2, a + 2]
>>> TestSuite(k).run()
```

Next we compute with a finite field of order 16:

```
sage: k16 = FiniteField(16, 'b', impl='pari_ffelt')
sage: z = k16.gen()
sage: z
b
sage: z.charpoly('x')
x^4 + x + 1
sage: k16.is_field()
True
sage: k16.characteristic()
2
sage: z.multiplicative_order()
15
```

```
>>> from sage.all import *
>>> k16 = FiniteField(Integer(16), 'b', impl='pari_ffelt')
>>> z = k16.gen()
>>> z
b
>>> z.charpoly('x')
x^4 + x + 1
>>> k16.is_field()
True
```

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```
>>> k16.characteristic()
2
>>> z.multiplicative_order()
15
```

Illustration of dumping and loading:

```
sage: K = FiniteField(7^10, 'b', impl='pari_ffelt')
sage: loads(K.dumps()) == K
True

sage: K = FiniteField(10007^10, 'a', impl='pari_ffelt')
sage: loads(K.dumps()) == K
True
```

```
>>> from sage.all import *
>>> K = FiniteField(Integer(7)**Integer(10), 'b', impl='pari_ffelt')
>>> loads(K.dumps()) == K
True

>>> K = FiniteField(Integer(10007)**Integer(10), 'a', impl='pari_ffelt')
>>> loads(K.dumps()) == K
True
```

Element

alias of *FiniteFieldElement_pari_ffelt*

characteristic()

Return the characteristic of `self`.

EXAMPLES:

```
sage: F = FiniteField(3^4, 'a', impl='pari_ffelt')
sage: F.characteristic()
3
```

```
>>> from sage.all import *
>>> F = FiniteField(Integer(3)**Integer(4), 'a', impl='pari_ffelt')
>>> F.characteristic()
3
```

degree()

Return the degree of `self` over its prime field.

EXAMPLES:

```
sage: F = FiniteField(3^20, 'a', impl='pari_ffelt')
sage: F.degree()
20
```

```
>>> from sage.all import *
>>> F = FiniteField(Integer(3)**Integer(20), 'a', impl='pari_ffelt')
```

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```
>>> F.degree()
20
```

gen(*n=0*)

Return a generator of `self` over its prime field, which is a root of `self.modulus()`.

INPUT:

- `n` – must be 0

OUTPUT:

An element a of `self` such that `self.modulus()(a) == 0`.

> **Warning**
>
> This generator is not guaranteed to be a generator for the multiplicative group. To obtain the latter, use `multiplicative_generator()` or use the `modulus="primitive"` option when constructing the field.

EXAMPLES:

```
sage: R.<x> = PolynomialRing(GF(2))
sage: FiniteField(2^4, 'b', impl='pari_ffelt').gen()
b
sage: k = FiniteField(3^4, 'alpha', impl='pari_ffelt')
sage: a = k.gen()
sage: a
alpha
sage: a^4
alpha^3 + 1
```

```
>>> from sage.all import *
>>> R = PolynomialRing(GF(Integer(2)), names=('x',)); (x,) = R._first_ngens(1)
>>> FiniteField(Integer(2)**Integer(4), 'b', impl='pari_ffelt').gen()
b
>>> k = FiniteField(Integer(3)**Integer(4), 'alpha', impl='pari_ffelt')
>>> a = k.gen()
>>> a
alpha
>>> a**Integer(4)
alpha^3 + 1
```

4.2 Finite field elements implemented via PARI's FFELT type

AUTHORS:

- Peter Bruin (June 2013): initial version, based on element_ext_pari.py by William Stein et al. and element_ntl_gf2e.pyx by Martin Albrecht.

class sage.rings.finite_rings.element_pari_ffelt.**FiniteFieldElement_pari_ffelt**

Bases: *FinitePolyExtElement*

An element of a finite field implemented using PARI.

EXAMPLES:

```
sage: K = FiniteField(10007^10, 'a', impl='pari_ffelt')
sage: a = K.gen(); a
a
sage: type(a)
<class 'sage.rings.finite_rings.element_pari_ffelt.FiniteFieldElement_pari_ffelt'>
```

```
>>> from sage.all import *
>>> K = FiniteField(Integer(10007)**Integer(10), 'a', impl='pari_ffelt')
>>> a = K.gen(); a
a
>>> type(a)
<class 'sage.rings.finite_rings.element_pari_ffelt.FiniteFieldElement_pari_ffelt'>
```

charpoly(*var='x'*)

Return the characteristic polynomial of `self`.

INPUT:

- `var` – string (default: `'x'`); variable name to use

EXAMPLES:

```
sage: R.<x> = PolynomialRing(FiniteField(3))
sage: F.<a> = FiniteField(3^2, modulus=x^2 + 1, impl='pari_ffelt')
sage: a.charpoly('y')
y^2 + 1
```

```
>>> from sage.all import *
>>> R = PolynomialRing(FiniteField(Integer(3)), names=('x',)); (x,) = R._
↪first_ngens(1)
>>> F = FiniteField(Integer(3)**Integer(2), modulus=x**Integer(2) +␣
↪Integer(1), impl='pari_ffelt', names=('a',)); (a,) = F._first_ngens(1)
>>> a.charpoly('y')
y^2 + 1
```

frobenius(*k=1*)

Return the (p^k)-th power of `self`, where p is the characteristic of the field.

INPUT:

- `k` – integer (default: 1); must fit in a C `int`

Note that if k is negative, then this computes the appropriate root.

is_one()

Return `True` if `self` equals 1.

EXAMPLES:

```
sage: F.<a> = FiniteField(5^3, impl='pari_ffelt')
sage: a.is_one()
False
sage: (a/a).is_one()
True
```

```
>>> from sage.all import *
>>> F = FiniteField(Integer(5)**Integer(3), impl='pari_ffelt', names=('a',));␣
↪(a,) = F._first_ngens(1)
>>> a.is_one()
False
>>> (a/a).is_one()
True
```

is_square()

Return `True` if and only if `self` is a square in the finite field.

EXAMPLES:

```
sage: k.<a> = FiniteField(3^2, impl='pari_ffelt')
sage: a.is_square()
False
sage: (a**2).is_square()
True

sage: k.<a> = FiniteField(2^2, impl='pari_ffelt')
sage: (a**2).is_square()
True

sage: k.<a> = FiniteField(17^5, impl='pari_ffelt')
sage: (a**2).is_square()
True
sage: a.is_square()
False
sage: k(0).is_square()
True
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(3)**Integer(2), impl='pari_ffelt', names=('a',));␣
↪(a,) = k._first_ngens(1)
>>> a.is_square()
False
>>> (a**Integer(2)).is_square()
True

>>> k = FiniteField(Integer(2)**Integer(2), impl='pari_ffelt', names=('a',));␣
↪(a,) = k._first_ngens(1)
>>> (a**Integer(2)).is_square()
True

>>> k = FiniteField(Integer(17)**Integer(5), impl='pari_ffelt', names=('a',));
↪ (a,) = k._first_ngens(1)
>>> (a**Integer(2)).is_square()
True
>>> a.is_square()
False
>>> k(Integer(0)).is_square()
True
```

is_unit()

Return `True` if `self` is nonzero.

EXAMPLES:

```
sage: F.<a> = FiniteField(5^3, impl='pari_ffelt')
sage: a.is_unit()
True
```

```
>>> from sage.all import *
>>> F = FiniteField(Integer(5)**Integer(3), impl='pari_ffelt', names=('a',));␣
↪(a,) = F._first_ngens(1)
>>> a.is_unit()
True
```

is_zero()

Return `True` if `self` equals 0.

EXAMPLES:

```
sage: F.<a> = FiniteField(5^3, impl='pari_ffelt')
sage: a.is_zero()
False
sage: (a - a).is_zero()
True
```

```
>>> from sage.all import *
>>> F = FiniteField(Integer(5)**Integer(3), impl='pari_ffelt', names=('a',));␣
↪(a,) = F._first_ngens(1)
>>> a.is_zero()
False
>>> (a - a).is_zero()
True
```

lift()

If `self` is an element of the prime field, return a lift of this element to an integer.

EXAMPLES:

```
sage: k = FiniteField(next_prime(10^10)^2, 'u', impl='pari_ffelt')
sage: a = k(17)/k(19)
sage: b = a.lift(); b
7894736858
sage: b.parent()
Integer Ring
```

```
>>> from sage.all import *
>>> k = FiniteField(next_prime(Integer(10)**Integer(10))**Integer(2), 'u',␣
↪impl='pari_ffelt')
>>> a = k(Integer(17))/k(Integer(19))
>>> b = a.lift(); b
7894736858
>>> b.parent()
Integer Ring
```

log(*base*, *order=None*, *check=False*)

Return a discrete logarithm of `self` with respect to the given base.

INPUT:

- `base` – nonzero field element
- `order` – integer (optional), the order of the base
- `check` – boolean (default: `False`); if set, test whether the given `order` is correct

OUTPUT:

An integer x such that `self` equals `base` raised to the power x. If no such x exists, a `ValueError` is raised.

EXAMPLES:

```
sage: F.<g> = FiniteField(2^10, impl='pari_ffelt')
sage: b = g; a = g^37
sage: a.log(b)
37
sage: b^37; a
g^8 + g^7 + g^4 + g + 1
g^8 + g^7 + g^4 + g + 1
```

```
>>> from sage.all import *
>>> F = FiniteField(Integer(2)**Integer(10), impl='pari_ffelt', names=('g',));
↪ (g,) = F._first_ngens(1)
>>> b = g; a = g**Integer(37)
>>> a.log(b)
37
>>> b**Integer(37); a
g^8 + g^7 + g^4 + g + 1
g^8 + g^7 + g^4 + g + 1
```

```
sage: F.<a> = FiniteField(5^2, impl='pari_ffelt')
sage: F(-1).log(F(2))
2
sage: F(1).log(a)
0
```

```
>>> from sage.all import *
>>> F = FiniteField(Integer(5)**Integer(2), impl='pari_ffelt', names=('a',));␣
↪(a,) = F._first_ngens(1)
>>> F(-Integer(1)).log(F(Integer(2)))
2
>>> F(Integer(1)).log(a)
0
```

```
sage: p = 2^127-1
sage: F.<t> = GF((p, 3))
sage: elt = F.random_element()^(p^2+p+1)
sage: (elt^2).log(elt, p-1)
2
```

```
>>> from sage.all import *
>>> p = Integer(2)**Integer(127)-Integer(1)
>>> F = GF((p, Integer(3)), names=('t',)); (t,) = F._first_ngens(1)
>>> elt = F.random_element()**(p**Integer(2)+p+Integer(1))
>>> (elt**Integer(2)).log(elt, p-Integer(1))
2
```

Passing the `order` argument can lead to huge speedups when factoring the order of the entire unit group is expensive but the order of the base element is much smaller:

```
sage: %timeit (elt^2).log(elt)       # not tested
6.18 s ± 85 ms per loop (mean ± std. dev. of 7 runs, 1 loop each)
sage: %timeit (elt^2).log(elt, p-1)  # not tested
147 ms ± 1.39 ms per loop (mean ± std. dev. of 7 runs, 10 loops each)
```

```
>>> from sage.all import *
>>> %timeit (elt**Integer(2)).log(elt)       # not tested
6.18 s ± 85 ms per loop (mean ± std. dev. of 7 runs, 1 loop each)
>>> %timeit (elt**Integer(2)).log(elt, p-Integer(1))  # not tested
147 ms ± 1.39 ms per loop (mean ± std. dev. of 7 runs, 10 loops each)
```

Some cases where the logarithm is not defined or does not exist:

```
sage: F.<a> = GF(3^10, impl='pari_ffelt')
sage: a.log(-1)
Traceback (most recent call last):
...
ArithmeticError: element a does not lie in group generated by 2
sage: a.log(0)
Traceback (most recent call last):
...
ArithmeticError: discrete logarithm with base 0 is not defined
sage: F(0).log(1)
Traceback (most recent call last):
...
ArithmeticError: discrete logarithm of 0 is not defined
```

```
>>> from sage.all import *
>>> F = GF(Integer(3)**Integer(10), impl='pari_ffelt', names=('a',)); (a,) =␣
↪F._first_ngens(1)
>>> a.log(-Integer(1))
Traceback (most recent call last):
...
ArithmeticError: element a does not lie in group generated by 2
>>> a.log(Integer(0))
Traceback (most recent call last):
...
ArithmeticError: discrete logarithm with base 0 is not defined
>>> F(Integer(0)).log(Integer(1))
Traceback (most recent call last):
...
ArithmeticError: discrete logarithm of 0 is not defined
```

minpoly(*var='x'*)

Return the minimal polynomial of `self`.

INPUT:

- `var` – string (default: `'x'`); variable name to use

EXAMPLES:

```
sage: R.<x> = PolynomialRing(FiniteField(3))
sage: F.<a> = FiniteField(3^2, modulus=x^2 + 1, impl='pari_ffelt')
sage: a.minpoly('y')
y^2 + 1
```

```
>>> from sage.all import *
>>> R = PolynomialRing(FiniteField(Integer(3)), names=('x',)); (x,) = R._
↪first_ngens(1)
>>> F = FiniteField(Integer(3)**Integer(2), modulus=x**Integer(2) +␣
↪Integer(1), impl='pari_ffelt', names=('a',)); (a,) = F._first_ngens(1)
>>> a.minpoly('y')
y^2 + 1
```

multiplicative_order()

Return the order of `self` in the multiplicative group.

EXAMPLES:

```
sage: a = FiniteField(5^3, 'a', impl='pari_ffelt').0
sage: a.multiplicative_order()
124
sage: a**124
1
```

```
>>> from sage.all import *
>>> a = FiniteField(Integer(5)**Integer(3), 'a', impl='pari_ffelt').gen(0)
>>> a.multiplicative_order()
124
>>> a**Integer(124)
1
```

polynomial(*name=None*)

Return the unique representative of `self` as a polynomial over the prime field whose degree is less than the degree of the finite field over its prime field.

INPUT:

- `name` – (optional) variable name

EXAMPLES:

```
sage: k.<a> = FiniteField(3^2, impl='pari_ffelt')
sage: pol = a.polynomial()
sage: pol
a
sage: parent(pol)
Univariate Polynomial Ring in a over Finite Field of size 3
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(3)**Integer(2), impl='pari_ffelt', names=('a',));␣
↪(a,) = k._first_ngens(1)
>>> pol = a.polynomial()
>>> pol
a
>>> parent(pol)
Univariate Polynomial Ring in a over Finite Field of size 3
```

```
sage: k = FiniteField(3^4, 'alpha', impl='pari_ffelt')
sage: a = k.gen()
sage: a.polynomial()
alpha
sage: (a**2 + 1).polynomial('beta')
beta^2 + 1
sage: (a**2 + 1).polynomial().parent()
Univariate Polynomial Ring in alpha over Finite Field of size 3
sage: (a**2 + 1).polynomial('beta').parent()
Univariate Polynomial Ring in beta over Finite Field of size 3
```

```
>>> from sage.all import *
>>> k = FiniteField(Integer(3)**Integer(4), 'alpha', impl='pari_ffelt')
>>> a = k.gen()
>>> a.polynomial()
alpha
>>> (a**Integer(2) + Integer(1)).polynomial('beta')
beta^2 + 1
>>> (a**Integer(2) + Integer(1)).polynomial().parent()
Univariate Polynomial Ring in alpha over Finite Field of size 3
>>> (a**Integer(2) + Integer(1)).polynomial('beta').parent()
Univariate Polynomial Ring in beta over Finite Field of size 3
```

pth_power(*k=1*)

Return the (p^k)-th power of `self`, where p is the characteristic of the field.

INPUT:

- `k` – integer (default: 1); must fit in a C `int`

Note that if k is negative, then this computes the appropriate root.

sqrt(*extend=False*, *all=False*)

Return a square root of `self`, if it exists.

INPUT:

- `extend` – boolean (default: `False`)

> **Warning**
>
> This option is not implemented.

- `all` – boolean (default: `False`)

OUTPUT:

A square root of `self`, if it exists. If `all` is `True`, a list containing all square roots of `self` (of length zero, one or two) is returned instead.

If `extend` is `True`, a square root is chosen in an extension field if necessary. If `extend` is `False`, a `ValueError` is raised if the element is not a square in the base field.

> **Warning**
>
> The `extend` option is not implemented (yet).

EXAMPLES:

```
sage: F = FiniteField(7^2, 'a', impl='pari_ffelt')
sage: F(2).sqrt()
4
sage: F(3).sqrt() in (2*F.gen() + 6, 5*F.gen() + 1)
True
sage: F(3).sqrt()**2
3
sage: F(4).sqrt(all=True)
[2, 5]

sage: K = FiniteField(7^3, 'alpha', impl='pari_ffelt')
sage: K(3).sqrt()
Traceback (most recent call last):
...
ValueError: element is not a square
sage: K(3).sqrt(all=True)
[]

sage: K.<a> = GF(3^17, impl='pari_ffelt')
sage: (a^3 - a - 1).sqrt()
a^16 + 2*a^15 + a^13 + 2*a^12 + a^10 + 2*a^9 + 2*a^8 + a^7 + a^6 + 2*a^5 + a^
↪4 + 2*a^2 + 2*a + 2
```

```
>>> from sage.all import *
>>> F = FiniteField(Integer(7)**Integer(2), 'a', impl='pari_ffelt')
>>> F(Integer(2)).sqrt()
4
>>> F(Integer(3)).sqrt() in (Integer(2)*F.gen() + Integer(6), Integer(5)*F.
↪gen() + Integer(1))
True
>>> F(Integer(3)).sqrt()**Integer(2)
3
>>> F(Integer(4)).sqrt(all=True)
[2, 5]

>>> K = FiniteField(Integer(7)**Integer(3), 'alpha', impl='pari_ffelt')
>>> K(Integer(3)).sqrt()
Traceback (most recent call last):
...
ValueError: element is not a square
```

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```
>>> K(Integer(3)).sqrt(all=True)
[]

>>> K = GF(Integer(3)**Integer(17), impl='pari_ffelt', names=('a',)); (a,) =␣
↪K._first_ngens(1)
>>> (a**Integer(3) - a - Integer(1)).sqrt()
a^16 + 2*a^15 + a^13 + 2*a^12 + a^10 + 2*a^9 + 2*a^8 + a^7 + a^6 + 2*a^5 + a^
↪4 + 2*a^2 + 2*a + 2
```

sage.rings.finite_rings.element_pari_ffelt.**unpickle_FiniteFieldElement_pari_ffelt**(*parent*, *elem*)

EXAMPLES:

```
sage: # needs sage.modules
sage: k.<a> = GF(2^20, impl='pari_ffelt')
sage: e = k.random_element()
sage: f = loads(dumps(e)) # indirect doctest
sage: e == f
True
```

```
>>> from sage.all import *
>>> # needs sage.modules
>>> k = GF(Integer(2)**Integer(20), impl='pari_ffelt', names=('a',)); (a,) = k._
↪first_ngens(1)
>>> e = k.random_element()
>>> f = loads(dumps(e)) # indirect doctest
>>> e == f
True
```

CHAPTER

FIVE

FINITE FIELDS USING GIVARO

5.1 Givaro finite fields

Finite fields that are implemented using Zech logs and the cardinality must be less than 2^{16}. By default, Conway polynomials are used as minimal polynomial.

class sage.rings.finite_rings.finite_field_givaro.**FiniteField_givaro**(*q*, *name='a'*, *modulus=None*, *repr='poly'*, *cache=False*)

Bases: *FiniteField*

Finite field implemented using Zech logs and the cardinality must be less than 2^{16}. By default, Conway polynomials are used as minimal polynomials.

INPUT:

- `q` – p^n (must be prime power)
- `name` – (default: `'a'`) variable used for `poly_repr()`
- `modulus` – a minimal polynomial to use for reduction
- `repr` – (default: `'poly'`) controls the way elements are printed to the user:
  - 'log': repr is `log_repr()`
  - 'int': repr is `int_repr()`
  - 'poly': repr is `poly_repr()`
- `cache` – boolean (default: `False`); if `True` a cache of all elements of this field is created. Thus, arithmetic does not create new elements which speeds calculations up. Also, if many elements are needed during a calculation this cache reduces the memory requirement as at most *order()* elements are created.

OUTPUT: Givaro finite field with characteristic p and cardinality p^n

EXAMPLES:

By default, Conway polynomials are used for extension fields:

```
sage: k.<a> = GF(2**8)
sage: -a ^ k.degree()
a^4 + a^3 + a^2 + 1
sage: f = k.modulus(); f
x^8 + x^4 + x^3 + x^2 + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8), names=('a',)); (a,) = k._first_ngens(1)
>>> -a ** k.degree()
a^4 + a^3 + a^2 + 1
>>> f = k.modulus(); f
x^8 + x^4 + x^3 + x^2 + 1
```

You may enforce a modulus:

```
sage: P.<x> = PolynomialRing(GF(2))
sage: f = x^8 + x^4 + x^3 + x + 1 # Rijndael Polynomial
sage: k.<a> = GF(2^8, modulus=f)
sage: k.modulus()
x^8 + x^4 + x^3 + x + 1
sage: a^(2^8)
a
```

```
>>> from sage.all import *
>>> P = PolynomialRing(GF(Integer(2)), names=('x',)); (x,) = P._first_ngens(1)
>>> f = x**Integer(8) + x**Integer(4) + x**Integer(3) + x + Integer(1) # Rijndael␣
↪Polynomial
>>> k = GF(Integer(2)**Integer(8), modulus=f, names=('a',)); (a,) = k._first_
↪ngens(1)
>>> k.modulus()
x^8 + x^4 + x^3 + x + 1
>>> a**(Integer(2)**Integer(8))
a
```

You may enforce a random modulus:

```
sage: k = GF(3**5, 'a', modulus='random')
sage: k.modulus() # random polynomial
x^5 + 2*x^4 + 2*x^3 + x^2 + 2
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(5), 'a', modulus='random')
>>> k.modulus() # random polynomial
x^5 + 2*x^4 + 2*x^3 + x^2 + 2
```

Three different representations are possible:

```
sage: FiniteField(9, 'a', impl='givaro', repr='poly').gen()
a
sage: FiniteField(9, 'a', impl='givaro', repr='int').gen()
3
sage: FiniteField(9, 'a', impl='givaro', repr='log').gen()
1
```

```
>>> from sage.all import *
>>> FiniteField(Integer(9), 'a', impl='givaro', repr='poly').gen()
a
>>> FiniteField(Integer(9), 'a', impl='givaro', repr='int').gen()
```

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```
3
>>> FiniteField(Integer(9), 'a', impl='givaro', repr='log').gen()
1
```

For prime fields, the default modulus is the polynomial $x - 1$, but you can ask for a different modulus:

```
sage: GF(1009, impl='givaro').modulus()
x + 1008
sage: GF(1009, impl='givaro', modulus='conway').modulus()
x + 998
```

```
>>> from sage.all import *
>>> GF(Integer(1009), impl='givaro').modulus()
x + 1008
>>> GF(Integer(1009), impl='givaro', modulus='conway').modulus()
x + 998
```

a_times_b_minus_c(*a*, *b*, *c*)

Return `a*b - c`.

INPUT:

- a, b, c – *FiniteField_givaroElement*

EXAMPLES:

```
sage: k.<a> = GF(3**3)
sage: k.a_times_b_minus_c(a,a,k(1))
a^2 + 2
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(3), names=('a',)); (a,) = k._first_ngens(1)
>>> k.a_times_b_minus_c(a,a,k(Integer(1)))
a^2 + 2
```

a_times_b_plus_c(*a*, *b*, *c*)

Return `a*b + c`. This is faster than multiplying `a` and `b` first and adding `c` to the result.

INPUT:

- a, b, c – *FiniteField_givaroElement*

EXAMPLES:

```
sage: k.<a> = GF(2**8)
sage: k.a_times_b_plus_c(a,a,k(1))
a^2 + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8), names=('a',)); (a,) = k._first_ngens(1)
>>> k.a_times_b_plus_c(a,a,k(Integer(1)))
a^2 + 1
```

c_minus_a_times_b(*a*, *b*, *c*)

Return `c - a*b`.

INPUT:

- a, b, c – *FiniteField_givaroElement*

EXAMPLES:

```
sage: k.<a> = GF(3**3)
sage: k.c_minus_a_times_b(a,a,k(1))
2*a^2 + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(3), names=('a',)); (a,) = k._first_ngens(1)
>>> k.c_minus_a_times_b(a,a,k(Integer(1)))
2*a^2 + 1
```

characteristic()

Return the characteristic of this field.

EXAMPLES:

```
sage: p = GF(19^5,'a').characteristic(); p
19
sage: type(p)
<class 'sage.rings.integer.Integer'>
```

```
>>> from sage.all import *
>>> p = GF(Integer(19)**Integer(5),'a').characteristic(); p
19
>>> type(p)
<class 'sage.rings.integer.Integer'>
```

degree()

If the cardinality of `self` is p^n, then this returns n.

OUTPUT: integer; the degree

EXAMPLES:

```
sage: GF(3^4,'a').degree()
4
```

```
>>> from sage.all import *
>>> GF(Integer(3)**Integer(4),'a').degree()
4
```

frobenius_endomorphism(*n=1*)

INPUT:

- n – integer (default: 1)

OUTPUT:

The n-th power of the absolute arithmetic Frobenius endomorphism on this finite field.

EXAMPLES:

```
sage: k.<t> = GF(3^5)
sage: Frob = k.frobenius_endomorphism(); Frob
Frobenius endomorphism t |--> t^3 on Finite Field in t of size 3^5

sage: a = k.random_element()
sage: Frob(a) == a^3
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(5), names=('t',)); (t,) = k._first_ngens(1)
>>> Frob = k.frobenius_endomorphism(); Frob
Frobenius endomorphism t |--> t^3 on Finite Field in t of size 3^5

>>> a = k.random_element()
>>> Frob(a) == a**Integer(3)
True
```

We can specify a power:

```
sage: k.frobenius_endomorphism(2)
Frobenius endomorphism t |--> t^(3^2) on Finite Field in t of size 3^5
```

```
>>> from sage.all import *
>>> k.frobenius_endomorphism(Integer(2))
Frobenius endomorphism t |--> t^(3^2) on Finite Field in t of size 3^5
```

The result is simplified if possible:

```
sage: k.frobenius_endomorphism(6)
Frobenius endomorphism t |--> t^3 on Finite Field in t of size 3^5
sage: k.frobenius_endomorphism(5)
Identity endomorphism of Finite Field in t of size 3^5
```

```
>>> from sage.all import *
>>> k.frobenius_endomorphism(Integer(6))
Frobenius endomorphism t |--> t^3 on Finite Field in t of size 3^5
>>> k.frobenius_endomorphism(Integer(5))
Identity endomorphism of Finite Field in t of size 3^5
```

Comparisons work:

```
sage: k.frobenius_endomorphism(6) == Frob
True
sage: from sage.categories.morphism import IdentityMorphism
sage: k.frobenius_endomorphism(5) == IdentityMorphism(k)
True
```

```
>>> from sage.all import *
>>> k.frobenius_endomorphism(Integer(6)) == Frob
True
>>> from sage.categories.morphism import IdentityMorphism
>>> k.frobenius_endomorphism(Integer(5)) == IdentityMorphism(k)
True
```

AUTHOR:

- Xavier Caruso (2012-06-29)

from_integer(*n*)

Given an integer n return a finite field element in `self` which equals n under the condition that *gen()* is set to *characteristic()*.

EXAMPLES:

```
sage: k.<a> = GF(2^8)
sage: k.from_integer(8)
a^3
sage: e = k.from_integer(151); e
a^7 + a^4 + a^2 + a + 1
sage: 2^7 + 2^4 + 2^2 + 2 + 1
151
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8), names=('a',)); (a,) = k._first_ngens(1)
>>> k.from_integer(Integer(8))
a^3
>>> e = k.from_integer(Integer(151)); e
a^7 + a^4 + a^2 + a + 1
>>> Integer(2)**Integer(7) + Integer(2)**Integer(4) + Integer(2)**Integer(2)␣
↪+ Integer(2) + Integer(1)
151
```

gen(*n=0*)

Return a generator of `self` over its prime field, which is a root of `self.modulus()`.

INPUT:

- `n` – must be 0

OUTPUT:

An element a of `self` such that `self.modulus()(a) == 0`.

Warning

This generator is not guaranteed to be a generator for the multiplicative group. To obtain the latter, use `multiplicative_generator()` or use the `modulus="primitive"` option when constructing the field.

EXAMPLES:

```
sage: k = GF(3^4, 'b'); k.gen()
b
sage: k.gen(1)
Traceback (most recent call last):
...
IndexError: only one generator
sage: F = FiniteField(31, impl='givaro')
sage: F.gen()
1
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(4), 'b'); k.gen()
b
>>> k.gen(Integer(1))
Traceback (most recent call last):
...
IndexError: only one generator
>>> F = FiniteField(Integer(31), impl='givaro')
>>> F.gen()
1
```

int_to_log(*n*)

Given an integer n this method returns i where i satisfies $g^i = n \mod p$ where g is the generator and p is the characteristic of `self`.

INPUT:

- `n` – integer representation of a finite field element

OUTPUT: log representation of `n`

EXAMPLES:

```
sage: k = GF(7**3, 'a')
sage: k.int_to_log(4)
228
sage: k.int_to_log(3)
57
sage: k.gen()^57
3
```

```
>>> from sage.all import *
>>> k = GF(Integer(7)**Integer(3), 'a')
>>> k.int_to_log(Integer(4))
228
>>> k.int_to_log(Integer(3))
57
>>> k.gen()**Integer(57)
3
```

log_to_int(*n*)

Given an integer n this method returns `i` where `i` satisfies $g^n = i$ where g is the generator of `self`; the result is interpreted as an integer.

INPUT:

- `n` – log representation of a finite field element

OUTPUT: integer representation of a finite field element

EXAMPLES:

```
sage: k = GF(2**8, 'a')
sage: k.log_to_int(4)
16
sage: k.log_to_int(20)
180
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8), 'a')
>>> k.log_to_int(Integer(4))
16
>>> k.log_to_int(Integer(20))
180
```

order()

Return the cardinality of this field.

OUTPUT: integer; the number of elements in `self`

EXAMPLES:

```
sage: n = GF(19^5,'a').order(); n
2476099
sage: type(n)
<class 'sage.rings.integer.Integer'>
```

```
>>> from sage.all import *
>>> n = GF(Integer(19)**Integer(5),'a').order(); n
2476099
>>> type(n)
<class 'sage.rings.integer.Integer'>
```

prime_subfield()

Return the prime subfield $\mathbf{F}_p$ of `self` if `self` is $\mathbf{F}_{p^n}$.

EXAMPLES:

```
sage: GF(3^4, 'b').prime_subfield()
Finite Field of size 3

sage: S.<b> = GF(5^2); S
Finite Field in b of size 5^2
sage: S.prime_subfield()
Finite Field of size 5
sage: type(S.prime_subfield())
<class 'sage.rings.finite_rings.finite_field_prime_modn.FiniteField_prime_
↪modn_with_category'>
```

```
>>> from sage.all import *
>>> GF(Integer(3)**Integer(4), 'b').prime_subfield()
Finite Field of size 3

>>> S = GF(Integer(5)**Integer(2), names=('b',)); (b,) = S._first_ngens(1); S
Finite Field in b of size 5^2
>>> S.prime_subfield()
Finite Field of size 5
>>> type(S.prime_subfield())
<class 'sage.rings.finite_rings.finite_field_prime_modn.FiniteField_prime_
↪modn_with_category'>
```

random_element(**args*, ***kwds*)

Return a random element of `self`.

EXAMPLES:

```
sage: k = GF(23**3, 'a')
sage: e = k.random_element()
sage: e.parent() is k
True
sage: type(e)
<class 'sage.rings.finite_rings.element_givaro.FiniteField_givaroElement'>

sage: P.<x> = PowerSeriesRing(GF(3^3, 'a'))
sage: P.random_element(5).parent() is P
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(23)**Integer(3), 'a')
>>> e = k.random_element()
>>> e.parent() is k
True
>>> type(e)
<class 'sage.rings.finite_rings.element_givaro.FiniteField_givaroElement'>

>>> P = PowerSeriesRing(GF(Integer(3)**Integer(3), 'a'), names=('x',)); (x,)␣
↪= P._first_ngens(1)
>>> P.random_element(Integer(5)).parent() is P
True
```

5.2 Givaro finite field elements

Sage includes the Givaro finite field library, for highly optimized arithmetic in finite fields.

Note

The arithmetic is performed by the Givaro C++ library which uses Zech logs internally to represent finite field elements. This implementation is the default finite extension field implementation in Sage for the cardinality less than 2^{16}, as it is a lot faster than the PARI implementation. Some functionality in this class however is implemented using PARI.

EXAMPLES:

```
sage: k = GF(5); type(k)
<class 'sage.rings.finite_rings.finite_field_prime_modn.FiniteField_prime_modn_with_
↪category'>
sage: k = GF(5^2,'c'); type(k)
<class 'sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro_with_category'>
sage: k = GF(2^16,'c'); type(k)
<class 'sage.rings.finite_rings.finite_field_ntl_gf2e.FiniteField_ntl_gf2e_with_
↪category'>
sage: k = GF(3^16,'c'); type(k)
<class 'sage.rings.finite_rings.finite_field_pari_ffelt.FiniteField_pari_ffelt_with_
```

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```
↪category'>

sage: n = previous_prime_power(2^16 - 1)
sage: while is_prime(n):
....:  n = previous_prime_power(n)
sage: factor(n)
251^2
sage: k = GF(n,'c'); type(k)
<class 'sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro_with_category'>
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)); type(k)
<class 'sage.rings.finite_rings.finite_field_prime_modn.FiniteField_prime_modn_with_
↪category'>
>>> k = GF(Integer(5)**Integer(2),'c'); type(k)
<class 'sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro_with_category'>
>>> k = GF(Integer(2)**Integer(16),'c'); type(k)
<class 'sage.rings.finite_rings.finite_field_ntl_gf2e.FiniteField_ntl_gf2e_with_
↪category'>
>>> k = GF(Integer(3)**Integer(16),'c'); type(k)
<class 'sage.rings.finite_rings.finite_field_pari_ffelt.FiniteField_pari_ffelt_with_
↪category'>

>>> n = previous_prime_power(Integer(2)**Integer(16) - Integer(1))
>>> while is_prime(n):
...  n = previous_prime_power(n)
>>> factor(n)
251^2
>>> k = GF(n,'c'); type(k)
<class 'sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro_with_category'>
```

AUTHORS:

- Martin Albrecht <malb@informatik.uni-bremen.de> (2006-06-05)
- William Stein (2006-12-07): editing, lots of docs, etc.
- Robert Bradshaw (2007-05-23): is_square/sqrt, pow.

class `sage.rings.finite_rings.element_givaro.`**`Cache_givaro`**

Bases: *`Cache_base`*

Finite Field.

These are implemented using Zech logs and the cardinality must be less than 2^{16}. By default Conway polynomials are used as minimal polynomial.

INPUT:

- `q` – p^n (must be prime power)
- `name` – variable used for poly_repr (default: `'a'`)
- `modulus` – a polynomial to use as modulus
- `repr` – (default: `'poly'`) controls the way elements are printed to the user:
  - 'log': repr is `log_repr()`

  - 'int': repr is `int_repr()`
  - 'poly': repr is `poly_repr()`

- `cache` – boolean (default: `False`); if `True` a cache of all elements of this field is created. Thus, arithmetic does not create new elements which speeds calculations up. Also, if many elements are needed during a calculation this cache reduces the memory requirement as at most `order()` elements are created.

OUTPUT: Givaro finite field with characteristic p and cardinality p^n

EXAMPLES:

By default Conway polynomials are used:

```
sage: k.<a> = GF(2**8)
sage: -a ^ k.degree()
a^4 + a^3 + a^2 + 1
sage: f = k.modulus(); f
x^8 + x^4 + x^3 + x^2 + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8), names=('a',)); (a,) = k._first_ngens(1)
>>> -a ** k.degree()
a^4 + a^3 + a^2 + 1
>>> f = k.modulus(); f
x^8 + x^4 + x^3 + x^2 + 1
```

You may enforce a modulus:

```
sage: P.<x> = PolynomialRing(GF(2))
sage: f = x^8 + x^4 + x^3 + x + 1 # Rijndael polynomial
sage: k.<a> = GF(2^8, modulus=f)
sage: k.modulus()
x^8 + x^4 + x^3 + x + 1
sage: a^(2^8)
a
```

```
>>> from sage.all import *
>>> P = PolynomialRing(GF(Integer(2)), names=('x',)); (x,) = P._first_ngens(1)
>>> f = x**Integer(8) + x**Integer(4) + x**Integer(3) + x + Integer(1) # Rijndael␣
↪polynomial
>>> k = GF(Integer(2)**Integer(8), modulus=f, names=('a',)); (a,) = k._first_
↪ngens(1)
>>> k.modulus()
x^8 + x^4 + x^3 + x + 1
>>> a**(Integer(2)**Integer(8))
a
```

You may enforce a random modulus:

```
sage: k = GF(3**5, 'a', modulus='random')
sage: k.modulus() # random polynomial
x^5 + 2*x^4 + 2*x^3 + x^2 + 2
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(5), 'a', modulus='random')
>>> k.modulus() # random polynomial
x^5 + 2*x^4 + 2*x^3 + x^2 + 2
```

For binary fields, you may ask for a minimal weight polynomial:

```
sage: k = GF(2**10, 'a', modulus='minimal_weight')
sage: k.modulus()
x^10 + x^3 + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(10), 'a', modulus='minimal_weight')
>>> k.modulus()
x^10 + x^3 + 1
```

a_times_b_minus_c(*a*, *b*, *c*)

Return `a*b - c`.

INPUT:

- a, b, c – *FiniteField_givaroElement*

EXAMPLES:

```
sage: k.<a> = GF(3**3)
sage: k._cache.a_times_b_minus_c(a,a,k(1))
a^2 + 2
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(3), names=('a',)); (a,) = k._first_ngens(1)
>>> k._cache.a_times_b_minus_c(a,a,k(Integer(1)))
a^2 + 2
```

a_times_b_plus_c(*a*, *b*, *c*)

Return `a*b + c`.

This is faster than multiplying `a` and `b` first and adding `c` to the result.

INPUT:

- a, b, c – *FiniteField_givaroElement*

EXAMPLES:

```
sage: k.<a> = GF(2**8)
sage: k._cache.a_times_b_plus_c(a,a,k(1))
a^2 + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8), names=('a',)); (a,) = k._first_ngens(1)
>>> k._cache.a_times_b_plus_c(a,a,k(Integer(1)))
a^2 + 1
```

c_minus_a_times_b(*a*, *b*, *c*)

Return `c - a*b`.

INPUT:

- a, b, c – *FiniteField_givaroElement*

EXAMPLES:

```
sage: k.<a> = GF(3**3)
sage: k._cache.c_minus_a_times_b(a,a,k(1))
2*a^2 + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(3), names=('a',)); (a,) = k._first_ngens(1)
>>> k._cache.c_minus_a_times_b(a,a,k(Integer(1)))
2*a^2 + 1
```

characteristic()

Return the characteristic of this field.

EXAMPLES:

```
sage: p = GF(19^3,'a')._cache.characteristic(); p
19
```

```
>>> from sage.all import *
>>> p = GF(Integer(19)**Integer(3),'a')._cache.characteristic(); p
19
```

element_from_data(*e*)

Coerces several data types to `self`.

INPUT:

- e – data to coerce in

EXAMPLES:

```
sage: k = GF(3^8, 'a')
sage: type(k)
<class 'sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro_with_
↪category'>
sage: e = k.vector_space(map=False).gen(1); e
(0, 1, 0, 0, 0, 0, 0, 0)
sage: k(e) #indirect doctest
a
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(8), 'a')
>>> type(k)
<class 'sage.rings.finite_rings.finite_field_givaro.FiniteField_givaro_with_
↪category'>
>>> e = k.vector_space(map=False).gen(Integer(1)); e
(0, 1, 0, 0, 0, 0, 0, 0)
```

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```
>>> k(e) #indirect doctest
a
```

exponent()

Return the degree of this field over $\mathbf{F}_p$.

EXAMPLES:

```
sage: K.<a> = GF(9); K._cache.exponent()
2
```

```
>>> from sage.all import *
>>> K = GF(Integer(9), names=('a',)); (a,) = K._first_ngens(1); K._cache.
↪exponent()
2
```

fetch_int(*number*)

Given an integer `n` return a finite field element in `self` which equals `n` under the condition that *gen()* is set to *characteristic()*.

EXAMPLES:

```
sage: k.<a> = GF(2^8)
sage: k._cache.fetch_int(8)
a^3
sage: e = k._cache.fetch_int(151); e
a^7 + a^4 + a^2 + a + 1
sage: 2^7 + 2^4 + 2^2 + 2 + 1
151
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8), names=('a',)); (a,) = k._first_ngens(1)
>>> k._cache.fetch_int(Integer(8))
a^3
>>> e = k._cache.fetch_int(Integer(151)); e
a^7 + a^4 + a^2 + a + 1
>>> Integer(2)**Integer(7) + Integer(2)**Integer(4) + Integer(2)**Integer(2)␣
↪+ Integer(2) + Integer(1)
151
```

gen()

Return a generator of the field.

EXAMPLES:

```
sage: K.<a> = GF(625)
sage: K._cache.gen()
a
```

```
>>> from sage.all import *
>>> K = GF(Integer(625), names=('a',)); (a,) = K._first_ngens(1)
>>> K._cache.gen()
a
```

int_to_log(*n*)

Given an integer n this method returns i where i satisfies $g^i = n \mod p$ where g is the generator and p is the characteristic of `self`.

INPUT:

- `n` – integer representation of a finite field element

OUTPUT: log representation of `n`

EXAMPLES:

```
sage: k = GF(7**3, 'a')
sage: k._cache.int_to_log(4)
228
sage: k._cache.int_to_log(3)
57
sage: k.gen()^57
3
```

```
>>> from sage.all import *
>>> k = GF(Integer(7)**Integer(3), 'a')
>>> k._cache.int_to_log(Integer(4))
228
>>> k._cache.int_to_log(Integer(3))
57
>>> k.gen()**Integer(57)
3
```

log_to_int(*n*)

Given an integer n this method returns i where i satisfies $g^n = i$ where g is the generator of `self`; the result is interpreted as an integer.

INPUT:

- `n` – log representation of a finite field element

OUTPUT: integer representation of a finite field element

EXAMPLES:

```
sage: k = GF(2**8, 'a')
sage: k._cache.log_to_int(4)
16
sage: k._cache.log_to_int(20)
180
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8), 'a')
>>> k._cache.log_to_int(Integer(4))
16
>>> k._cache.log_to_int(Integer(20))
180
```

order()

Return the order of this field.

EXAMPLES:

```
sage: K.<a> = GF(9)
sage: K._cache.order()
9
```

```
>>> from sage.all import *
>>> K = GF(Integer(9), names=('a',)); (a,) = K._first_ngens(1)
>>> K._cache.order()
9
```

order_c()

Return the order of this field.

EXAMPLES:

```
sage: K.<a> = GF(9)
sage: K._cache.order_c()
9
```

```
>>> from sage.all import *
>>> K = GF(Integer(9), names=('a',)); (a,) = K._first_ngens(1)
>>> K._cache.order_c()
9
```

random_element(**args*, ***kwds*)

Return a random element of `self`.

EXAMPLES:

```
sage: k = GF(23**3, 'a')
sage: e = k._cache.random_element()
sage: e.parent() is k
True
sage: type(e)
<class 'sage.rings.finite_rings.element_givaro.FiniteField_givaroElement'>

sage: P.<x> = PowerSeriesRing(GF(3^3, 'a'))
sage: P.random_element(5).parent() is P
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(23)**Integer(3), 'a')
>>> e = k._cache.random_element()
>>> e.parent() is k
True
>>> type(e)
<class 'sage.rings.finite_rings.element_givaro.FiniteField_givaroElement'>

>>> P = PowerSeriesRing(GF(Integer(3)**Integer(3), 'a'), names=('x',)); (x,)␣
↪= P._first_ngens(1)
>>> P.random_element(Integer(5)).parent() is P
True
```

repr

class sage.rings.finite_rings.element_givaro.**FiniteField_givaroElement**

Bases: *FinitePolyExtElement*

An element of a (Givaro) finite field.

is_one()

Return `True` if `self == k(1)`.

EXAMPLES:

```
sage: k.<a> = GF(3^4); k
Finite Field in a of size 3^4
sage: a.is_one()
False
sage: k(1).is_one()
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(4), names=('a',)); (a,) = k._first_ngens(1); k
Finite Field in a of size 3^4
>>> a.is_one()
False
>>> k(Integer(1)).is_one()
True
```

is_square()

Return `True` if `self` is a square in `self.parent()`.

ALGORITHM:

Elements are stored as powers of generators, so we simply check to see if it is an even power of a generator.

EXAMPLES:

```
sage: k.<a> = GF(9); k
Finite Field in a of size 3^2
sage: a.is_square()
False
sage: v = set([x^2 for x in k])
sage: [x.is_square() for x in v]
[True, True, True, True, True]
sage: [x.is_square() for x in k if not x in v]
[False, False, False, False]
```

```
>>> from sage.all import *
>>> k = GF(Integer(9), names=('a',)); (a,) = k._first_ngens(1); k
Finite Field in a of size 3^2
>>> a.is_square()
False
>>> v = set([x**Integer(2) for x in k])
>>> [x.is_square() for x in v]
[True, True, True, True, True]
>>> [x.is_square() for x in k if not x in v]
[False, False, False, False]
```

is_unit()

Return `True` if `self` is nonzero, so it is a unit as an element of the finite field.

EXAMPLES:

```
sage: k.<a> = GF(3^4); k
Finite Field in a of size 3^4
sage: a.is_unit()
True
sage: k(0).is_unit()
False
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(4), names=('a',)); (a,) = k._first_ngens(1); k
Finite Field in a of size 3^4
>>> a.is_unit()
True
>>> k(Integer(0)).is_unit()
False
```

log(*base*, *order=None*, *check=False*)

Return the log to the base b of `self`, i.e., an integer n such that $b^n =$ `self`.

INPUT:

- `base` – non-zero field element
- `order` – integer (optional), multiple of order of `base`
- `check` – boolean (default: `False`): If set, test whether the given `order` is correct.

Warning

TODO – This is currently implemented by solving the discrete log problem – which shouldn't be needed because of how finite field elements are represented.

EXAMPLES:

```
sage: k.<b> = GF(5^2); k
Finite Field in b of size 5^2
sage: a = b^7
sage: a.log(b)
7
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(2), names=('b',)); (b,) = k._first_ngens(1); k
Finite Field in b of size 5^2
>>> a = b**Integer(7)
>>> a.log(b)
7
```

multiplicative_order()

Return the multiplicative order of this field element.

EXAMPLES:

```
sage: S.<b> = GF(5^2); S
Finite Field in b of size 5^2
sage: b.multiplicative_order()
24
sage: (b^6).multiplicative_order()
4
```

```
>>> from sage.all import *
>>> S = GF(Integer(5)**Integer(2), names=('b',)); (b,) = S._first_ngens(1); S
Finite Field in b of size 5^2
>>> b.multiplicative_order()
24
>>> (b**Integer(6)).multiplicative_order()
4
```

polynomial(*name=None*)

Return `self` viewed as a polynomial over `self.parent().prime_subfield()`.

EXAMPLES:

```
sage: k.<b> = GF(5^2); k
Finite Field in b of size 5^2
sage: f = (b^2+1).polynomial(); f
b + 4
sage: type(f)
<class 'sage.rings.polynomial.polynomial_zmod_flint.Polynomial_zmod_flint'>
sage: parent(f)
Univariate Polynomial Ring in b over Finite Field of size 5
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(2), names=('b',)); (b,) = k._first_ngens(1); k
Finite Field in b of size 5^2
>>> f = (b**Integer(2)+Integer(1)).polynomial(); f
b + 4
>>> type(f)
<class 'sage.rings.polynomial.polynomial_zmod_flint.Polynomial_zmod_flint'>
>>> parent(f)
Univariate Polynomial Ring in b over Finite Field of size 5
```

sqrt(*extend=False*, *all=False*)

Return a square root of this finite field element in its parent, if there is one. Otherwise, raise a `ValueError`.

INPUT:

- `extend` – boolean (default: `True`); if `True`, return a square root in an extension ring, if necessary. Otherwise, raise a `ValueError` if the root is not in the base ring.

  Warning

  this option is not implemented!

- `all` – boolean (default: `False`); if `True`, return all square roots of `self`, instead of just one

> **Warning**
>
> The `extend` option is not implemented (yet).

ALGORITHM:

`self` is stored as a^k for some generator a. Return $a^{k/2}$ for even k.

EXAMPLES:

```
sage: k.<a> = GF(7^2)
sage: k(2).sqrt()
3
sage: k(3).sqrt()
2*a + 6
sage: k(3).sqrt()**2
3
sage: k(4).sqrt()
2
sage: k.<a> = GF(7^3)
sage: k(3).sqrt()
Traceback (most recent call last):
...
ValueError: must be a perfect square.
```

```
>>> from sage.all import *
>>> k = GF(Integer(7)**Integer(2), names=('a',)); (a,) = k._first_ngens(1)
>>> k(Integer(2)).sqrt()
3
>>> k(Integer(3)).sqrt()
2*a + 6
>>> k(Integer(3)).sqrt()**Integer(2)
3
>>> k(Integer(4)).sqrt()
2
>>> k = GF(Integer(7)**Integer(3), names=('a',)); (a,) = k._first_ngens(1)
>>> k(Integer(3)).sqrt()
Traceback (most recent call last):
...
ValueError: must be a perfect square.
```

class sage.rings.finite_rings.element_givaro.**FiniteField_givaro_iterator**

Bases: `object`

Iterator over `FiniteField_givaro` elements. We iterate multiplicatively, as powers of a fixed internal generator.

EXAMPLES:

```
sage: for x in GF(2^2,'a'): print(x)
0
a
a + 1
1
```

```
>>> from sage.all import *
>>> for x in GF(Integer(2)**Integer(2),'a'): print(x)
0
a
a + 1
1
```

sage.rings.finite_rings.element_givaro.**unpickle_Cache_givaro**(*parent*, *p*, *k*, *modulus*, *rep*, *cache*)

EXAMPLES:

```
sage: k = GF(3**7, 'a')
sage: loads(dumps(k)) == k # indirect doctest
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(3)**Integer(7), 'a')
>>> loads(dumps(k)) == k # indirect doctest
True
```

sage.rings.finite_rings.element_givaro.**unpickle_FiniteField_givaroElement**(*parent*, *x*)

5.3 Givaro finite field morphisms

Special implementation for givaro finite fields of:

- embeddings between finite fields
- frobenius endomorphisms

SEEALSO:

```
:mod:`sage.rings.finite_rings.hom_finite_field`
```

AUTHOR:

- Xavier Caruso (2012-06-29)

class sage.rings.finite_rings.hom_finite_field_givaro.**FiniteFieldHomomorphism_givaro**

Bases: *FiniteFieldHomomorphism_generic*

class sage.rings.finite_rings.hom_finite_field_givaro.**FrobeniusEndomorphism_givaro**

Bases: *FrobeniusEndomorphism_finite_field*

fixed_field()

Return the fixed field of `self`.

OUTPUT:

- a tuple (K, e), where K is the subfield of the domain consisting of elements fixed by `self` and e is an embedding of K into the domain.

> **Note**
>
> The name of the variable used for the subfield (if it is not a prime subfield) is suffixed by `_fixed`.

EXAMPLES:

```
sage: k.<t> = GF(5^6)
sage: f = k.frobenius_endomorphism(2)
sage: kfixed, embed = f.fixed_field()
sage: kfixed
Finite Field in t_fixed of size 5^2
sage: embed # random
Ring morphism:
  From: Finite Field in t_fixed of size 5^2
  To:   Finite Field in t of size 5^6
  Defn: t_fixed |--> 4*t^5 + 2*t^4 + 4*t^2 + t

sage: tfixed = kfixed.gen()
sage: embed(tfixed) # random
4*t^5 + 2*t^4 + 4*t^2 + t
```

```
>>> from sage.all import *
>>> k = GF(Integer(5)**Integer(6), names=('t',)); (t,) = k._first_ngens(1)
>>> f = k.frobenius_endomorphism(Integer(2))
>>> kfixed, embed = f.fixed_field()
>>> kfixed
Finite Field in t_fixed of size 5^2
>>> embed # random
Ring morphism:
  From: Finite Field in t_fixed of size 5^2
  To:   Finite Field in t of size 5^6
  Defn: t_fixed |--> 4*t^5 + 2*t^4 + 4*t^2 + t

>>> tfixed = kfixed.gen()
>>> embed(tfixed) # random
4*t^5 + 2*t^4 + 4*t^2 + t
```

class sage.rings.finite_rings.hom_finite_field_givaro.**SectionFiniteFieldHomomorphism_givaro**

Bases: *SectionFiniteFieldHomomorphism_generic*

CHAPTER

SIX

FINITE FIELDS OF CHARACTERISTIC 2 USING NTL

6.1 Finite fields of characteristic 2

class sage.rings.finite_rings.finite_field_ntl_gf2e.**FiniteField_ntl_gf2e**(*q*, *names='a'*, *modulus=None*, *repr='poly'*)

Bases: *FiniteField*

Finite Field of characteristic 2 and order 2^n.

INPUT:

- `q` – 2^n (must be 2 power)
- `names` – variable used for poly_repr (default: `'a'`)
- `modulus` – a minimal polynomial to use for reduction
- **`repr` – controls the way elements are printed to the user:** (default: `'poly'`)
  - `'poly'` – polynomial representation

OUTPUT: finite field with characteristic 2 and cardinality 2^n

EXAMPLES:

```
sage: k.<a> = GF(2^16)
sage: type(k)
<class 'sage.rings.finite_rings.finite_field_ntl_gf2e.FiniteField_ntl_gf2e_with_
↪category'>
sage: k.<a> = GF(2^1024)
sage: k.modulus()
x^1024 + x^19 + x^6 + x + 1
sage: set_random_seed(6397)
sage: k.<a> = GF(2^17, modulus='random')
sage: k.modulus()
x^17 + x^16 + x^15 + x^10 + x^8 + x^6 + x^4 + x^3 + x^2 + x + 1
sage: k.modulus().is_irreducible()
True
sage: k.<a> = GF(2^211, modulus='minimal_weight')
sage: k.modulus()
x^211 + x^11 + x^10 + x^8 + 1
sage: k.<a> = GF(2^211, modulus='conway')
```

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```
sage: k.modulus()
x^211 + x^9 + x^6 + x^5 + x^3 + x + 1
sage: k.<a> = GF(2^23, modulus='conway')
sage: a.multiplicative_order() == k.order() - 1
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(16), names=('a',)); (a,) = k._first_ngens(1)
>>> type(k)
<class 'sage.rings.finite_rings.finite_field_ntl_gf2e.FiniteField_ntl_gf2e_with_
↪category'>
>>> k = GF(Integer(2)**Integer(1024), names=('a',)); (a,) = k._first_ngens(1)
>>> k.modulus()
x^1024 + x^19 + x^6 + x + 1
>>> set_random_seed(Integer(6397))
>>> k = GF(Integer(2)**Integer(17), modulus='random', names=('a',)); (a,) = k._
↪first_ngens(1)
>>> k.modulus()
x^17 + x^16 + x^15 + x^10 + x^8 + x^6 + x^4 + x^3 + x^2 + x + 1
>>> k.modulus().is_irreducible()
True
>>> k = GF(Integer(2)**Integer(211), modulus='minimal_weight', names=('a',)); (a,
↪) = k._first_ngens(1)
>>> k.modulus()
x^211 + x^11 + x^10 + x^8 + 1
>>> k = GF(Integer(2)**Integer(211), modulus='conway', names=('a',)); (a,) = k._
↪first_ngens(1)
>>> k.modulus()
x^211 + x^9 + x^6 + x^5 + x^3 + x + 1
>>> k = GF(Integer(2)**Integer(23), modulus='conway', names=('a',)); (a,) = k._
↪first_ngens(1)
>>> a.multiplicative_order() == k.order() - Integer(1)
True
```

characteristic()

Return the characteristic of `self` which is 2.

EXAMPLES:

```
sage: k.<a> = GF(2^16,modulus='random')
sage: k.characteristic()
2
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(16),modulus='random', names=('a',)); (a,) = k._
↪first_ngens(1)
>>> k.characteristic()
2
```

degree()

If this field has cardinality 2^n this method returns n.

EXAMPLES:

```
sage: k.<a> = GF(2^64)
sage: k.degree()
64
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(64), names=('a',)); (a,) = k._first_ngens(1)
>>> k.degree()
64
```

from_integer(*number*)

Given an integer n less than `cardinality()` with base 2 representation $a_0 + 2 \cdot a_1 + \cdots + 2^k a_k$, returns $a_0 + a_1 \cdot x + \cdots + a_k x^k$, where x is the generator of this finite field.

INPUT:

- `number` – integer

EXAMPLES:

```
sage: k.<a> = GF(2^48)
sage: k.from_integer(2^43 + 2^15 + 1)
a^43 + a^15 + 1
sage: k.from_integer(33793)
a^15 + a^10 + 1
sage: 33793.digits(2) # little endian
[1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1]
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(48), names=('a',)); (a,) = k._first_ngens(1)
>>> k.from_integer(Integer(2)**Integer(43) + Integer(2)**Integer(15) +␣
↪Integer(1))
a^43 + a^15 + 1
>>> k.from_integer(Integer(33793))
a^15 + a^10 + 1
>>> Integer(33793).digits(Integer(2)) # little endian
[1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1]
```

gen(*n=0*)

Return a generator of `self` over its prime field, which is a root of `self.modulus()`.

INPUT:

- `n` – must be 0

OUTPUT:

An element a of `self` such that `self.modulus()(a) == 0`.

> **Warning**
>
> This generator is not guaranteed to be a generator for the multiplicative group. To obtain the latter, use `multiplicative_generator()` or use the `modulus="primitive"` option when constructing the field.

EXAMPLES:

```
sage: k.<a> = GF(2^19)
sage: k.gen() == a
True
sage: a
a
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(19), names=('a',)); (a,) = k._first_ngens(1)
>>> k.gen() == a
True
>>> a
a
```

order()

Return the cardinality of this field.

EXAMPLES:

```
sage: k.<a> = GF(2^64)
sage: k.order()
18446744073709551616
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(64), names=('a',)); (a,) = k._first_ngens(1)
>>> k.order()
18446744073709551616
```

prime_subfield()

Return the prime subfield $\mathbf{F}_p$ of `self` if `self` is $\mathbf{F}_{p^n}$.

EXAMPLES:

```
sage: F.<a> = GF(2^16)
sage: F.prime_subfield()
Finite Field of size 2
```

```
>>> from sage.all import *
>>> F = GF(Integer(2)**Integer(16), names=('a',)); (a,) = F._first_ngens(1)
>>> F.prime_subfield()
Finite Field of size 2
```

sage.rings.finite_rings.finite_field_ntl_gf2e.**late_import**()

Imports various modules after startup.

EXAMPLES:

```
sage: sage.rings.finite_rings.finite_field_ntl_gf2e.late_import()
sage: sage.rings.finite_rings.finite_field_ntl_gf2e.GF2 is None # indirect doctest
False
```

```
>>> from sage.all import *
>>> sage.rings.finite_rings.finite_field_ntl_gf2e.late_import()
>>> sage.rings.finite_rings.finite_field_ntl_gf2e.GF2 is None # indirect doctest
False
```

6.2 Elements of finite fields of characteristic 2

This implementation uses NTL's GF2E class to perform the arithmetic and is the standard implementation for `GF(2^n)` for `n >= 16`.

AUTHORS:

- Martin Albrecht <malb@informatik.uni-bremen.de> (2007-10)

class `sage.rings.finite_rings.element_ntl_gf2e.`**`Cache_ntl_gf2e`**

Bases: *`Cache_base`*

This class stores information for an NTL finite field in a Cython class so that elements can access it quickly.

It's modeled on *`NativeIntStruct`*, but includes many functions that were previously included in the parent (see Issue #12062).

degree()

If the field has cardinality 2^n this method returns n.

EXAMPLES:

```
sage: k.<a> = GF(2^64)
sage: k._cache.degree()
64
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(64), names=('a',)); (a,) = k._first_ngens(1)
>>> k._cache.degree()
64
```

fetch_int(*number*)

Given an integer less than p^n with base 2 representation $a_0 + a_1 \cdot 2 + \cdots + a_k 2^k$, this returns $a_0 + a_1 x + \cdots + a_k x^k$, where x is the generator of this finite field.

INPUT:

- `number` – integer; of size less than the cardinality

EXAMPLES:

```
sage: k.<a> = GF(2^48)
sage: k._cache.fetch_int(2^33 + 2 + 1)
a^33 + a + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(48), names=('a',)); (a,) = k._first_ngens(1)
>>> k._cache.fetch_int(Integer(2)**Integer(33) + Integer(2) + Integer(1))
a^33 + a + 1
```

import_data(*e*)

EXAMPLES:

```
sage: k.<a> = GF(2^17)
sage: V = k.vector_space(map=False)
sage: v = [1,0,0,0,0,1,0,0,1,0,0,0,0,1,0,0,0]
sage: k._cache.import_data(v)
```

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```
a^13 + a^8 + a^5 + 1
sage: k._cache.import_data(V(v))
a^13 + a^8 + a^5 + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(17), names=('a',)); (a,) = k._first_ngens(1)
>>> V = k.vector_space(map=False)
>>> v = [Integer(1),Integer(0),Integer(0),Integer(0),Integer(0),Integer(1),
↪Integer(0),Integer(0),Integer(1),Integer(0),Integer(0),Integer(0),
↪Integer(0),Integer(1),Integer(0),Integer(0),Integer(0)]
>>> k._cache.import_data(v)
a^13 + a^8 + a^5 + 1
>>> k._cache.import_data(V(v))
a^13 + a^8 + a^5 + 1
```

order()

Return the cardinality of the field.

EXAMPLES:

```
sage: k.<a> = GF(2^64)
sage: k._cache.order()
18446744073709551616
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(64), names=('a',)); (a,) = k._first_ngens(1)
>>> k._cache.order()
18446744073709551616
```

polynomial()

Return the list of 0s and 1s giving the defining polynomial of the field.

EXAMPLES:

```
sage: k.<a> = GF(2^20,modulus='minimal_weight')
sage: k._cache.polynomial()
[1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1]
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(20),modulus='minimal_weight', names=('a',));␣
↪(a,) = k._first_ngens(1)
>>> k._cache.polynomial()
[1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1]
```

class sage.rings.finite_rings.element_ntl_gf2e.**FiniteField_ntl_gf2eElement**

Bases: *FinitePolyExtElement*

An element of an NTL:GF2E finite field.

charpoly(*var='x'*)

Return the characteristic polynomial of `self` as a polynomial in var over the prime subfield.

INPUT:

- `var` – string (default: `'x'`)

OUTPUT: polynomial

EXAMPLES:

```
sage: k.<a> = GF(2^8, impl='ntl')
sage: b = a^3 + a
sage: b.minpoly()
x^4 + x^3 + x^2 + x + 1
sage: b.charpoly()
x^8 + x^6 + x^4 + x^2 + 1
sage: b.charpoly().factor()
(x^4 + x^3 + x^2 + x + 1)^2
sage: b.charpoly('Z')
Z^8 + Z^6 + Z^4 + Z^2 + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(8), impl='ntl', names=('a',)); (a,) = k._first_
↪ngens(1)
>>> b = a**Integer(3) + a
>>> b.minpoly()
x^4 + x^3 + x^2 + x + 1
>>> b.charpoly()
x^8 + x^6 + x^4 + x^2 + 1
>>> b.charpoly().factor()
(x^4 + x^3 + x^2 + x + 1)^2
>>> b.charpoly('Z')
Z^8 + Z^6 + Z^4 + Z^2 + 1
```

is_one()

Return `True` if `self == k(1)`.

Equivalent to `self != k(0)`.

EXAMPLES:

```
sage: k.<a> = GF(2^20)
sage: a.is_one() # indirect doctest
False
sage: k(1).is_one()
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(20), names=('a',)); (a,) = k._first_ngens(1)
>>> a.is_one() # indirect doctest
False
>>> k(Integer(1)).is_one()
True
```

is_square()

Return `True` as every element in $\mathbf{F}_{2^n}$ is a square.

EXAMPLES:

```
sage: k.<a> = GF(2^18)
sage: e = k.random_element()
sage: e.parent() is k
True
sage: e.is_square()
True
sage: e.sqrt()^2 == e
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(18), names=('a',)); (a,) = k._first_ngens(1)
>>> e = k.random_element()
>>> e.parent() is k
True
>>> e.is_square()
True
>>> e.sqrt()**Integer(2) == e
True
```

is_unit()

Return `True` if `self` is nonzero, so it is a unit as an element of the finite field.

EXAMPLES:

```
sage: k.<a> = GF(2^17)
sage: a.is_unit()
True
sage: k(0).is_unit()
False
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(17), names=('a',)); (a,) = k._first_ngens(1)
>>> a.is_unit()
True
>>> k(Integer(0)).is_unit()
False
```

log(*base*, *order=None*, *check=False*)

Compute an integer x such that $b^x = a$, where a is `self` and b is `base`.

INPUT:

- `base` – finite field element
- `order` – integer (optional), the order of the base
- `check` – boolean (default: `False`): If set, test whether the given `order` is correct.

OUTPUT:

Integer x such that $a^x = b$, if it exists. Raises a `ValueError` exception if no such x exists.

ALGORITHM: pari:fflog

EXAMPLES:

```
sage: F = FiniteField(2^10, 'a')
sage: g = F.gen()
sage: b = g; a = g^37
sage: a.log(b)
37
sage: b^37; a
a^8 + a^7 + a^4 + a + 1
a^8 + a^7 + a^4 + a + 1
```

```
>>> from sage.all import *
>>> F = FiniteField(Integer(2)**Integer(10), 'a')
>>> g = F.gen()
>>> b = g; a = g**Integer(37)
>>> a.log(b)
37
>>> b**Integer(37); a
a^8 + a^7 + a^4 + a + 1
a^8 + a^7 + a^4 + a + 1
```

Big instances used to take a very long time before Issue #32842:

```
sage: g = GF(2^61).gen()
sage: g.log(g^7)
1976436865040309101
```

```
>>> from sage.all import *
>>> g = GF(Integer(2)**Integer(61)).gen()
>>> g.log(g**Integer(7))
1976436865040309101
```

AUTHORS:

- David Joyner and William Stein (2005-11)
- Lorenz Panny (2021-11): use PARI's pari:fflog instead of `sage.groups.generic.discrete_log()`

minpoly(*var='x'*)

Return the minimal polynomial of `self`, which is the smallest degree polynomial $f \in \mathbf{F}_2[x]$ such that `f(self) == 0`.

INPUT:

- `var` – string (default: `'x'`)

OUTPUT: polynomial

EXAMPLES:

```
sage: K.<a> = GF(2^100)
sage: f = a.minpoly(); f
x^100 + x^57 + x^56 + x^55 + x^52 + x^48 + x^47 + x^46 + x^45 + x^44 + x^43 +␣
↪x^41 + x^37 + x^36 + x^35 + x^34 + x^31 + x^30 + x^27 + x^25 + x^24 + x^22␣
↪+ x^20 + x^19 + x^16 + x^15 + x^11 + x^9 + x^8 + x^6 + x^5 + x^3 + 1
sage: f(a)
0
```

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```
sage: g = K.random_element()
sage: g.minpoly()(g)
0
```

```
>>> from sage.all import *
>>> K = GF(Integer(2)**Integer(100), names=('a',)); (a,) = K._first_ngens(1)
>>> f = a.minpoly(); f
x^100 + x^57 + x^56 + x^55 + x^52 + x^48 + x^47 + x^46 + x^45 + x^44 + x^43 +␣
↪x^41 + x^37 + x^36 + x^35 + x^34 + x^31 + x^30 + x^27 + x^25 + x^24 + x^22␣
↪+ x^20 + x^19 + x^16 + x^15 + x^11 + x^9 + x^8 + x^6 + x^5 + x^3 + 1
>>> f(a)
0
>>> g = K.random_element()
>>> g.minpoly()(g)
0
```

polynomial(*name=None*)

Return `self` viewed as a polynomial over `self.parent().prime_subfield()`.

INPUT:

- `name` – (optional) variable name

EXAMPLES:

```
sage: k.<a> = GF(2^17)
sage: e = a^15 + a^13 + a^11 + a^10 + a^9 + a^8 + a^7 + a^6 + a^4 + a + 1
sage: e.polynomial()
a^15 + a^13 + a^11 + a^10 + a^9 + a^8 + a^7 + a^6 + a^4 + a + 1

sage: from sage.rings.polynomial.polynomial_element import Polynomial
sage: isinstance(e.polynomial(), Polynomial)
True

sage: e.polynomial('x')
x^15 + x^13 + x^11 + x^10 + x^9 + x^8 + x^7 + x^6 + x^4 + x + 1
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(17), names=('a',)); (a,) = k._first_ngens(1)
>>> e = a**Integer(15) + a**Integer(13) + a**Integer(11) + a**Integer(10) +␣
↪a**Integer(9) + a**Integer(8) + a**Integer(7) + a**Integer(6) +␣
↪a**Integer(4) + a + Integer(1)
>>> e.polynomial()
a^15 + a^13 + a^11 + a^10 + a^9 + a^8 + a^7 + a^6 + a^4 + a + 1

>>> from sage.rings.polynomial.polynomial_element import Polynomial
>>> isinstance(e.polynomial(), Polynomial)
True

>>> e.polynomial('x')
x^15 + x^13 + x^11 + x^10 + x^9 + x^8 + x^7 + x^6 + x^4 + x + 1
```

sqrt(*all=False*, *extend=False*)

Return a square root of this finite field element in its parent.

EXAMPLES:

```
sage: k.<a> = GF(2^20)
sage: a.is_square()
True
sage: a.sqrt()
a^19 + a^15 + a^14 + a^12 + a^9 + a^7 + a^4 + a^3 + a + 1
sage: a.sqrt()^2 == a
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(20), names=('a',)); (a,) = k._first_ngens(1)
>>> a.is_square()
True
>>> a.sqrt()
a^19 + a^15 + a^14 + a^12 + a^9 + a^7 + a^4 + a^3 + a + 1
>>> a.sqrt()**Integer(2) == a
True
```

This failed before Issue #4899:

```
sage: GF(2^16,'a')(1).sqrt()
1
```

```
>>> from sage.all import *
>>> GF(Integer(2)**Integer(16),'a')(Integer(1)).sqrt()
1
```

trace()

Return the trace of `self`.

EXAMPLES:

```
sage: K.<a> = GF(2^25)
sage: a.trace()
0
sage: a.charpoly()
x^25 + x^8 + x^6 + x^2 + 1
sage: parent(a.trace())
Finite Field of size 2

sage: b = a+1
sage: b.trace()
1
sage: b.charpoly()[1]
1
```

```
>>> from sage.all import *
>>> K = GF(Integer(2)**Integer(25), names=('a',)); (a,) = K._first_ngens(1)
>>> a.trace()
0
>>> a.charpoly()
x^25 + x^8 + x^6 + x^2 + 1
```

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```
>>> parent(a.trace())
Finite Field of size 2

>>> b = a+Integer(1)
>>> b.trace()
1
>>> b.charpoly()[Integer(1)]
1
```

weight()

Return the number of nonzero coefficients in the polynomial representation of `self`.

EXAMPLES:

```
sage: K.<a> = GF(2^21)
sage: a.weight()
1
sage: (a^5+a^2+1).weight()
3
sage: b = 1/(a+1); b
a^20 + a^19 + a^18 + a^17 + a^16 + a^15 + a^14 + a^13 + a^12 + a^11 + a^10 +␣
↪a^9 + a^8 + a^7 + a^6 + a^4 + a^3 + a^2
sage: b.weight()
18
```

```
>>> from sage.all import *
>>> K = GF(Integer(2)**Integer(21), names=('a',)); (a,) = K._first_ngens(1)
>>> a.weight()
1
>>> (a**Integer(5)+a**Integer(2)+Integer(1)).weight()
3
>>> b = Integer(1)/(a+Integer(1)); b
a^20 + a^19 + a^18 + a^17 + a^16 + a^15 + a^14 + a^13 + a^12 + a^11 + a^10 +␣
↪a^9 + a^8 + a^7 + a^6 + a^4 + a^3 + a^2
>>> b.weight()
18
```

sage.rings.finite_rings.element_ntl_gf2e.**unpickleFiniteField_ntl_gf2eElement**(*parent*, *elem*)

EXAMPLES:

```
sage: k.<a> = GF(2^20)
sage: e = k.random_element()
sage: f = loads(dumps(e)) # indirect doctest
sage: e == f
True
```

```
>>> from sage.all import *
>>> k = GF(Integer(2)**Integer(20), names=('a',)); (a,) = k._first_ngens(1)
>>> e = k.random_element()
>>> f = loads(dumps(e)) # indirect doctest
>>> e == f
True
```

CHAPTER

SEVEN

MISCELLANEOUS

7.1 Finite residue fields

We can take the residue field of maximal ideals in the ring of integers of number fields. We can also take the residue field of irreducible polynomials over $\mathbf{F}_p$.

EXAMPLES:

```
sage: # needs sage.rings.number_field
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^3 - 7)
sage: P = K.ideal(29).factor()[0][0]
sage: k = K.residue_field(P); k
Residue field in abar of Fractional ideal (2*a^2 + 3*a - 10)
sage: k.order()
841
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) - Integer(7), names=('a',)); (a,) = K._first_
↪ngens(1)
>>> P = K.ideal(Integer(29)).factor()[Integer(0)][Integer(0)]
>>> k = K.residue_field(P); k
Residue field in abar of Fractional ideal (2*a^2 + 3*a - 10)
>>> k.order()
841
```

We reduce mod a prime for which the ring of integers is not monogenic (i.e., 2 is an essential discriminant divisor):

```
sage: # needs sage.rings.number_field
sage: K.<a> = NumberField(x^3 + x^2 - 2*x + 8)
sage: F = K.factor(2); F
(Fractional ideal (-1/2*a^2 + 1/2*a - 1)) * (Fractional ideal (a^2 - 2*a + 3)) *␣
↪(Fractional ideal (-3/2*a^2 + 5/2*a - 4))
sage: F[0][0].residue_field()
Residue field of Fractional ideal (-1/2*a^2 + 1/2*a - 1)
sage: F[1][0].residue_field()
Residue field of Fractional ideal (a^2 - 2*a + 3)
sage: F[2][0].residue_field()
Residue field of Fractional ideal (-3/2*a^2 + 5/2*a - 4)
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> K = NumberField(x**Integer(3) + x**Integer(2) - Integer(2)*x + Integer(8), names=(
↪'a',)); (a,) = K._first_ngens(1)
>>> F = K.factor(Integer(2)); F
(Fractional ideal (-1/2*a^2 + 1/2*a - 1)) * (Fractional ideal (a^2 - 2*a + 3)) *␣
↪(Fractional ideal (-3/2*a^2 + 5/2*a - 4))
>>> F[Integer(0)][Integer(0)].residue_field()
Residue field of Fractional ideal (-1/2*a^2 + 1/2*a - 1)
>>> F[Integer(1)][Integer(0)].residue_field()
Residue field of Fractional ideal (a^2 - 2*a + 3)
>>> F[Integer(2)][Integer(0)].residue_field()
Residue field of Fractional ideal (-3/2*a^2 + 5/2*a - 4)
```

We can also form residue fields from **Z**:

```
sage: ZZ.residue_field(17)
Residue field of Integers modulo 17
```

```
>>> from sage.all import *
>>> ZZ.residue_field(Integer(17))
Residue field of Integers modulo 17
```

And for polynomial rings over finite fields:

```
sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(5)[]
sage: I = R.ideal(t^2 + 2)
sage: k = ResidueField(I); k
Residue field in tbar of Principal ideal (t^2 + 2) of
 Univariate Polynomial Ring in t over Finite Field of size 5
```

```
>>> from sage.all import *
>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(5))['t']; (t,) = R._first_ngens(1)
>>> I = R.ideal(t**Integer(2) + Integer(2))
>>> k = ResidueField(I); k
Residue field in tbar of Principal ideal (t^2 + 2) of
 Univariate Polynomial Ring in t over Finite Field of size 5
```

AUTHORS:

- David Roe (2007-10-3): initial version
- William Stein (2007-12): bug fixes
- John Cremona (2008-9): extend reduction maps to the whole valuation ring add support for residue fields of ZZ
- David Roe (2009-12): added support for $GF(p)(t)$ and moved to new coercion framework.

class sage.rings.finite_rings.residue_field.**LiftingMap**

Bases: Section

Lifting map from residue class field to number field.

EXAMPLES:

```
sage: # needs sage.rings.number_field
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^3 + 2)
sage: F = K.factor(5)[0][0].residue_field()
sage: F.degree()
2
sage: L = F.lift_map(); L
Lifting map:
  From: Residue field in abar of Fractional ideal (a^2 + 2*a - 1)
  To:   Maximal Order generated by a in Number Field in a with defining␣
↪polynomial x^3 + 2
sage: L(F.0^2)
3*a + 1
sage: L(3*a + 1) == F.0^2
True

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(13)[]
sage: P = R.ideal(8*t^12 + 9*t^11 + 11*t^10 + 2*t^9 + 11*t^8
....:             + 3*t^7 + 12*t^6 + t^4 + 7*t^3 + 5*t^2 + 12*t + 1)
sage: k.<a> = P.residue_field()
sage: k.lift_map()
Lifting map:
  From: Residue field in a of Principal ideal (t^12 + 6*t^11 + 3*t^10
        + 10*t^9 + 3*t^8 + 2*t^7 + 8*t^6 + 5*t^4 + 9*t^3 + 12*t^2 + 8*t + 5) of
        Univariate Polynomial Ring in t over Finite Field of size 13
  To:   Univariate Polynomial Ring in t over Finite Field of size 13
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) + Integer(2), names=('a',)); (a,) = K._first_
↪ngens(1)
>>> F = K.factor(Integer(5))[Integer(0)][Integer(0)].residue_field()
>>> F.degree()
2
>>> L = F.lift_map(); L
Lifting map:
  From: Residue field in abar of Fractional ideal (a^2 + 2*a - 1)
  To:   Maximal Order generated by a in Number Field in a with defining␣
↪polynomial x^3 + 2
>>> L(F.gen(0)**Integer(2))
3*a + 1
>>> L(Integer(3)*a + Integer(1)) == F.gen(0)**Integer(2)
True

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(13))['t']; (t,) = R._first_ngens(1)
>>> P = R.ideal(Integer(8)*t**Integer(12) + Integer(9)*t**Integer(11) +␣
↪Integer(11)*t**Integer(10) + Integer(2)*t**Integer(9) +␣
↪Integer(11)*t**Integer(8)
...             + Integer(3)*t**Integer(7) + Integer(12)*t**Integer(6) +␣
```

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```
↪t**Integer(4) + Integer(7)*t**Integer(3) + Integer(5)*t**Integer(2) +␣
↪Integer(12)*t + Integer(1))
>>> k = P.residue_field(names=('a',)); (a,) = k._first_ngens(1)
>>> k.lift_map()
Lifting map:
  From: Residue field in a of Principal ideal (t^12 + 6*t^11 + 3*t^10
        + 10*t^9 + 3*t^8 + 2*t^7 + 8*t^6 + 5*t^4 + 9*t^3 + 12*t^2 + 8*t + 5) of
        Univariate Polynomial Ring in t over Finite Field of size 13
  To:   Univariate Polynomial Ring in t over Finite Field of size 13
```

class sage.rings.finite_rings.residue_field.**ReductionMap**

Bases: Map

A reduction map from a (subset) of a number field or function field to this residue class field.

It will be defined on those elements of the field with nonnegative valuation at the specified prime.

EXAMPLES:

```
sage: # needs sage.rings.number_field sage.symbolic
sage: I = QQ[sqrt(17)].factor(5)[0][0]; I
Fractional ideal (5)
sage: k = I.residue_field(); k
Residue field in sqrt17bar of Fractional ideal (5)
sage: R = k.reduction_map(); R
Partially defined reduction map:
  From: Number Field in sqrt17 with defining polynomial x^2 - 17
        with sqrt17 = 4.123105625617660?
  To:   Residue field in sqrt17bar of Fractional ideal (5)

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(next_prime(2^20))[]; P = R.ideal(t^2 + t + 1)
sage: k = P.residue_field()
sage: k.reduction_map()
Partially defined reduction map:
  From: Fraction Field of
        Univariate Polynomial Ring in t over Finite Field of size 1048583
  To:   Residue field in tbar of Principal ideal (t^2 + t + 1) of
        Univariate Polynomial Ring in t over Finite Field of size 1048583
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field sage.symbolic
>>> I = QQ[sqrt(Integer(17))].factor(Integer(5))[Integer(0)][Integer(0)]; I
Fractional ideal (5)
>>> k = I.residue_field(); k
Residue field in sqrt17bar of Fractional ideal (5)
>>> R = k.reduction_map(); R
Partially defined reduction map:
  From: Number Field in sqrt17 with defining polynomial x^2 - 17
        with sqrt17 = 4.123105625617660?
  To:   Residue field in sqrt17bar of Fractional ideal (5)

>>> # needs sage.rings.finite_rings
```

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```
>>> R = GF(next_prime(Integer(2)**Integer(20)))['t']; (t,) = R._first_ngens(1); P␣
↪= R.ideal(t**Integer(2) + t + Integer(1))
>>> k = P.residue_field()
>>> k.reduction_map()
Partially defined reduction map:
  From: Fraction Field of
        Univariate Polynomial Ring in t over Finite Field of size 1048583
  To:   Residue field in tbar of Principal ideal (t^2 + t + 1) of
        Univariate Polynomial Ring in t over Finite Field of size 1048583
```

section()

Compute a section of the map, namely a map that lifts elements of the residue field to elements of the field.

EXAMPLES:

```
sage: # needs sage.rings.number_field
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^5 - 5*x + 2)
sage: P = K.ideal(47).factor()[0][0]
sage: k = K.residue_field(P)
sage: f = k.convert_map_from(K)
sage: s = f.section(); s
Lifting map:
  From: Residue field in abar of Fractional ideal (14*a^4 - 24*a^3 - 26*a^2 +␣
↪58*a - 15)
  To:   Number Field in a with defining polynomial x^5 - 5*x + 2
sage: s(k.gen())
a
sage: L.<b> = NumberField(x^5 + 17*x + 1)
sage: P = L.factor(53)[0][0]
sage: l = L.residue_field(P)
sage: g = l.convert_map_from(L)
sage: s = g.section(); s
Lifting map:
  From: Residue field in bbar of Fractional ideal (53, b^2 + 23*b + 8)
  To:   Number Field in b with defining polynomial x^5 + 17*x + 1
sage: s(l.gen()).parent()
Number Field in b with defining polynomial x^5 + 17*x + 1

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(2)[]; h = t^5 + t^2 + 1
sage: k.<a> = R.residue_field(h)
sage: K = R.fraction_field()
sage: f = k.convert_map_from(K)
sage: f.section()                                                          #␣
↪needs sage.libs.ntl
Lifting map:
  From: Residue field in a of Principal ideal (t^5 + t^2 + 1) of
        Univariate Polynomial Ring in t over Finite Field of size 2 (using␣
↪GF2X)
  To:   Fraction Field of
        Univariate Polynomial Ring in t over Finite Field of size 2 (using␣
↪GF2X)
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(5) - Integer(5)*x + Integer(2), names=('a',));␣
↪(a,) = K._first_ngens(1)
>>> P = K.ideal(Integer(47)).factor()[Integer(0)][Integer(0)]
>>> k = K.residue_field(P)
>>> f = k.convert_map_from(K)
>>> s = f.section(); s
Lifting map:
  From: Residue field in abar of Fractional ideal (14*a^4 - 24*a^3 - 26*a^2 +␣
↪58*a - 15)
  To:   Number Field in a with defining polynomial x^5 - 5*x + 2
>>> s(k.gen())
a
>>> L = NumberField(x**Integer(5) + Integer(17)*x + Integer(1), names=('b',));
↪ (b,) = L._first_ngens(1)
>>> P = L.factor(Integer(53))[Integer(0)][Integer(0)]
>>> l = L.residue_field(P)
>>> g = l.convert_map_from(L)
>>> s = g.section(); s
Lifting map:
  From: Residue field in bbar of Fractional ideal (53, b^2 + 23*b + 8)
  To:   Number Field in b with defining polynomial x^5 + 17*x + 1
>>> s(l.gen()).parent()
Number Field in b with defining polynomial x^5 + 17*x + 1

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(2))['t']; (t,) = R._first_ngens(1); h = t**Integer(5) +␣
↪t**Integer(2) + Integer(1)
>>> k = R.residue_field(h, names=('a',)); (a,) = k._first_ngens(1)
>>> K = R.fraction_field()
>>> f = k.convert_map_from(K)
>>> f.section()                                                          #␣
↪needs sage.libs.ntl
Lifting map:
  From: Residue field in a of Principal ideal (t^5 + t^2 + 1) of
        Univariate Polynomial Ring in t over Finite Field of size 2 (using␣
↪GF2X)
  To:   Fraction Field of
        Univariate Polynomial Ring in t over Finite Field of size 2 (using␣
↪GF2X)
```

class sage.rings.finite_rings.residue_field.**ResidueFieldFactory**

Bases: UniqueFactory

A factory that returns the residue class field of a prime ideal p of the ring of integers of a number field, or of a polynomial ring over a finite field.

INPUT:

- p – a prime ideal of an order in a number field
- names – the variable name for the finite field created; defaults to the name of the number field variable but with bar placed after it

- `check` – whether or not to check if p is prime

OUTPUT: the residue field at the prime p

EXAMPLES:

```
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^3 - 7)                                          #␣
↪needs sage.rings.number_field
sage: P = K.ideal(29).factor()[0][0]                                        #␣
↪needs sage.rings.number_field
sage: ResidueField(P)                                                       #␣
↪needs sage.rings.number_field
Residue field in abar of Fractional ideal (2*a^2 + 3*a - 10)
```

```
>>> from sage.all import *
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) - Integer(7), names=('a',)); (a,) = K._first_
↪ngens(1)# needs sage.rings.number_field
>>> P = K.ideal(Integer(29)).factor()[Integer(0)][Integer(0)]                 ␣
↪                        # needs sage.rings.number_field
>>> ResidueField(P)                                                        #␣
↪needs sage.rings.number_field
Residue field in abar of Fractional ideal (2*a^2 + 3*a - 10)
```

The result is cached:

```
sage: ResidueField(P) is ResidueField(P)                                    #␣
↪needs sage.rings.number_field
True
sage: k = K.residue_field(P); k                                             #␣
↪needs sage.rings.number_field
Residue field in abar of Fractional ideal (2*a^2 + 3*a - 10)
sage: k.order()                                                             #␣
↪needs sage.rings.number_field
841
```

```
>>> from sage.all import *
>>> ResidueField(P) is ResidueField(P)                                     #␣
↪needs sage.rings.number_field
True
>>> k = K.residue_field(P); k                                              #␣
↪needs sage.rings.number_field
Residue field in abar of Fractional ideal (2*a^2 + 3*a - 10)
>>> k.order()                                                              #␣
↪needs sage.rings.number_field
841
```

It also works for polynomial rings:

```
sage: R.<t> = GF(31)[]
sage: P = R.ideal(t^5 + 2*t + 11)
sage: ResidueField(P)                                                       #␣
↪needs sage.rings.finite_rings
```

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```
Residue field in tbar of Principal ideal (t^5 + 2*t + 11) of
 Univariate Polynomial Ring in t over Finite Field of size 31

sage: ResidueField(P) is ResidueField(P)                                    #␣
↪needs sage.rings.finite_rings
True
sage: k = ResidueField(P); k.order()                                        #␣
↪needs sage.rings.finite_rings
28629151
```

```
>>> from sage.all import *
>>> R = GF(Integer(31))['t']; (t,) = R._first_ngens(1)
>>> P = R.ideal(t**Integer(5) + Integer(2)*t + Integer(11))
>>> ResidueField(P)                                                        #␣
↪needs sage.rings.finite_rings
Residue field in tbar of Principal ideal (t^5 + 2*t + 11) of
 Univariate Polynomial Ring in t over Finite Field of size 31

>>> ResidueField(P) is ResidueField(P)                                     #␣
↪needs sage.rings.finite_rings
True
>>> k = ResidueField(P); k.order()                                         #␣
↪needs sage.rings.finite_rings
28629151
```

An example where the generator of the number field doesn't generate the residue class field:

```
sage: # needs sage.rings.number_field
sage: K.<a> = NumberField(x^3 - 875)
sage: P = K.ideal(5).factor()[0][0]; k = K.residue_field(P); k
Residue field in abar of Fractional ideal (5, 1/25*a^2 - 2/5*a - 1)
sage: k.polynomial()
abar^2 + 3*abar + 4
sage: k.0^3 - 875
2
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> K = NumberField(x**Integer(3) - Integer(875), names=('a',)); (a,) = K._first_
↪ngens(1)
>>> P = K.ideal(Integer(5)).factor()[Integer(0)][Integer(0)]; k = K.residue_
↪field(P); k
Residue field in abar of Fractional ideal (5, 1/25*a^2 - 2/5*a - 1)
>>> k.polynomial()
abar^2 + 3*abar + 4
>>> k.gen(0)**Integer(3) - Integer(875)
2
```

An example where the residue class field is large but of degree 1:

```
sage: # needs sage.rings.number_field
sage: K.<a> = NumberField(x^3 - 875)
```

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```
sage: P = K.ideal(2007).factor()[2][0]; k = K.residue_field(P); k
Residue field of Fractional ideal (223, 1/5*a + 11)
sage: k(a)
168
sage: k(a)^3 - 875
0
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> K = NumberField(x**Integer(3) - Integer(875), names=('a',)); (a,) = K._first_
↪ngens(1)
>>> P = K.ideal(Integer(2007)).factor()[Integer(2)][Integer(0)]; k = K.residue_
↪field(P); k
Residue field of Fractional ideal (223, 1/5*a + 11)
>>> k(a)
168
>>> k(a)**Integer(3) - Integer(875)
0
```

And for polynomial rings:

```
sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(next_prime(2^18))[]
sage: P = R.ideal(t - 5)
sage: k = ResidueField(P); k
Residue field of Principal ideal (t + 262142) of
 Univariate Polynomial Ring in t over Finite Field of size 262147
sage: k(t)
5
```

```
>>> from sage.all import *
>>> # needs sage.rings.finite_rings
>>> R = GF(next_prime(Integer(2)**Integer(18)))['t']; (t,) = R._first_ngens(1)
>>> P = R.ideal(t - Integer(5))
>>> k = ResidueField(P); k
Residue field of Principal ideal (t + 262142) of
 Univariate Polynomial Ring in t over Finite Field of size 262147
>>> k(t)
5
```

In this example, 2 is an inessential discriminant divisor, so divides the index of `ZZ[a]` in the maximal order for all `a`:

```
sage: # needs sage.rings.number_field
sage: K.<a> = NumberField(x^3 + x^2 - 2*x + 8)
sage: P = K.ideal(2).factor()[0][0]; P
Fractional ideal (-1/2*a^2 + 1/2*a - 1)
sage: F = K.residue_field(P); F
Residue field of Fractional ideal (-1/2*a^2 + 1/2*a - 1)
sage: F(a)
0
sage: B = K.maximal_order().basis(); B
```

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```
[1, 1/2*a^2 + 1/2*a, a^2]
sage: F(B[1])
1
sage: F(B[2])
0
sage: F
Residue field of Fractional ideal (-1/2*a^2 + 1/2*a - 1)
sage: F.degree()
1
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> K = NumberField(x**Integer(3) + x**Integer(2) - Integer(2)*x + Integer(8),␣
↪names=('a',)); (a,) = K._first_ngens(1)
>>> P = K.ideal(Integer(2)).factor()[Integer(0)][Integer(0)]; P
Fractional ideal (-1/2*a^2 + 1/2*a - 1)
>>> F = K.residue_field(P); F
Residue field of Fractional ideal (-1/2*a^2 + 1/2*a - 1)
>>> F(a)
0
>>> B = K.maximal_order().basis(); B
[1, 1/2*a^2 + 1/2*a, a^2]
>>> F(B[Integer(1)])
1
>>> F(B[Integer(2)])
0
>>> F
Residue field of Fractional ideal (-1/2*a^2 + 1/2*a - 1)
>>> F.degree()
1
```

create_key_and_extra_args(*p*, *names=None*, *check=True*, *impl=None*, ***kwds*)

Return a tuple containing the key (uniquely defining data) and any extra arguments.

EXAMPLES:

```
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^3 - 7)                                          #␣
↪needs sage.rings.number_field
sage: ResidueField(K.ideal(29).factor()[0][0])  # indirect doctest          #␣
↪needs sage.rings.number_field
Residue field in abar of Fractional ideal (2*a^2 + 3*a - 10)
```

```
>>> from sage.all import *
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) - Integer(7), names=('a',)); (a,) = K._
↪first_ngens(1)# needs sage.rings.number_field
>>> ResidueField(K.ideal(Integer(29)).factor()[Integer(0)][Integer(0)])  #␣
↪indirect doctest          # needs sage.rings.number_field
Residue field in abar of Fractional ideal (2*a^2 + 3*a - 10)
```

create_object(*version*, *key*, ***kwds*)

Create the object from the key and extra arguments. This is only called if the object was not found in the

cache.

EXAMPLES:

```
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^3 - 7)                                          #␣
↪needs sage.rings.number_field
sage: P = K.ideal(29).factor()[0][0]                                        #␣
↪needs sage.rings.number_field
sage: ResidueField(P) is ResidueField(P)  # indirect doctest               #␣
↪needs sage.rings.number_field
True
```

```
>>> from sage.all import *
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) - Integer(7), names=('a',)); (a,) = K._
↪first_ngens(1)# needs sage.rings.number_field
>>> P = K.ideal(Integer(29)).factor()[Integer(0)][Integer(0)]                 ␣
↪                        # needs sage.rings.number_field
>>> ResidueField(P) is ResidueField(P)  # indirect doctest                 #␣
↪needs sage.rings.number_field
True
```

class sage.rings.finite_rings.residue_field.**ResidueFieldHomomorphism_global**

Bases: RingHomomorphism

The class representing a homomorphism from the order of a number field or function field to the residue field at a given prime.

EXAMPLES:

```
sage: # needs sage.rings.number_field
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^3 - 7)
sage: P  = K.ideal(29).factor()[0][0]
sage: k  = K.residue_field(P)
sage: OK = K.maximal_order()
sage: abar = k(OK.1); abar
abar
sage: (1+abar)^179
24*abar + 12

sage: # needs sage.rings.number_field
sage: phi = k.coerce_map_from(OK); phi
Ring morphism:
  From: Maximal Order generated by a in Number Field in a with defining␣
↪polynomial x^3 - 7
  To:   Residue field in abar of Fractional ideal (2*a^2 + 3*a - 10)
sage: phi in Hom(OK,k)
True
sage: phi(OK.1)
abar

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(19)[]; P = R.ideal(t^2 + 5)
```

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```
sage: k.<a> = R.residue_field(P)
sage: f = k.coerce_map_from(R); f
Ring morphism:
  From: Univariate Polynomial Ring in t over Finite Field of size 19
  To:   Residue field in a of Principal ideal (t^2 + 5) of
        Univariate Polynomial Ring in t over Finite Field of size 19
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) - Integer(7), names=('a',)); (a,) = K._first_
↪ngens(1)
>>> P  = K.ideal(Integer(29)).factor()[Integer(0)][Integer(0)]
>>> k  = K.residue_field(P)
>>> OK = K.maximal_order()
>>> abar = k(OK.gen(1)); abar
abar
>>> (Integer(1)+abar)**Integer(179)
24*abar + 12

>>> # needs sage.rings.number_field
>>> phi = k.coerce_map_from(OK); phi
Ring morphism:
  From: Maximal Order generated by a in Number Field in a with defining␣
↪polynomial x^3 - 7
  To:   Residue field in abar of Fractional ideal (2*a^2 + 3*a - 10)
>>> phi in Hom(OK,k)
True
>>> phi(OK.gen(1))
abar

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(19))['t']; (t,) = R._first_ngens(1); P = R.ideal(t**Integer(2)␣
↪+ Integer(5))
>>> k = R.residue_field(P, names=('a',)); (a,) = k._first_ngens(1)
>>> f = k.coerce_map_from(R); f
Ring morphism:
  From: Univariate Polynomial Ring in t over Finite Field of size 19
  To:   Residue field in a of Principal ideal (t^2 + 5) of
        Univariate Polynomial Ring in t over Finite Field of size 19
```

lift(x)

Return a lift of `x` to the Order, returning a "polynomial" in the generator with coefficients between 0 and $p-1$.

EXAMPLES:

```
sage: # needs sage.rings.number_field
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^3 - 7)
sage: P = K.ideal(29).factor()[0][0]
sage: k = K.residue_field(P)
```

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```
sage: OK = K.maximal_order()
sage: f = k.coerce_map_from(OK)
sage: c = OK(a)
sage: b = k(a)
sage: f.lift(13*b + 5)
13*a + 5
sage: f.lift(12821*b + 918)
3*a + 19

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(17)[]; P = R.ideal(t^3 + t^2 + 7)
sage: k.<a> = P.residue_field(); f = k.coerce_map_from(R)
sage: f.lift(a^2 + 5*a + 1)
t^2 + 5*t + 1
sage: f(f.lift(a^2 + 5*a + 1)) == a^2 + 5*a + 1                                #␣
↪needs sage.modules
True
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) - Integer(7), names=('a',)); (a,) = K._
↪first_ngens(1)
>>> P = K.ideal(Integer(29)).factor()[Integer(0)][Integer(0)]
>>> k = K.residue_field(P)
>>> OK = K.maximal_order()
>>> f = k.coerce_map_from(OK)
>>> c = OK(a)
>>> b = k(a)
>>> f.lift(Integer(13)*b + Integer(5))
13*a + 5
>>> f.lift(Integer(12821)*b + Integer(918))
3*a + 19

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(17))['t']; (t,) = R._first_ngens(1); P = R.
↪ideal(t**Integer(3) + t**Integer(2) + Integer(7))
>>> k = P.residue_field(names=('a',)); (a,) = k._first_ngens(1); f = k.coerce_
↪map_from(R)
>>> f.lift(a**Integer(2) + Integer(5)*a + Integer(1))
t^2 + 5*t + 1
>>> f(f.lift(a**Integer(2) + Integer(5)*a + Integer(1))) == a**Integer(2) +␣
↪Integer(5)*a + Integer(1)                                # needs sage.modules
True
```

section()

Compute a section of the map, namely a map that lifts elements of the residue field to elements of the ring of integers.

EXAMPLES:

```
sage: # needs sage.rings.number_field
```

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```
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^5 - 5*x + 2)
sage: P = K.ideal(47).factor()[0][0]
sage: k = K.residue_field(P)
sage: f = k.coerce_map_from(K.ring_of_integers())
sage: s = f.section(); s
Lifting map:
  From: Residue field in abar of Fractional ideal (14*a^4 - 24*a^3 - 26*a^2 +␣
↪58*a - 15)
  To:   Maximal Order generated by a in Number Field in a with defining␣
↪polynomial x^5 - 5*x + 2
sage: s(k.gen())
a
sage: L.<b> = NumberField(x^5 + 17*x + 1)
sage: P = L.factor(53)[0][0]
sage: l = L.residue_field(P)
sage: g = l.coerce_map_from(L.ring_of_integers())
sage: s = g.section(); s
Lifting map:
  From: Residue field in bbar of Fractional ideal (53, b^2 + 23*b + 8)
  To:   Maximal Order generated by b in Number Field in b
        with defining polynomial x^5 + 17*x + 1
sage: s(l.gen()).parent()
Maximal Order generated by b in Number Field in b with defining polynomial x^
↪5 + 17*x + 1

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(17)[]; P = R.ideal(t^3 + t^2 + 7)
sage: k.<a> = P.residue_field()
sage: f = k.coerce_map_from(R)
sage: f.section()
(map internal to coercion system -- copy before use)
Lifting map:
  From: Residue field in a of Principal ideal (t^3 + t^2 + 7) of
        Univariate Polynomial Ring in t over Finite Field of size 17
  To:   Univariate Polynomial Ring in t over Finite Field of size 17
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(5) - Integer(5)*x + Integer(2), names=('a',));␣
↪(a,) = K._first_ngens(1)
>>> P = K.ideal(Integer(47)).factor()[Integer(0)][Integer(0)]
>>> k = K.residue_field(P)
>>> f = k.coerce_map_from(K.ring_of_integers())
>>> s = f.section(); s
Lifting map:
  From: Residue field in abar of Fractional ideal (14*a^4 - 24*a^3 - 26*a^2 +␣
↪58*a - 15)
  To:   Maximal Order generated by a in Number Field in a with defining␣
↪polynomial x^5 - 5*x + 2
>>> s(k.gen())
```

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```
a
>>> L = NumberField(x**Integer(5) + Integer(17)*x + Integer(1), names=('b',));
↪ (b,) = L._first_ngens(1)
>>> P = L.factor(Integer(53))[Integer(0)][Integer(0)]
>>> l = L.residue_field(P)
>>> g = l.coerce_map_from(L.ring_of_integers())
>>> s = g.section(); s
Lifting map:
  From: Residue field in bbar of Fractional ideal (53, b^2 + 23*b + 8)
  To:   Maximal Order generated by b in Number Field in b
        with defining polynomial x^5 + 17*x + 1
>>> s(l.gen()).parent()
Maximal Order generated by b in Number Field in b with defining polynomial x^
↪5 + 17*x + 1

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(17))['t']; (t,) = R._first_ngens(1); P = R.
↪ideal(t**Integer(3) + t**Integer(2) + Integer(7))
>>> k = P.residue_field(names=('a',)); (a,) = k._first_ngens(1)
>>> f = k.coerce_map_from(R)
>>> f.section()
(map internal to coercion system -- copy before use)
Lifting map:
  From: Residue field in a of Principal ideal (t^3 + t^2 + 7) of
        Univariate Polynomial Ring in t over Finite Field of size 17
  To:   Univariate Polynomial Ring in t over Finite Field of size 17
```

class sage.rings.finite_rings.residue_field.**ResidueField_generic**(*p*)

Bases: Field

The class representing a generic residue field.

EXAMPLES:

```
sage: # needs sage.rings.number_field
sage: I = QQ[i].factor(2)[0][0]; I
Fractional ideal (I - 1)
sage: k = I.residue_field(); k
Residue field of Fractional ideal (I - 1)
sage: type(k)
<class 'sage.rings.finite_rings.residue_field.ResidueFiniteField_prime_modn_with_
↪category'>

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(29)[]; P = R.ideal(t^2 + 2); k.<a> = ResidueField(P); k
Residue field in a of Principal ideal (t^2 + 2) of
 Univariate Polynomial Ring in t over Finite Field of size 29
sage: type(k)                                                            #␣
↪needs sage.libs.linbox
<class 'sage.rings.finite_rings.residue_field_givaro.ResidueFiniteField_givaro_
↪with_category'>
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> I = QQ[i].factor(Integer(2))[Integer(0)][Integer(0)]; I
Fractional ideal (I - 1)
>>> k = I.residue_field(); k
Residue field of Fractional ideal (I - 1)
>>> type(k)
<class 'sage.rings.finite_rings.residue_field.ResidueFiniteField_prime_modn_with_
↪category'>

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(29))['t']; (t,) = R._first_ngens(1); P = R.ideal(t**Integer(2)␣
↪+ Integer(2)); k = ResidueField(P, names=('a',)); (a,) = k._first_ngens(1); k
Residue field in a of Principal ideal (t^2 + 2) of
 Univariate Polynomial Ring in t over Finite Field of size 29
>>> type(k)                                                                  #␣
↪needs sage.libs.linbox
<class 'sage.rings.finite_rings.residue_field_givaro.ResidueFiniteField_givaro_
↪with_category'>
```

construction()

Construction of this residue field.

OUTPUT:

An `AlgebraicExtensionFunctor` and the number field that this residue field has been obtained from.

The residue field is determined by a prime (fractional) ideal in a number field. If this ideal can be coerced into a different number field, then the construction functor applied to this number field will return the corresponding residue field. See Issue #15223.

EXAMPLES:

```
sage: # needs sage.rings.number_field
sage: K.<z> = CyclotomicField(7)
sage: P = K.factor(17)[0][0]
sage: k = K.residue_field(P); k
Residue field in zbar of Fractional ideal (17)
sage: F, R = k.construction()
sage: F
AlgebraicExtensionFunctor
sage: R
Cyclotomic Field of order 7 and degree 6
sage: F(R) is k
True
sage: F(ZZ)
Residue field of Integers modulo 17
sage: F(CyclotomicField(49))
Residue field in zbar of Fractional ideal (17)
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> K = CyclotomicField(Integer(7), names=('z',)); (z,) = K._first_ngens(1)
>>> P = K.factor(Integer(17))[Integer(0)][Integer(0)]
```

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```
>>> k = K.residue_field(P); k
Residue field in zbar of Fractional ideal (17)
>>> F, R = k.construction()
>>> F
AlgebraicExtensionFunctor
>>> R
Cyclotomic Field of order 7 and degree 6
>>> F(R) is k
True
>>> F(ZZ)
Residue field of Integers modulo 17
>>> F(CyclotomicField(Integer(49)))
Residue field in zbar of Fractional ideal (17)
```

ideal()

Return the maximal ideal that this residue field is the quotient by.

EXAMPLES:

```
sage: # needs sage.rings.number_field
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^3 + x + 1)
sage: P = K.ideal(29).factor()[0][0]
sage: k = K.residue_field(P)  # indirect doctest
sage: k.ideal() is P
True
sage: p = next_prime(2^40); p
1099511627791
sage: k = K.residue_field(K.prime_above(p))
sage: k.ideal().norm() == p
True

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(17)[]; P = R.ideal(t^3 + t^2 + 7)
sage: k.<a> = R.residue_field(P)
sage: k.ideal()
Principal ideal (t^3 + t^2 + 7) of
 Univariate Polynomial Ring in t over Finite Field of size 17
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) + x + Integer(1), names=('a',)); (a,) = K._
↪first_ngens(1)
>>> P = K.ideal(Integer(29)).factor()[Integer(0)][Integer(0)]
>>> k = K.residue_field(P)  # indirect doctest
>>> k.ideal() is P
True
>>> p = next_prime(Integer(2)**Integer(40)); p
1099511627791
>>> k = K.residue_field(K.prime_above(p))
>>> k.ideal().norm() == p
```

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```
True

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(17))['t']; (t,) = R._first_ngens(1); P = R.
↪ideal(t**Integer(3) + t**Integer(2) + Integer(7))
>>> k = R.residue_field(P, names=('a',)); (a,) = k._first_ngens(1)
>>> k.ideal()
Principal ideal (t^3 + t^2 + 7) of
 Univariate Polynomial Ring in t over Finite Field of size 17
```

lift(x)

Return a lift of `x` to the Order, returning a "polynomial" in the generator with coefficients between 0 and $p-1$.

EXAMPLES:

```
sage: # needs sage.rings.number_field
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^3 - 7)
sage: P = K.ideal(29).factor()[0][0]
sage: k = K.residue_field(P)
sage: OK = K.maximal_order()
sage: c = OK(a)
sage: b = k(a)
sage: k.lift(13*b + 5)
13*a + 5
sage: k.lift(12821*b + 918)
3*a + 19

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(17)[]; P = R.ideal(t^3 + t^2 + 7)
sage: k.<a> = P.residue_field()
sage: k.lift(a^2 + 5)
t^2 + 5
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) - Integer(7), names=('a',)); (a,) = K._
↪first_ngens(1)
>>> P = K.ideal(Integer(29)).factor()[Integer(0)][Integer(0)]
>>> k = K.residue_field(P)
>>> OK = K.maximal_order()
>>> c = OK(a)
>>> b = k(a)
>>> k.lift(Integer(13)*b + Integer(5))
13*a + 5
>>> k.lift(Integer(12821)*b + Integer(918))
3*a + 19

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(17))['t']; (t,) = R._first_ngens(1); P = R.
```

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```
↪ideal(t**Integer(3) + t**Integer(2) + Integer(7))
>>> k = P.residue_field(names=('a',)); (a,) = k._first_ngens(1)
>>> k.lift(a**Integer(2) + Integer(5))
t^2 + 5
```

lift_map()

Return the standard map from this residue field up to the ring of integers lifting the canonical projection.

EXAMPLES:

```
sage: # needs sage.rings.number_field sage.symbolic
sage: I = QQ[3^(1/3)].factor(5)[1][0]; I
Fractional ideal (a - 2)
sage: k = I.residue_field(); k
Residue field of Fractional ideal (a - 2)
sage: f = k.lift_map(); f
Lifting map:
  From: Residue field of Fractional ideal (a - 2)
  To:   Maximal Order generated by a in Number Field in a
        with defining polynomial x^3 - 3 with a = 1.442249570307409?
sage: f.domain()
Residue field of Fractional ideal (a - 2)
sage: f.codomain()
Maximal Order generated by a in Number Field in a
 with defining polynomial x^3 - 3 with a = 1.442249570307409?
sage: f(k.0)
1

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(17)[]; P = R.ideal(t^3 + t^2 + 7)
sage: k.<a> = P.residue_field()
sage: f = k.lift_map(); f
(map internal to coercion system -- copy before use)
Lifting map:
  From: Residue field in a of Principal ideal (t^3 + t^2 + 7) of
        Univariate Polynomial Ring in t over Finite Field of size 17
  To:   Univariate Polynomial Ring in t over Finite Field of size 17
sage: f(a^2 + 5)
t^2 + 5
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field sage.symbolic
>>> I = QQ[Integer(3)**(Integer(1)/Integer(3))].
↪factor(Integer(5))[Integer(1)][Integer(0)]; I
Fractional ideal (a - 2)
>>> k = I.residue_field(); k
Residue field of Fractional ideal (a - 2)
>>> f = k.lift_map(); f
Lifting map:
  From: Residue field of Fractional ideal (a - 2)
  To:   Maximal Order generated by a in Number Field in a
        with defining polynomial x^3 - 3 with a = 1.442249570307409?
```

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```
>>> f.domain()
Residue field of Fractional ideal (a - 2)
>>> f.codomain()
Maximal Order generated by a in Number Field in a
 with defining polynomial x^3 - 3 with a = 1.442249570307409?
>>> f(k.gen(0))
1

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(17))['t']; (t,) = R._first_ngens(1); P = R.
↪ideal(t**Integer(3) + t**Integer(2) + Integer(7))
>>> k = P.residue_field(names=('a',)); (a,) = k._first_ngens(1)
>>> f = k.lift_map(); f
(map internal to coercion system -- copy before use)
Lifting map:
  From: Residue field in a of Principal ideal (t^3 + t^2 + 7) of
        Univariate Polynomial Ring in t over Finite Field of size 17
  To:   Univariate Polynomial Ring in t over Finite Field of size 17
>>> f(a**Integer(2) + Integer(5))
t^2 + 5
```

reduction_map()

Return the partially defined reduction map from the number field to this residue class field.

EXAMPLES:

```
sage: # needs sage.rings.number_field sage.symbolic
sage: I = QQ[2^(1/3)].factor(2)[0][0]; I
Fractional ideal (a)
sage: k = I.residue_field(); k
Residue field of Fractional ideal (a)
sage: pi = k.reduction_map(); pi
Partially defined reduction map:
  From: Number Field in a with defining polynomial x^3 - 2
        with a = 1.259921049894873?
  To:   Residue field of Fractional ideal (a)
sage: pi.domain()
Number Field in a with defining polynomial x^3 - 2 with a = 1.259921049894873?
sage: pi.codomain()
Residue field of Fractional ideal (a)

sage: # needs sage.rings.number_field
sage: x = polygen(ZZ, 'x')
sage: K.<a> = NumberField(x^3 + x^2 - 2*x + 32)
sage: F = K.factor(2)[0][0].residue_field()
sage: F.reduction_map().domain()
Number Field in a with defining polynomial x^3 + x^2 - 2*x + 32
sage: K.<a> = NumberField(x^3 + 128)
sage: F = K.factor(2)[0][0].residue_field()
sage: F.reduction_map().codomain()
Residue field of Fractional ideal (1/4*a)
```

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```
sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(17)[]; P = R.ideal(t^3 + t^2 + 7)
sage: k.<a> = P.residue_field(); f = k.reduction_map(); f
Partially defined reduction map:
  From: Fraction Field of Univariate Polynomial Ring in t
        over Finite Field of size 17
  To:   Residue field in a of Principal ideal (t^3 + t^2 + 7) of
        Univariate Polynomial Ring in t over Finite Field of size 17
sage: f(1/t)
12*a^2 + 12*a
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field sage.symbolic
>>> I = QQ[Integer(2)**(Integer(1)/Integer(3))].
↪factor(Integer(2))[Integer(0)][Integer(0)]; I
Fractional ideal (a)
>>> k = I.residue_field(); k
Residue field of Fractional ideal (a)
>>> pi = k.reduction_map(); pi
Partially defined reduction map:
  From: Number Field in a with defining polynomial x^3 - 2
        with a = 1.259921049894873?
  To:   Residue field of Fractional ideal (a)
>>> pi.domain()
Number Field in a with defining polynomial x^3 - 2 with a = 1.259921049894873?
>>> pi.codomain()
Residue field of Fractional ideal (a)

>>> # needs sage.rings.number_field
>>> x = polygen(ZZ, 'x')
>>> K = NumberField(x**Integer(3) + x**Integer(2) - Integer(2)*x +␣
↪Integer(32), names=('a',)); (a,) = K._first_ngens(1)
>>> F = K.factor(Integer(2))[Integer(0)][Integer(0)].residue_field()
>>> F.reduction_map().domain()
Number Field in a with defining polynomial x^3 + x^2 - 2*x + 32
>>> K = NumberField(x**Integer(3) + Integer(128), names=('a',)); (a,) = K._
↪first_ngens(1)
>>> F = K.factor(Integer(2))[Integer(0)][Integer(0)].residue_field()
>>> F.reduction_map().codomain()
Residue field of Fractional ideal (1/4*a)

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(17))['t']; (t,) = R._first_ngens(1); P = R.
↪ideal(t**Integer(3) + t**Integer(2) + Integer(7))
>>> k = P.residue_field(names=('a',)); (a,) = k._first_ngens(1); f = k.
↪reduction_map(); f
Partially defined reduction map:
  From: Fraction Field of Univariate Polynomial Ring in t
        over Finite Field of size 17
  To:   Residue field in a of Principal ideal (t^3 + t^2 + 7) of
        Univariate Polynomial Ring in t over Finite Field of size 17
>>> f(Integer(1)/t)
```

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```
12*a^2 + 12*a
```

class sage.rings.finite_rings.residue_field.**ResidueFiniteField_prime_modn**(*p*, *name*, *intp*, *to_vs*, *to_order*, *PB*)

Bases: *ResidueField_generic*, *FiniteField_prime_modn*

The class representing residue fields of number fields that have prime order.

EXAMPLES:

```
sage: # needs sage.rings.number_field
sage: R.<x> = QQ[]
sage: K.<a> = NumberField(x^3 - 7)
sage: P = K.ideal(29).factor()[1][0]
sage: k = ResidueField(P); k
Residue field of Fractional ideal (-a^2 - 2*a - 2)
sage: k.order()
29
sage: OK = K.maximal_order()
sage: c = OK(a)
sage: b = k(a)
sage: k.coerce_map_from(OK)(c)
16
sage: k(4)
4
sage: k(c + 5)
21
sage: b + c
3

sage: # needs sage.rings.finite_rings
sage: R.<t> = GF(7)[]; P = R.ideal(2*t + 3)
sage: k = P.residue_field(); k
Residue field of Principal ideal (t + 5) of
 Univariate Polynomial Ring in t over Finite Field of size 7
sage: k(t^2)
4
sage: k.order()
7
```

```
>>> from sage.all import *
>>> # needs sage.rings.number_field
>>> R = QQ['x']; (x,) = R._first_ngens(1)
>>> K = NumberField(x**Integer(3) - Integer(7), names=('a',)); (a,) = K._first_
↪ngens(1)
>>> P = K.ideal(Integer(29)).factor()[Integer(1)][Integer(0)]
>>> k = ResidueField(P); k
Residue field of Fractional ideal (-a^2 - 2*a - 2)
>>> k.order()
29
>>> OK = K.maximal_order()
>>> c = OK(a)
```

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```
>>> b = k(a)
>>> k.coerce_map_from(OK)(c)
16
>>> k(Integer(4))
4
>>> k(c + Integer(5))
21
>>> b + c
3

>>> # needs sage.rings.finite_rings
>>> R = GF(Integer(7))['t']; (t,) = R._first_ngens(1); P = R.ideal(Integer(2)*t +␣
↪Integer(3))
>>> k = P.residue_field(); k
Residue field of Principal ideal (t + 5) of
 Univariate Polynomial Ring in t over Finite Field of size 7
>>> k(t**Integer(2))
4
>>> k.order()
7
```

7.2 Algebraic closures of finite fields

Let $\mathbf{F}$ be a finite field, and let $\overline{\mathbf{F}}$ be an algebraic closure of $\mathbf{F}$; this is unique up to (non-canonical) isomorphism. For every $n \geq 1$, there is a unique subfield $\mathbf{F}_n$ of $\overline{\mathbf{F}}$ such that $\mathbf{F} \subset \mathbf{F}_n$ and $[\mathbf{F}_n : \mathbf{F}] = n$.

In Sage, algebraic closures of finite fields are implemented using compatible systems of finite fields. The resulting Sage object keeps track of a finite lattice of the subfields $\mathbf{F}_n$ and the embeddings between them. This lattice is extended as necessary.

The Sage class corresponding to $\overline{\mathbf{F}}$ can be constructed from the finite field $\mathbf{F}$ by using the `algebraic_closure()` method.

The Sage class for elements of $\overline{\mathbf{F}}$ is `AlgebraicClosureFiniteFieldElement`. Such an element is represented as an element of one of the $\mathbf{F}_n$. This means that each element $x \in \mathbf{F}$ has infinitely many different representations, one for each n such that x is in $\mathbf{F}_n$.

Note

Only prime finite fields are currently accepted as base fields for algebraic closures. To obtain an algebraic closure of a non-prime finite field $\mathbf{F}$, take an algebraic closure of the prime field of $\mathbf{F}$ and embed $\mathbf{F}$ into this.

Algebraic closures of finite fields are currently implemented using (pseudo-)Conway polynomials; see `AlgebraicClosureFiniteField_pseudo_conway` and the module `conway_polynomials`. Other implementations may be added by creating appropriate subclasses of `AlgebraicClosureFiniteField_generic`.

In the current implementation, algebraic closures do not satisfy the unique parent condition. Moreover, there is no coercion map between different algebraic closures of the same finite field. There is a conceptual reason for this, namely that the definition of pseudo-Conway polynomials only determines an algebraic closure up to *non-unique* isomorphism. This means in particular that different algebraic closures, and their respective elements, never compare equal.

AUTHORS:

- Peter Bruin (August 2013): initial version
- Vincent Delecroix (November 2013): additional methods

`sage.rings.algebraic_closure_finite_field.`**`AlgebraicClosureFiniteField`**(*base_ring*, *name*, *category=None*, *implementation=None*, ***kwds*)

Construct an algebraic closure of a finite field.

The recommended way to use this functionality is by calling the `algebraic_closure()` method of the finite field.

> **Note**
>
> Algebraic closures of finite fields in Sage do not have the unique representation property, because they are not determined up to unique isomorphism by their defining data.

EXAMPLES:

```
sage: from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
sage: F = GF(2).algebraic_closure()
sage: F1 = AlgebraicClosureFiniteField(GF(2), 'z')
sage: F1 is F
False
```

```
>>> from sage.all import *
>>> from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
>>> F = GF(Integer(2)).algebraic_closure()
>>> F1 = AlgebraicClosureFiniteField(GF(Integer(2)), 'z')
>>> F1 is F
False
```

In the pseudo-Conway implementation, non-identical instances never compare equal:

```
sage: F1 == F
False
sage: loads(dumps(F)) == F
False
```

```
>>> from sage.all import *
>>> F1 == F
False
>>> loads(dumps(F)) == F
False
```

This is to ensure that the result of comparing two instances cannot change with time.

class `sage.rings.algebraic_closure_finite_field.`**`AlgebraicClosureFiniteFieldElement`**(*parent*, *value*)

Bases: `FieldElement`

Element of an algebraic closure of a finite field.

EXAMPLES:

```
sage: F = GF(3).algebraic_closure()
sage: F.gen(2)
z2
sage: type(F.gen(2))
<class 'sage.rings.algebraic_closure_finite_field.AlgebraicClosureFiniteField_
↪pseudo_conway_with_category.element_class'>
```

```
>>> from sage.all import *
>>> F = GF(Integer(3)).algebraic_closure()
>>> F.gen(Integer(2))
z2
>>> type(F.gen(Integer(2)))
<class 'sage.rings.algebraic_closure_finite_field.AlgebraicClosureFiniteField_
↪pseudo_conway_with_category.element_class'>
```

as_finite_field_element(*minimal=False*)

Return `self` as a finite field element.

INPUT:

- `minimal` – boolean (default: `False`); if `True`, always return the smallest subfield containing `self`

OUTPUT:

- a triple (`field`, `element`, `morphism`) where `field` is a finite field, `element` an element of `field` and `morphism` a morphism from `field` to `self.parent()`.

EXAMPLES:

```
sage: F = GF(3).algebraic_closure('t')
sage: t = F.gen(5)
sage: t.as_finite_field_element()
(Finite Field in t5 of size 3^5,
 t5,
 Ring morphism:
  From: Finite Field in t5 of size 3^5
  To:   Algebraic closure of Finite Field of size 3
  Defn: t5 |--> t5)
```

```
>>> from sage.all import *
>>> F = GF(Integer(3)).algebraic_closure('t')
>>> t = F.gen(Integer(5))
>>> t.as_finite_field_element()
(Finite Field in t5 of size 3^5,
 t5,
 Ring morphism:
  From: Finite Field in t5 of size 3^5
  To:   Algebraic closure of Finite Field of size 3
  Defn: t5 |--> t5)
```

By default, `field` is not necessarily minimal. We can force it to be minimal using the `minimal` option:

```
sage: s = t + 1 - t
sage: s.as_finite_field_element()[0]
Finite Field in t5 of size 3^5
sage: s.as_finite_field_element(minimal=True)[0]
Finite Field of size 3
```

```
>>> from sage.all import *
>>> s = t + Integer(1) - t
>>> s.as_finite_field_element()[Integer(0)]
Finite Field in t5 of size 3^5
>>> s.as_finite_field_element(minimal=True)[Integer(0)]
Finite Field of size 3
```

This also works when the element has to be converted between two non-trivial finite subfields (see Issue #16509):

```
sage: K = GF(5).algebraic_closure()
sage: z = K.gen(5) - K.gen(5) + K.gen(2)
sage: z.as_finite_field_element(minimal=True)
(Finite Field in z2 of size 5^2, z2, Ring morphism:
   From: Finite Field in z2 of size 5^2
   To:   Algebraic closure of Finite Field of size 5
   Defn: z2 |--> z2)
```

```
>>> from sage.all import *
>>> K = GF(Integer(5)).algebraic_closure()
>>> z = K.gen(Integer(5)) - K.gen(Integer(5)) + K.gen(Integer(2))
>>> z.as_finite_field_element(minimal=True)
(Finite Field in z2 of size 5^2, z2, Ring morphism:
   From: Finite Field in z2 of size 5^2
   To:   Algebraic closure of Finite Field of size 5
   Defn: z2 |--> z2)
```

There are automatic coercions between the various subfields:

```
sage: a = K.gen(2) + 1
sage: _,b,_ = a.as_finite_field_element()
sage: K4 = K.subfield(4)[0]
sage: K4(b)
z4^3 + z4^2 + z4 + 4
sage: b.minimal_polynomial() == K4(b).minimal_polynomial()
True
sage: K(K4(b)) == K(b)
True
```

```
>>> from sage.all import *
>>> a = K.gen(Integer(2)) + Integer(1)
>>> _,b,_ = a.as_finite_field_element()
>>> K4 = K.subfield(Integer(4))[Integer(0)]
>>> K4(b)
z4^3 + z4^2 + z4 + 4
>>> b.minimal_polynomial() == K4(b).minimal_polynomial()
```

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```
True
>>> K(K4(b)) == K(b)
True
```

You can also use the inclusions that are implemented at the level of the algebraic closure:

```
sage: f = K.inclusion(2,4); f
Ring morphism:
  From: Finite Field in z2 of size 5^2
  To:   Finite Field in z4 of size 5^4
  Defn: z2 |--> z4^3 + z4^2 + z4 + 3
sage: f(b)
z4^3 + z4^2 + z4 + 4
```

```
>>> from sage.all import *
>>> f = K.inclusion(Integer(2),Integer(4)); f
Ring morphism:
  From: Finite Field in z2 of size 5^2
  To:   Finite Field in z4 of size 5^4
  Defn: z2 |--> z4^3 + z4^2 + z4 + 3
>>> f(b)
z4^3 + z4^2 + z4 + 4
```

change_level(*n*)

Return a representation of `self` as an element of the subfield of degree n of the parent, if possible.

EXAMPLES:

```
sage: F = GF(3).algebraic_closure()
sage: z = F.gen(4)
sage: (z^10).change_level(6)
2*z6^5 + 2*z6^3 + z6^2 + 2*z6 + 2
sage: z.change_level(6)
Traceback (most recent call last):
...
ValueError: z4 is not in the image of Ring morphism:
  From: Finite Field in z2 of size 3^2
  To:   Finite Field in z4 of size 3^4
  Defn: z2 |--> 2*z4^3 + 2*z4^2 + 1

sage: a = F(1).change_level(3); a
1
sage: a.change_level(2)
1
sage: F.gen(3).change_level(1)
Traceback (most recent call last):
...
ValueError: z3 is not in the image of Ring morphism:
  From: Finite Field of size 3
  To:   Finite Field in z3 of size 3^3
  Defn: 1 |--> 1
```

```
>>> from sage.all import *
>>> F = GF(Integer(3)).algebraic_closure()
>>> z = F.gen(Integer(4))
>>> (z**Integer(10)).change_level(Integer(6))
2*z6^5 + 2*z6^3 + z6^2 + 2*z6 + 2
>>> z.change_level(Integer(6))
Traceback (most recent call last):
...
ValueError: z4 is not in the image of Ring morphism:
  From: Finite Field in z2 of size 3^2
  To:   Finite Field in z4 of size 3^4
  Defn: z2 |--> 2*z4^3 + 2*z4^2 + 1

>>> a = F(Integer(1)).change_level(Integer(3)); a
1
>>> a.change_level(Integer(2))
1
>>> F.gen(Integer(3)).change_level(Integer(1))
Traceback (most recent call last):
...
ValueError: z3 is not in the image of Ring morphism:
  From: Finite Field of size 3
  To:   Finite Field in z3 of size 3^3
  Defn: 1 |--> 1
```

is_square()

Return `True` if `self` is a square.

This always returns `True`.

EXAMPLES:

```
sage: F = GF(3).algebraic_closure()
sage: F.gen(2).is_square()
True
```

```
>>> from sage.all import *
>>> F = GF(Integer(3)).algebraic_closure()
>>> F.gen(Integer(2)).is_square()
True
```

minimal_polynomial()

Return the minimal polynomial of `self` over the prime field.

EXAMPLES:

```
sage: F = GF(11).algebraic_closure()
sage: F.gen(3).minpoly()
x^3 + 2*x + 9
```

```
>>> from sage.all import *
>>> F = GF(Integer(11)).algebraic_closure()
>>> F.gen(Integer(3)).minpoly()
x^3 + 2*x + 9
```

minpoly()

Return the minimal polynomial of `self` over the prime field.

EXAMPLES:

```
sage: F = GF(11).algebraic_closure()
sage: F.gen(3).minpoly()
x^3 + 2*x + 9
```

```
>>> from sage.all import *
>>> F = GF(Integer(11)).algebraic_closure()
>>> F.gen(Integer(3)).minpoly()
x^3 + 2*x + 9
```

multiplicative_order()

Return the multiplicative order of `self`.

EXAMPLES:

```
sage: K = GF(7).algebraic_closure()
sage: K.gen(5).multiplicative_order()
16806
sage: (K.gen(1) + K.gen(2) + K.gen(3)).multiplicative_order()
7353
```

```
>>> from sage.all import *
>>> K = GF(Integer(7)).algebraic_closure()
>>> K.gen(Integer(5)).multiplicative_order()
16806
>>> (K.gen(Integer(1)) + K.gen(Integer(2)) + K.gen(Integer(3))).
↪multiplicative_order()
7353
```

nth_root(n)

Return an n-th root of `self`.

EXAMPLES:

```
sage: F = GF(5).algebraic_closure()
sage: t = F.gen(2) + 1
sage: s = t.nth_root(15); s
4*z6^5 + 3*z6^4 + 2*z6^3 + 2*z6^2 + 4
sage: s**15 == t
True
```

```
>>> from sage.all import *
>>> F = GF(Integer(5)).algebraic_closure()
>>> t = F.gen(Integer(2)) + Integer(1)
>>> s = t.nth_root(Integer(15)); s
4*z6^5 + 3*z6^4 + 2*z6^3 + 2*z6^2 + 4
>>> s**Integer(15) == t
True
```

Todo

This function could probably be made faster.

pth_power(*k=1*)

Return the p^k-th power of `self`, where p is the characteristic of `self.parent()`.

EXAMPLES:

```
sage: K = GF(13).algebraic_closure('t')
sage: t3 = K.gen(3)
sage: s = 1 + t3 + t3**2
sage: s.pth_power()
10*t3^2 + 6*t3
sage: s.pth_power(2)
2*t3^2 + 6*t3 + 11
sage: s.pth_power(3)
t3^2 + t3 + 1
sage: s.pth_power(3).parent() is K
True
```

```
>>> from sage.all import *
>>> K = GF(Integer(13)).algebraic_closure('t')
>>> t3 = K.gen(Integer(3))
>>> s = Integer(1) + t3 + t3**Integer(2)
>>> s.pth_power()
10*t3^2 + 6*t3
>>> s.pth_power(Integer(2))
2*t3^2 + 6*t3 + 11
>>> s.pth_power(Integer(3))
t3^2 + t3 + 1
>>> s.pth_power(Integer(3)).parent() is K
True
```

pth_root(*k=1*)

Return the unique p^k-th root of `self`, where p is the characteristic of `self.parent()`.

EXAMPLES:

```
sage: K = GF(13).algebraic_closure('t')
sage: t3 = K.gen(3)
sage: s = 1 + t3 + t3**2
sage: s.pth_root()
2*t3^2 + 6*t3 + 11
sage: s.pth_root(2)
10*t3^2 + 6*t3
sage: s.pth_root(3)
t3^2 + t3 + 1
sage: s.pth_root(2).parent() is K
True
```

```
>>> from sage.all import *
>>> K = GF(Integer(13)).algebraic_closure('t')
```

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```
>>> t3 = K.gen(Integer(3))
>>> s = Integer(1) + t3 + t3**Integer(2)
>>> s.pth_root()
2*t3^2 + 6*t3 + 11
>>> s.pth_root(Integer(2))
10*t3^2 + 6*t3
>>> s.pth_root(Integer(3))
t3^2 + t3 + 1
>>> s.pth_root(Integer(2)).parent() is K
True
```

sqrt(*all=False*)

Return a square root of `self`.

If the optional keyword argument `all` is set to `True`, return a list of all square roots of `self` instead.

EXAMPLES:

```
sage: F = GF(3).algebraic_closure()
sage: F.gen(2).sqrt()
z4^3 + z4 + 1
sage: F.gen(2).sqrt(all=True)
[z4^3 + z4 + 1, 2*z4^3 + 2*z4 + 2]
sage: (F.gen(2)^2).sqrt()
z2
sage: (F.gen(2)^2).sqrt(all=True)
[z2, 2*z2]
```

```
>>> from sage.all import *
>>> F = GF(Integer(3)).algebraic_closure()
>>> F.gen(Integer(2)).sqrt()
z4^3 + z4 + 1
>>> F.gen(Integer(2)).sqrt(all=True)
[z4^3 + z4 + 1, 2*z4^3 + 2*z4 + 2]
>>> (F.gen(Integer(2))**Integer(2)).sqrt()
z2
>>> (F.gen(Integer(2))**Integer(2)).sqrt(all=True)
[z2, 2*z2]
```

class sage.rings.algebraic_closure_finite_field.**AlgebraicClosureFiniteField_generic**(*base_ring*, *name*, *category=None*)

Bases: `Field`

Algebraic closure of a finite field.

Element

alias of `AlgebraicClosureFiniteFieldElement`

algebraic_closure()

Return an algebraic closure of `self`.

This always returns `self`.

EXAMPLES:

```
sage: from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
sage: F = AlgebraicClosureFiniteField(GF(5), 'z')
sage: F.algebraic_closure() is F
True
```

```
>>> from sage.all import *
>>> from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
>>> F = AlgebraicClosureFiniteField(GF(Integer(5)), 'z')
>>> F.algebraic_closure() is F
True
```

characteristic()

Return the characteristic of `self`.

EXAMPLES:

```
sage: from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
sage: p = next_prime(1000)
sage: F = AlgebraicClosureFiniteField(GF(p), 'z')
sage: F.characteristic() == p
True
```

```
>>> from sage.all import *
>>> from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
>>> p = next_prime(Integer(1000))
>>> F = AlgebraicClosureFiniteField(GF(p), 'z')
>>> F.characteristic() == p
True
```

gen(n)

Return the n-th generator of `self`.

EXAMPLES:

```
sage: from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
sage: F = AlgebraicClosureFiniteField(GF(5), 'z')
sage: F.gen(2)
z2
```

```
>>> from sage.all import *
>>> from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
>>> F = AlgebraicClosureFiniteField(GF(Integer(5)), 'z')
>>> F.gen(Integer(2))
z2
```

gens()

Return a family of generators of `self`.

OUTPUT:

- a `Family`, indexed by the positive integers, whose n-th element is `self.gen(n)`.

EXAMPLES:

```
sage: from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
sage: F = AlgebraicClosureFiniteField(GF(5), 'z')
sage: g = F.gens(); g
Lazy family (...(i))_{i in Positive integers}
sage: g[3]
z3
```

```
>>> from sage.all import *
>>> from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
>>> F = AlgebraicClosureFiniteField(GF(Integer(5)), 'z')
>>> g = F.gens(); g
Lazy family (...(i))_{i in Positive integers}
>>> g[Integer(3)]
z3
```

inclusion(m, n)

Return the canonical inclusion map from the subfield of degree m to the subfield of degree n.

EXAMPLES:

```
sage: F = GF(3).algebraic_closure()
sage: F.inclusion(1, 2)
Ring morphism:
  From: Finite Field of size 3
  To:   Finite Field in z2 of size 3^2
  Defn: 1 |--> 1
sage: F.inclusion(2, 4)
Ring morphism:
  From: Finite Field in z2 of size 3^2
  To:   Finite Field in z4 of size 3^4
  Defn: z2 |--> 2*z4^3 + 2*z4^2 + 1
```

```
>>> from sage.all import *
>>> F = GF(Integer(3)).algebraic_closure()
>>> F.inclusion(Integer(1), Integer(2))
Ring morphism:
  From: Finite Field of size 3
  To:   Finite Field in z2 of size 3^2
  Defn: 1 |--> 1
>>> F.inclusion(Integer(2), Integer(4))
Ring morphism:
  From: Finite Field in z2 of size 3^2
  To:   Finite Field in z4 of size 3^4
  Defn: z2 |--> 2*z4^3 + 2*z4^2 + 1
```

ngens()

Return the number of generators of `self`, which is infinity.

EXAMPLES:

```
sage: from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
sage: AlgebraicClosureFiniteField(GF(5), 'z').ngens()
+Infinity
```

```
>>> from sage.all import *
>>> from sage.rings.algebraic_closure_finite_field import␣
↪AlgebraicClosureFiniteField
>>> AlgebraicClosureFiniteField(GF(Integer(5)), 'z').ngens()
+Infinity
```

some_elements()

Return some elements of this field.

EXAMPLES:

```
sage: F = GF(7).algebraic_closure()
sage: F.some_elements()
(1, z2, z3 + 1)
```

```
>>> from sage.all import *
>>> F = GF(Integer(7)).algebraic_closure()
>>> F.some_elements()
(1, z2, z3 + 1)
```

subfield(n)

Return the unique subfield of degree n of `self` together with its canonical embedding into `self`.

EXAMPLES:

```
sage: F = GF(3).algebraic_closure()
sage: F.subfield(1)
(Finite Field of size 3,
 Ring morphism:
   From: Finite Field of size 3
   To:   Algebraic closure of Finite Field of size 3
   Defn: 1 |--> 1)
sage: F.subfield(4)
(Finite Field in z4 of size 3^4,
 Ring morphism:
   From: Finite Field in z4 of size 3^4
   To:   Algebraic closure of Finite Field of size 3
   Defn: z4 |--> z4)
```

```
>>> from sage.all import *
>>> F = GF(Integer(3)).algebraic_closure()
>>> F.subfield(Integer(1))
(Finite Field of size 3,
 Ring morphism:
```

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```
  From: Finite Field of size 3
  To:   Algebraic closure of Finite Field of size 3
  Defn: 1 |--> 1)
>>> F.subfield(Integer(4))
(Finite Field in z4 of size 3^4,
 Ring morphism:
  From: Finite Field in z4 of size 3^4
  To:   Algebraic closure of Finite Field of size 3
  Defn: z4 |--> z4)
```

class sage.rings.algebraic_closure_finite_field.**AlgebraicClosureFiniteField_pseudo_conway**(*base_ring*, *name*, *category=None*, *lattice=None*, *use_database=*

Bases: `WithEqualityById`, *`AlgebraicClosureFiniteField_generic`*

Algebraic closure of a finite field, constructed using pseudo-Conway polynomials.

EXAMPLES:

```
sage: F = GF(5).algebraic_closure(implementation='pseudo_conway')
sage: F.cardinality()
+Infinity
sage: F.algebraic_closure() is F
True
sage: x = F(3).nth_root(12); x
z4^3 + z4^2 + 4*z4
sage: x**12
3
```

```
>>> from sage.all import *
>>> F = GF(Integer(5)).algebraic_closure(implementation='pseudo_conway')
>>> F.cardinality()
+Infinity
>>> F.algebraic_closure() is F
True
>>> x = F(Integer(3)).nth_root(Integer(12)); x
z4^3 + z4^2 + 4*z4
>>> x**Integer(12)
3
```

7.3 Routines for Conway and pseudo-Conway polynomials

AUTHORS:

- David Roe
- Jean-Pierre Flori
- Peter Bruin

class sage.rings.finite_rings.conway_polynomials.**PseudoConwayLattice**(*p*, *use_database=True*)

Bases: WithEqualityById, SageObject

A pseudo-Conway lattice over a given finite prime field.

The Conway polynomial f_n of degree n over $\mathbf{F}_p$ is defined by the following four conditions:

- f_n is irreducible.
- In the quotient field $\mathbf{F}_p[x]/(f_n)$, the element x mod f_n generates the multiplicative group.
- The minimal polynomial of $(x \bmod f_n)^{\frac{p^n-1}{p^m-1}}$ equals the Conway polynomial f_m, for every divisor m of n.
- f_n is lexicographically least among all such polynomials, under a certain ordering.

The final condition is needed only in order to make the Conway polynomial unique. We define a pseudo-Conway lattice to be any family of polynomials, indexed by the positive integers, satisfying the first three conditions.

INPUT:

- p – prime number
- use_database – boolean. If True, use actual Conway polynomials whenever they are available in the database. If False, always compute pseudo-Conway polynomials.

EXAMPLES:

```
sage: # needs sage.rings.finite_rings
sage: from sage.rings.finite_rings.conway_polynomials import PseudoConwayLattice
sage: PCL = PseudoConwayLattice(2, use_database=False)
sage: PCL.polynomial(3)    # random
x^3 + x + 1
```

```
>>> from sage.all import *
>>> # needs sage.rings.finite_rings
>>> from sage.rings.finite_rings.conway_polynomials import PseudoConwayLattice
>>> PCL = PseudoConwayLattice(Integer(2), use_database=False)
>>> PCL.polynomial(Integer(3))    # random
x^3 + x + 1
```

check_consistency(*n*)

Check that the pseudo-Conway polynomials of degree dividing n in this lattice satisfy the required compatibility conditions.

EXAMPLES:

```
sage: # needs sage.rings.finite_rings
sage: from sage.rings.finite_rings.conway_polynomials import␣
↪PseudoConwayLattice
sage: PCL = PseudoConwayLattice(2, use_database=False)
sage: PCL.check_consistency(6)
sage: PCL.check_consistency(60)  # long time
```

```
>>> from sage.all import *
>>> # needs sage.rings.finite_rings
>>> from sage.rings.finite_rings.conway_polynomials import PseudoConwayLattice
>>> PCL = PseudoConwayLattice(Integer(2), use_database=False)
>>> PCL.check_consistency(Integer(6))
>>> PCL.check_consistency(Integer(60))  # long time
```

polynomial(*n*)

Return the pseudo-Conway polynomial of degree n in this lattice.

INPUT:

- n – positive integer

OUTPUT: a pseudo-Conway polynomial of degree n for the prime p

ALGORITHM:

Uses an algorithm described in [HL1999], modified to find pseudo-Conway polynomials rather than Conway polynomials. The major difference is that we stop as soon as we find a primitive polynomial.

EXAMPLES:

```
sage: # needs sage.rings.finite_rings
sage: from sage.rings.finite_rings.conway_polynomials import␣
↪PseudoConwayLattice
sage: PCL = PseudoConwayLattice(2, use_database=False)
sage: PCL.polynomial(3)    # random
x^3 + x + 1
sage: PCL.polynomial(4)    # random
x^4 + x^3 + 1
sage: PCL.polynomial(60)   # random
x^60 + x^59 + x^58 + x^55 + x^54 + x^53 + x^52 + x^51 + x^48 + x^46 + x^45 +␣
↪x^42 + x^41 + x^39 + x^38 + x^37 + x^35 + x^32 + x^31 + x^30 + x^28 + x^24␣
↪+ x^22 + x^21 + x^18 + x^17 + x^16 + x^15 + x^14 + x^10 + x^8 + x^7 + x^5 +␣
↪x^3 + x^2 + x + 1
```

```
>>> from sage.all import *
>>> # needs sage.rings.finite_rings
>>> from sage.rings.finite_rings.conway_polynomials import PseudoConwayLattice
>>> PCL = PseudoConwayLattice(Integer(2), use_database=False)
>>> PCL.polynomial(Integer(3))    # random
x^3 + x + 1
>>> PCL.polynomial(Integer(4))    # random
x^4 + x^3 + 1
>>> PCL.polynomial(Integer(60))   # random
x^60 + x^59 + x^58 + x^55 + x^54 + x^53 + x^52 + x^51 + x^48 + x^46 + x^45 +␣
↪x^42 + x^41 + x^39 + x^38 + x^37 + x^35 + x^32 + x^31 + x^30 + x^28 + x^24␣
↪+ x^22 + x^21 + x^18 + x^17 + x^16 + x^15 + x^14 + x^10 + x^8 + x^7 + x^5 +␣
↪x^3 + x^2 + x + 1
```

sage.rings.finite_rings.conway_polynomials.**conway_polynomial**(*p*, *n*)

Return the Conway polynomial of degree n over GF(p).

If the requested polynomial is not known, this function raises a RuntimeError exception.

INPUT:

- p – prime number
- n – positive integer

OUTPUT:

- the Conway polynomial of degree n over the finite field GF(p), loaded from a table.

> **Note**
>
> The first time this function is called a table is read from disk, which takes a fraction of a second. Subsequent calls do not require reloading the table.

See also the `ConwayPolynomials()` object, which is the table of Conway polynomials used by this function.

EXAMPLES:

```
sage: conway_polynomial(2,5)                                                    #␣
↪needs conway_polynomials
x^5 + x^2 + 1
sage: conway_polynomial(101,5)                                                  #␣
↪needs conway_polynomials
x^5 + 2*x + 99
sage: conway_polynomial(97,101)                                                 #␣
↪needs conway_polynomials
Traceback (most recent call last):
...
RuntimeError: requested Conway polynomial not in database.
```

```
>>> from sage.all import *
>>> conway_polynomial(Integer(2),Integer(5))                                    ␣
↪              # needs conway_polynomials
x^5 + x^2 + 1
>>> conway_polynomial(Integer(101),Integer(5))                                  ␣
↪              # needs conway_polynomials
x^5 + 2*x + 99
>>> conway_polynomial(Integer(97),Integer(101))                                 ␣
↪              # needs conway_polynomials
Traceback (most recent call last):
...
RuntimeError: requested Conway polynomial not in database.
```

`sage.rings.finite_rings.conway_polynomials.`**`exists_conway_polynomial`**(p, n)

Check whether the Conway polynomial of degree n over `GF(p)` is known.

INPUT:

- `p` – prime number
- `n` – positive integer

OUTPUT:

- boolean: `True` if the Conway polynomial of degree n over `GF(p)` is in the database, `False` otherwise.

If the Conway polynomial is in the database, it can be obtained using the command `conway_polynomial(p,n)`.

EXAMPLES:

```
sage: exists_conway_polynomial(2,3)                                             #␣
↪needs conway_polynomials
True
sage: exists_conway_polynomial(2,-1)
False
```

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```
sage: exists_conway_polynomial(97,200)
False
sage: exists_conway_polynomial(6,6)
False
```

```
>>> from sage.all import *
>>> exists_conway_polynomial(Integer(2),Integer(3))
↪              # needs conway_polynomials
True
>>> exists_conway_polynomial(Integer(2),-Integer(1))
False
>>> exists_conway_polynomial(Integer(97),Integer(200))
False
>>> exists_conway_polynomial(Integer(6),Integer(6))
False
```

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